UNIVERSITÄT PADERBORN
Die Universität der Informationsgesellschaft

Faculty for Computer Science, Electrical Engineering
and Software Innovation Lab

Bachelor's Thesis
Submitted to the Faculty for Computer Science, Electrical Engineering and Mathematics
Software Innovation Lab
in Partial Fulfillment of the Requirements for the Degree of
Bachelor of Science

Behind the VR Wheel: Investigating the Impact of Assistance Systems on Driver Behavior

by
Salah Seidan

Supervisors:
Dr. Enes Yigitbas

Examiner:
1st: Dr. Enes Yigitbas
2nd: Dr. Simon Oberthür

Paderborn, September 9, 2024

Eidesstattliche Versicherung

Nachname: Seidan Vorname: Salah

Matrikelnr.: 6956862 Studiengang: Bachelor Informatik

☑ Bachelorarbeit ☐ Masterarbeit

Titel der Arbeit: Behind the VR Wheel: Investigating the Impact of Assistance Systems on Driver Behavior

☐ Die elektronische Fassung ist der Abschlussarbeit beigefügt.

☑ Die elektronische Fassung sende ich an die/den erste/n Prüfenden bzw. habe ich an die/den erste/n Prüfenden gesendet.

Ich versichere hiermit an Eides statt, dass ich die vorliegende Abschlussarbeit (Ausarbeitung inkl. Tabellen, Zeichnungen, etc.) selbstständig und ohne unzulässige fremde Hilfe erbracht habe. Ich habe keine anderen als die angegebenen Quellen und Hilfsmittel benutzt sowie wörtliche und sinngemäße Zitate kenntlich gemacht. Die Abschlussarbeit hat in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegen. Die elektronische Fassung entspricht der gedruckten und gebundenen Fassung.

Belehrung

Wer vorsätzlich gegen eine die Täuschung über Prüfungsleistungen betreffende Regelung einer Hochschulprüfungsordnung verstößt, handelt ordnungswidrig. Die Ordnungswidrigkeit kann mit einer Geldbuße von bis zu 50.000,00 € geahndet werden. Zuständige Verwaltungsbehörde für die Verfolgung und Ahndung von Ordnungswidrigkeiten ist die Vizepräsidentin / der Vizepräsident für Wirtschafts- und Personalverwaltung der Universität Paderborn. Im Falle eines mehrfachen oder sonstigen schwerwiegenden Täuschungsversuches kann der Prüfling zudem exmatrikuliert werden. (§ 63 Abs. 5 Hochschulgesetz NRW in der aktuellen Fassung).

Die Universität Paderborn wird ggf. eine elektronische Überprüfung der Abschlussarbeit durchführen, um eine Täuschung festzustellen.

Ich habe die oben genannten Belehrungen gelesen und verstanden und bestätige dieses mit meiner Unterschrift.

Ort: Paderborn Datum: September 9, 2024

Unterschrift:

Datenschutzhinweis

Die o.g. Daten werden aufgrund der geltenden Prüfungsordnung (Paragraph zur Abschlussarbeit) i.V.m. § 63 Abs. 5 Hochschulgesetz NRW erhoben. Auf Grundlage der übermittelten Daten (Name, Vorname, Matrikelnummer, Studiengang, Art und Thema der Abschlussarbeit) wird bei Plagiaten bzw. Täuschung der/die Prüfende und der Prüfungsausschuss Ihres Studienganges über Konsequenzen gemäß Prüfungsordnung i.V.m. Hochschulgesetz NRW entscheiden. Die Daten werden nach Abschluss des Prüfungsverfahrens gelöscht. Eine Weiterleitung der Daten kann an die/den Prüfende/n und den Prüfungsausschuss erfolgen. Falls der Prüfungsausschuss entscheidet, eine Geldbuße zu verhängen, werden die Daten an die Vizepräsidentin für Wirtschafts- und Personalverwaltung weitergeleitet. Verantwortlich für die Verarbeitung im regulären Verfahren ist der Prüfungsausschuss Ihres Studiengangs der Universität Paderborn, für die Verfolgung und Ahndung der Geldbuße ist die Vizepräsidentin für Wirtschafts- und Personalverwaltung.

Abstract

Virtual reality (VR) Technology has potential to be a significant innovation in our century, especially as technology rapidly transforms various aspects of our lives. This thesis addresses the problem of conventional driving schools being costly and time-consuming. Additionally, new drivers often lack opportunities to practice after obtaining their licenses, which may result in accidents. To tackle these issues, this study proposes the development of a realistic and affordable VR driving simulator. The solution aims to enhance immersive capabilities compared to traditional methods, providing a safe platform for novice drivers to improve their skills. Advanced assistance systems are integrated into this virtual system to guide novices through potentially hazardous conditions. The VR simulator offers a unified package that combines VR technology with Advanced Driver Assistance Systems (ADAS), making learning to drive more accessible, effective, and safe. The outcome of this research is a significant improvement in learning engagement and effective understanding through simulated driving scenarios. The results demonstrate that the VR-based driving simulator can enhance driver behavior monitoring and the development of advanced driving assistance systems, ultimately contributing to improved road safety. This work opens new possibilities in driver education through the use of VR, addressing the various shortcomings of traditional driving schools and non-VR simulators.

Contents

1 Introduction **1**
- 1.1 Motivation . . . 1
- 1.2 Problem Statement . . . 2
- 1.3 Solution Overview . . . 6
- 1.4 Thesis Structure . . . 8

2 Foundations **11**
- 2.1 Virtual Reality (VR) Technology . . . 11
- 2.2 Driving Simulators . . . 12
- 2.3 Advanced Driving Assistance System . . . 13
- 2.4 Development in the Unity Game Engine . . . 14
- 2.5 Modelling in the Blender 3D Modelling Software . . . 15
- 2.6 Motion Sickness within the VR Environment . . . 17
- 2.7 Usability Evaluation in VR . . . 17

3 Related Work **20**
- 3.1 Review Research Questions . . . 20
- 3.2 Previous Work . . . 20
  - 3.2.1 VR-OOM: Virtual Reality On-road driving Simulation . . . 21
  - 3.2.2 DReyeVR: Democratizing Virtual Reality Driving Simulation for Behavioral and Interaction Research . . . 22
  - 3.2.3 Automated driving using Virtual Reality . . . 23
  - 3.2.4 Augmented Reality-Based Advanced Driver-Assistance System for Connected Vehicles . . . 24
- 3.3 Results . . . 26
  - 3.3.1 Immersive Mixed Reality Support . . . 27
  - 3.3.2 Driving Scenarios and Challenges . . . 28
  - 3.3.3 Driving Assistance System Integration . . . 29
  - 3.3.4 Detection of Driving Hazards . . . 29
  - 3.3.5 Development of Low-Cost Simulator . . . 31
- 3.4 Discussion . . . 32
  - 3.4.1 Findings . . . 32
  - 3.4.2 Threats to validity . . . 34
- 3.5 Summary . . . 34

4 Conception **36**
- 4.1 Solution Overview . . . 36
- 4.2 Virtual Environment . . . 39
  - 4.2.1 Environment Modes . . . 39
  - 4.2.2 Environment Daytime . . . 42

4.2.3 Non-VR vs VR mode . . . 42
4.3 Driving Assistance System . . . 43
4.3.1 System Architecture . . . 44
4.3.2 System Elements . . . 46
4.4 Realistic Vehicle Functions . . . 48
4.4.1 Car Engine Start . . . 48
4.4.2 Gear Shifting . . . 48
4.4.3 Clutch Control . . . 48
4.4.4 Accelerating and Braking . . . 49
4.4.5 Turning Signals . . . 49
4.4.6 Lights Control . . . 49
4.4.7 Force Feedback . . . 49
4.4.8 Side and Rear Mirrors . . . 50

5 Implementation **52**
5.1 Implementation Overview . . . 52
5.2 VR Headset . . . 53
5.3 Driving Kit . . . 54
5.4 Virtual Environment . . . 55
5.4.1 Training Area . . . 55
5.4.2 Paderborn City . . . 57
5.4.3 Highway . . . 57
5.4.4 Daytime Settings . . . 61
5.4.5 Non-VR VS VR Mode . . . 61
5.5 Driving Assistance System . . . 62
5.5.1 System Architecture . . . 62
5.5.2 System Elements . . . 65
5.6 Realistic Vehicle Functions . . . 68

6 Evaluation **72**
6.1 Case Study . . . 72
6.2 Method . . . 73
6.2.1 General Design . . . 74
6.2.2 Usability Measurements . . . 76
6.2.3 Participant Acquisition . . . 78
6.2.4 Execution Procedure . . . 79
6.3 Results . . . 79
6.3.1 Demographics . . . 80
6.3.2 Efficiency . . . 83
6.3.3 Effectiveness . . . 87
6.3.4 User Satisfaction . . . 91
6.3.5 User Immersion and Presence in Virtual Environments . . . 94
6.3.6 User Workload in Virtual Environments . . . 95
6.3.7 General Questionnaire . . . 97
6.3.8 Open Ended Questions . . . 101

6.4 Discussion 104
6.4.1 Findings 104
6.4.2 Threats to Validity 105

7 Conclusion **108**
7.1 Summary 108
7.2 Comparative Analysis 109
7.3 Future Work 110

Bibliography **113**

Appendix Structure **120**
A.1 Prototype Code 120
A.2 Install Unity 121
A.3 Build Settings 121

B Usability Study Questionnaire **125**

1 Introduction

This chapter will discuss the main goals, problems, and solutions of this thesis. In Section 1.1 Motivation, the reasons that sparked this research will be presented, highlighting its importance and relevance. Moving on to Section 1.2 Problem Statement, we will define the core problem this thesis aims to address. Section 1.3 Solution Overview, will provide a brief declaration of the solution implemented to solve the problems discussed before. Finally, Section 1.4 Thesis Structure, gives an overview of the thesis structure, guiding through the upcoming chapters and what to expect.

1.1 Motivation

With modern society and the rapid development of technologies, solutions for lots of problems that earlier were urgent are appearing; they make life easier day by day and provide new experiences of learning in multiple ways. Virtual reality is an advanced human-to-computer interface that imitates a real-life environment and allows its members to collaborate inside a digital environment. Unlike traditional interfaces, in VR one can look around freely and gain an understanding of the environment from different perspectives and control elements without resorting to conventional approaches [1]. It was the first generation launched in 1989 by Jaron Lannier, which underwent many changes and improvements, thereby enabling the production of any virtual environment that one can think of [2]. driving schools give the necessary training to aspiring drivers but come at a cost that is unaffordable for many. The process of learning to drive and getting a license to do so for novice drivers is usually very long and expensive. It normally takes about 45 hours of riding classes and 22 hours of exercise before novices can acquire the key competencies [2]. Besides, once drivers obtain their licenses, they are seldom given chances to practice. The number of traffic crashes involving inexperienced drivers is higher. To cope with that, driving simulators come as a safer option. Driving simulators make use of intuitive interfaces, cognitive processes, and behavioral feedback to develop skills in a controlled manner. Among various types of driving simulators, VR-based systems offer realistic experiences similar to natural driving scenarios [3]. Despite the prevalence of VR-based flight simulators, the provision of VR simulators for car use stays restricted, especially those compatible with HMDs. While conventional simulators using multiple screens exist, studies indicate that they are much less powerful for research [2]. This work discusses the implementation and evaluation of a virtual reality (VR)-based driving simulator as a prospective alternative to conventional driving simulators. It features an integrated advanced assistance system that helps the driver overcome some driving hazards and pay more attention to the surrounding environment. Driving schools are important for teaching new drivers

how to drive a car, but they can be too expensive for many people. Moreover, after receiving their driver's license, they don't have much opportunity to practice, and all that can make them prone to causing dangerous accidents. They need to find a digital driving simulator that has low-cost equipment and an immersive drive system where drivers will feel immersion and presence, like driving a real car. This is a reason why, in the future, driving simulators could be a probable alternative. These are designed to be user-friendly in handling and giving feedback, which will help in learning how to drive. VR driving simulators provide a realistic driving experience using advanced VR technologies. It would make them feel that their hands are on the steering wheel, making turns, braking, accelerating, and even using the clutch in manual mode. Even more so, several conditions could also be designed in the virtual environment regarding weather, road, and traffic situations that can be encountered. This will help a driver improve on logical thinking and decision-making skills, which are crucial to avoid severe accidents. These advantages have made VR-based driving simulators popular, and they find their wide applicability in several institutions for teaching the basics of driving to beginners. Monitoring a driver's behavior while operating a vehicle has attained importance, and several technologies have been developed to achieve it [4]. These technologies log critical driver decisions, various vehicle functions and gauges, fatigue values, and sometimes even video captured just before accidents, all of which could prove to be very valuable for insurance companies or law enforcement [4]. However, capturing and collecting data on a driver's behavior alone is not sufficient to prevent accidents and train the drivers effectively [5]. For this reason, many companies use extra technologies in order to guide drivers and give them instructions while driving. The most well-known technology in such applications is one called the Advanced Driver Assistance System, ADAS. ADAS is an integrated system that is mainly applied in vehicles; it provides the main driving instructions, automates repetitive tasks, prevents fatal accidents, and improves traffic safety. This technology has also enhanced driving for both novice and experienced drivers [5]. Given the numerous benefits of ADAS, it is expected to become increasingly important in the future. The primary objective of this thesis is to implement and evaluate a virtual reality-based driving simulator, which could serve as a cost-effective, immersive, and highly realistic alternative to traditional driving schools or non-VR driving simulators that rely on expensive equipment. The thesis will first address the challenges faced by researchers in this field and then propose improved solutions and technologies to overcome these challenges. These solutions must meet several criteria to achieve the thesis's goals, including being cost-effective, user-friendly, safe for health, immersive, present, and educational.

1.2 Problem Statement

This study will be aimed at solving many of the problems that usually face the conventional driving simulator used by so many driving schools or any individual who wants to improve his driving skills and avoid as many accidents on the roads as possible. In this regard, a VR-based driving assistance system will be imple-

mented and assessed with many study cases that test the improvement in driving performance of the users and potentially replace the role of a driving instructor, which decreases the costs of driving licenses. Many technical features make VR Head-Mounted-Displays more realistic and, therefore, closer to the real-world scenario. Amongst the technological capabilities that make VR-based applications differ from traditional desktop applications are the stereoscopic 3D images of VR-based HMD [6]. 3D stereoscopic images give users the feeling of being in a 3D virtual environment, which is very important to simulate real-world conditions and make it an attractive feature to test many studies in different fields [6]. In driving simulators using 2D desktop displays makes it harder to navigate between different views and observe the environment around the vehicle. Otherwise, our real world is in 3D, not in 2D, which makes the 3D environment in many simulators more immersive and realistic [7]. Another technology, that the VR-based HMD has is the Six Degrees of Freedom or 6-DOF, it gives users the ability to rotate their head in 3 axes X, Y, and Z which is impossible in conventional simulators. Additionally to the three rotational axes it can also track user's positional movements on three axes X, Y and Z, that means overall the user inside VR applications will have Six Degrees of Freedom of movement and rotation, simulating the real-world movement [8]. 3D stereoscopic and 6-DOF features of VR-HMD allow it to be a better solution than the traditional Non-VR simulators, due to the simulation of the real movement, that drivers make inside a real car, which will make the driving experience inside these simulators more realistic, presence and immersive [9]. Another challenge that can face us while implementing and evaluating driving simulators is choosing a suitable immersive driving setup. Selecting a suitable driving simulator depends on the needs and purposes of the project [10]. Driving Simulators vary from basic and compact configurations to advanced, expensive systems [9]. The first type is the advanced Auto-Sim simulator and an example for it is the AS 1200-6. It is constructed using the Opel Astra IV vehicle body kit. It employs a motion platform with 6 degrees of freedom to imitate the driver's movements and accelerations [11]. The second type is the basic simulator. It has been developed to provide effective education and training on ADAS. Referred to as the ADAS operation training simulator, it comprises three 40-inch displays, a driver's seat, a steering wheel, and operation pedals, with one display in front of the driver's seat and the others extending at 45-degree angles from the right and left of the front display. The simulator's steering wheel controller incorporates a force feedback function and can mimic the behavior of the Lane Keep Assist System [12]. The last type is a compact driving simulator. We depend on this simulator type in this study because it integrates the VR-based HMD. It is a low-cost simulator in comparison with other types and uses a Head-Mounted-Display (HMD) combined with a steering wheel, gearbox, and pedal set from Logitech. This project utilizes a portable mixed-reality HMD, the Meta Quest 3, with the Logitech G923 driving set. All these devices are connected to a high-end PC equipped with an NVIDIA Geforce RTX 4080 graphic card, allows a realistic and immersive VR experience with effective and efficient driving training [10]. A Further challenge is to be able to capture and Analyze driver's Behavior and Interaction Patterns. Monitoring and analyzing drivers' behavior is crucial, particularly in driving simulators. This helps

us understand how drivers with different skill levels handle their vehicles in different scenarios [13]. Tracking drivers' behavior has always been essential for police and insurance companies to determine accident responsibility and identify dangerous driving actions. Also, collection of drivers' behavioral data in a virtual environment differs from that in real life. In practical conditions, cameras, sensors, or devices recording vital statistics of drivers while driving may be used by researchers to gather the data [14]. Contrarily, in a controlled virtual environment by driving simulators, we can capture drivers' behavior by recording inputs from the steering wheel, acceleration, braking actions, lane-changing frequency, interactions with traffic signals, adhere to the traffic lights, other cars and pedestrians, control the car speed according to the allowed speed limits in different streets. Driver Behaviour Metrics. After capturing and analyzing the data related to the driver's behavior, it is important to realize an appropriate system that checks if the drivers' actions are safe and appropriate regarding the current situation. The assistance system shall also give corrective guidance in case of necessity. Many accidents take place due to unavailability or inappropriate instruction systems inside vehicles or at the road, such as signs and markings [15]. According to research, most driving errors and hazards occur due to natural distractions. Diminishing these, the following modern technologies of connected vehicles, in-vehicle screens, cooperative systems, and ADAS will be implemented and utilized for this study. The last significant problem of VR driving simulators is Simulator Sickness. A kind of motion sickness within simulated environments, SSN can arise in the absence of any real motion and may appear at any instant of the exposure. Thus, the aftermath of the effects can be very long and is especially dangerous in driving a car. One of the negative impacts that VR imparts on the human body is VR sickness, also referred to as cybersickness or visually induced motion sickness. Symptoms include vomiting, nausea, and dizziness, often discouraging users from using VR again. The primary cause of VR sickness is the sensory conflict theory, which describes the discomfort VR users feel due to the mismatch between the actual motion perceived by the brain and the virtual motion experienced through a head-mounted display (HMD) [16]. Overcoming previously discussed challenges require an implementation of a driving simulator with an integrated assistance system, which will be used in many use cases, which will be discussed in Evaluation section 6.1 case study, The results will be evaluated to answer the main two research questions of this thesis:

RQ1: How much can an assistance system in a VR-based driving simulator enhance driving performance and potentially replace the role of a driving instructor?

RQ2: How does the transition from non-VR to VR advanced driving assistance system affect the performance scores and the usability criteria in a simulated environment?

The above problems imply certain challenges for the implementation of the prototype. Since the prototype is designed to leverage the potential of driving assistance systems in virtual reality, to enhance driving skills, and avoid accidents on the road, a range of features had to be implemented that make use of the discussed capa-

bilities of VR. This requires the incorporation of various features that utilize the identified capabilities of VR. Below is a list of essential features that stem from these capabilities, although it may not be comprehensive. These features were identified based on a thorough review of existing literature and previous works in the field of VR-based driving simulators, some of these works will be discussed and presented in Section 3.2. The primary challenges encountered during the implementation of the prototype involved ensuring that these features were developed to enhance the naturalness of collaboration and the usability of the system during the modeling and development process. These challenges are elaborated in the subsequent sections:

1. Immersive Virtual Reality Support
   - 1.1 Support for Head-Mounted Displays
   - 1.2 Head Movement with Six Degrees of Freedom
2. Driving Scenarios and Challenges
   - 2.1 Parking Training
   - 2.2 Highway Driving
   - 2.3 City Driving
   - 2.4 Night Driving
3. Driving Assistance System Integration
4. Detection of Driving Hazards
   - 4.1 Violations of Traffic Signals
   - 4.2 Detection of Lane Changes
   - 4.3 Adherence to Speed Limits
   - 4.4 Prevention of Accidents
5. Development of Low-Cost Simulator

Immersive Virtual Reality Support

Support for Head-Mounted Displays: This challenge involves ensuring compatibility with various head-mounted displays to provide an immersive experience. **Head Movement with Six Degrees of Freedom:** Implementing six degrees of freedom for head movement is crucial for a natural and intuitive user experience. Studies have indicated that this feature is essential for accurate spatial orientation and interaction within the VR environment.

Driving Scenarios and Challenges

Parking Training: This situation is vital for mimicking actual parking conditions, with research indicating that cities allocate around 50% to 70% of their land for parking [17]. Furthermore, the worldwide parking management industry is expected to grow to $12.91 billion by 2032, highlighting the crucial importance and influence

of effective parking strategies [18].
Highway Driving, City Driving, Night Driving: These scenarios cover a wide range of driving conditions that users may encounter. Each scenario presents unique challenges that need to be addressed to ensure comprehensive training. For example in night driving, the driver must rely on headlights and streetlights to navigate the road, and there is an increased chance of drowsy driving during the late-night hours [19]. In city driving, drivers must navigate through complex intersections, pedestrian crossings, and traffic signals, which require quick decision-making and attention to details [20]. Highway driving involves high-speed driving and lane changes, which can be challenging for novice drivers [21].

Driving Assistance System Integration

Integrating driving assistance systems such as lane-keeping assist and adaptive cruise control is essential for providing a realistic driving experience.

Detection of Driving Hazards

Violations of Traffic Signals, Detection of Lane Changes, Adherence to Speed Limits, Prevention of Accidents: These features are crucial for simulating real-world driving conditions and ensuring that users can practice safe driving habits. Studies have shown that hazard detection and response training can significantly reduce accident rates.

Development of Low-Cost Simulator

Developing a low-cost simulator is important for making VR-based training accessible to a wider audience. Previous works have explored various cost-effective solutions for creating high-quality simulators.

1.3 Solution Overview

The main goal of this thesis is to create a car driving training simulator in a VR and Non-VR environment with an integrated advanced assistance system using affordable and realistic modern driving equipment. In Section 1.2, we discussed various challenges that researchers might face when studying or implementing driving simulators, whether with or without VR technologies. Several solutions will be outlined in this section to address these issues.
To be able to answer the research questions presented in Section 1.2 a driving simulator prototype with the ability to choose between many options in the main menu has been developed. As shown in Figure 1, the user first has to choose his favorite car, our prototype has three different virtual 3D car models: BMW E92, Fiat Spider, and Volkswagen MK1. After that, the user opens the daytime menu to choose between two options day or night and this is important to test the driving simulator in different realistic daytime. Next, the user chooses between three maps designed especially for this study in Blender 3D modeling software. These maps

contain Training Area, which is the first map to start with to learn how to control the virtual 3D car model and learn how the advanced assistance system works. After that, the user can choose the Paderborn City map, which is also designed in Blender to simulate one-quarter of Paderborn City in Germany. The last map is the Highway map, which has been designed in Blender to Imitate the real German highway structure to let users learn how to drive on highways and pay attention to different rules there. Finally, the user chooses between two modes: VR mode and non-VR mode. In VR mode, the user has to use the VR headset in combination with the steering wheel, gearbox, and pedal gaming setup. In this study, we used the Meta Quest 3 and G923 Logitech steering wheel. Meta Quest 3 solves the challenges discussed in the previous section 1.2 cause it has six degrees of freedom, uses 110 Field of view to show enough space in the 3D environment, and also uses 3D stereoscopic images [22].

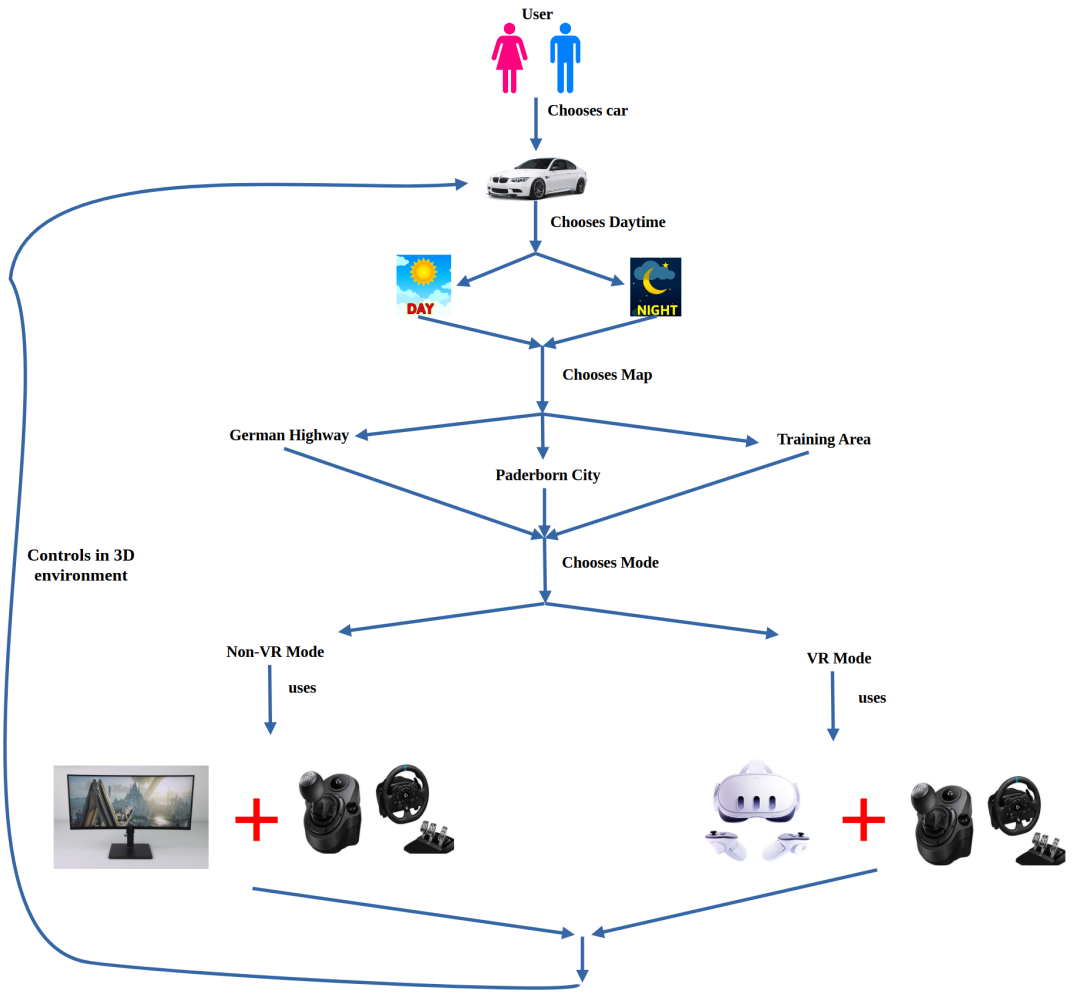


Figure 1: Basic overview of the driving simulator solution.

The steering wheel gaming setup used in this study is the G923 Logitech steering wheel and pedals with a gearbox to clone the real car setup and be able to control the virtual 3D car model more realistic [23]. In non-VR mode, the user uses a PC 3D display with G923 Logitech setup to be able to study the difference between VR and non-VR mode, and choose the best setup to use in the future.
As discussed in problem statement Section 1.2 capturing and analyzing drivers' behavior is a crucial aspect, even when studying or implementing a virtual driving simulator. However, this task is not straightforward. To achieve this, researchers aim to collect logging data, signals, and information generated in the development environment [24]. The use of VR simplifies this challenge due to the various sensors integrated into different HMDs and controllers, as well as localization technologies that capture head pose and hand position in a spatial environment. Additionally, VR devices often include sound recording inputs. This thesis will explore various data-collection methods to gather comprehensive information from participants during driving sessions. This includes recording inputs from the steering wheel, acceleration, brake pedals, and clutch, and according to the recorded data, the advanced assistance system which is integrated into the prototype will give users feedback on what did they do wrong and how to correct these actions. the assistance system will provide a static review of the participant's driving behavior. It will also collect data on how participants handle different scenarios, such as changing lanes, paying attention to pedestrians crossing the street and other cars, or controlling the car speed according to different speed limits. This will enable the analysis of usability criteria such as efficiency, effectiveness, user satisfaction, error rate, and learnability by comparing the different scores in different situations. As already mentioned, one of the major problems that users face while using driving simulators is motion sickness. So, specific techniques must be used in the driving simulator scene in order to reduce its effects. According to the findings presented in [16], with the support of specific visual traffic models like buildings, road textures, and camera field of view, such a problem can be handled and its risks minimized. This will help reduce the possibility of motion sickness due to VR HMDs, and thus increase acceptance for VR-based driving simulators.

1.4 Thesis Structure

In the second chapter 2, we will introduce and explore the key terms and concepts that form the foundation of this thesis. The third chapter 3 will delve into related research, which our work aims to build upon. Chapters four 4 and five 5 will detail the development of the prototype, starting with its conception and followed by its implementation. The sixth chapter 6 will present the usability study conducted with the prototype and analyze its findings. Finally, the seventh chapter 7 will summarize the thesis and suggest potential directions for future research. Figure 2 shows the structure of the thesis.

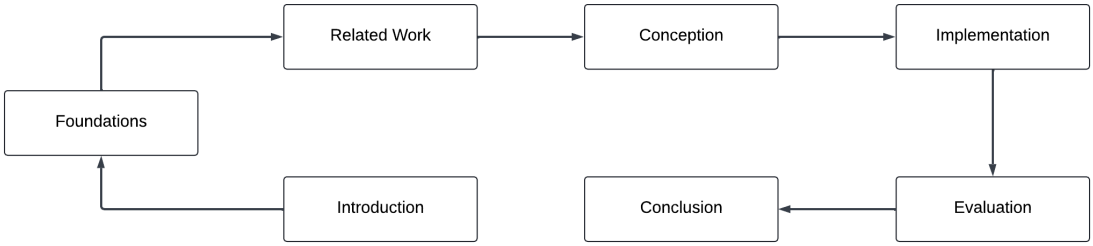


Figure 2: Thesis structure.

2 Foundations

2.1 Virtual Reality (VR) Technology

In recent years, virtual reality (VR) has emerged as a prominent subject within the information technology sector. The adaptation of Ernest Cline's widely acclaimed novel, Ready Player One, into a film by Steven Spielberg, introduced VR to a vast audience [25]. However, it is noteworthy that the foundational technology of VR has existed since the 1960s [26]. The high costs associated with the necessary equipment and the limitations in quality have historically impeded its widespread acceptance [27]. In the early stages of virtual reality (VR) technology, applications were mainly oriented towards serious fields such as flight simulation and military training, rather than gaming. The initial consumer head-mounted displays (HMDs), including Sega VR and Nintendo's Virtual Boy, emerged in the 1990s [28]. Nonetheless, these early VR systems were characterized by limited graphical performance and frequently induced motion sickness, leading to their lack of commercial success. To address these shortcomings, the CAVE Automatic Virtual Environment was introduced, enhancing both resolution and system latency [29]. However, the requirement for specialized rooms and costly projectors has restricted the adoption of CAVEs to professional environments, including engineering, military training, scientific research, and healthcare [30]. Approximately two decades later, the gaming sector saw the introduction of advanced HMDs such as Oculus Rift, HTC VIVE, and PlayStation VR, which facilitated the integration of VR into domestic settings. In addition to HMDs, contemporary VR technology generally incorporates audio devices, such as headphones or speakers, and controllers for haptic feedback. More advanced systems may also feature haptic gloves, suits, multi-dimensional treadmills, or other equipment designed to enhance the immersive experience [25]. The advent of wireless, fully autonomous virtual reality systems such as the Oculus Quest and HTC VIVE Focus, which operate independently of high-performance computers, has further diminished the barriers to home utilization. Consequently, contemporary virtual reality technologies are characterized as complex technologies that replace real-world sensory information with synthetic stimuli such as 3D visual imagery, spatialized sound, and force or tactile feedback [31]. A virtual environment is defined as "software representations of real (or imagined) agents, objects, and processes; and a human–computer interface for displaying and interacting with these models" [32]. In contrast, a virtual world is a particular category of virtual environment that is characterized by shared, simulated spaces that are inhabited and shaped by their inhabitants, who are represented as avatars [33]. The presence of customizable avatars and the capacity for multi-user interactions have been utilized to differentiate virtual worlds from virtual reality [34]. However, numerous multi-user VR applications currently available, such as AltspaceVR, enable users to

create and personalize their avatars. Nevertheless, both virtual environments and virtual worlds generally refer to software that operates on hardware that may or may not be VR-based [25]. Many researchers have delineated virtual reality (VR) through three fundamental attributes: telepresence, interactivity, and immersion [35]. Presence is generally defined as the sensation of being in a location distinct from one's actual physical surroundings [36]. Interactivity, which affects the sense of presence, pertains to the degree to which users can engage with and alter their virtual environment in real time [37]. Although there is considerable agreement among researchers regarding the definitions of presence and interactivity, the notion of immersion has been conceptualized in various ways within the literature [38], warranting further exploration. Some scholars have approached immersion through the lens of technological attributes that can be quantitatively assessed [36]. These attributes include inclusiveness (the degree to which reality is omitted), extensiveness (the variety of sensory modalities engaged), surrounding (the dimensions of the field of view), vividness (the quality, resolution, or richness of the visual displays), and matching (the alignment of preoperative feedback with visual information regarding body movements) [39]. Conversely, other researchers have framed immersion as a subjective experience [38], which can manifest in various forms, such as cognitive immersion (experienced when users tackle complex challenges), emotional immersion (felt during the unfolding of narrative elements), sensory-motoric immersion (arising from feedback on physical movements), and spatial immersion (experienced during extensive navigational activities) [40]. In addition to its applications in gaming and entertainment, virtual reality (VR) is gaining traction in the field of education. This trend is particularly evident in corporate training programs, but it is also being utilized to a certain degree by university students and in primary and secondary education settings [41]. Numerous companies have begun to develop VR applications tailored for classroom use, such as VR Immersive Education and Google Expeditions [42], which encompass a variety of subjects, including anatomy, geography, history, physics, and chemistry.

2.2 Driving Simulators

Driving simulators were initially created to train a significant number of personnel in the tactical operation of military vehicles during World War II. By the early 1960s, their application expanded into research, focusing on driver behavior and the interaction between drivers, vehicles, and the surrounding road environment [43]. By 1975, advancements in visual display and computer technology had led to the operation of at least sixteen driving simulators across the United States, employing various techniques for visual field generation, alongside two in Europe [44]. Following 1985, the development of driving simulators has increasingly aimed to meet the specific requirements of distinct groups, including the automotive industry, private research institutions, and academic entities. Presently, the primary applications of driving simulators involve examining the acceptability of innovative transportation elements such as road design and in-vehicle devices, assessing safety concepts (including potential accident increases due to new road designs

and in-vehicle technologies), and evaluating the credibility and transferability of simulator findings to real-world scenarios [45]. Driving simulators have been invaluable research tools in such disparate fields as civil engineering, transportation, psychology, and ergonomics, while their applications include innovative road design, testing of new tunnel designs, highway innovations, traffic calming measures, intelligent transport systems that develop new in-vehicle navigation systems, Head-Up Displays, and active pedals [45]. The advantage of driving simulators primarily pertains to offering a safe environment where driving research is conducted, and in the way that it can be set up in order to investigate various human factors issues at reasonable cost. Driving simulators offer more control over the variables in their experiment to the experimenter compared to field studies, and changes are easily made to conditions. These are integrated with a digital computer system that offers real-time processing, organization, and storage of data, and permits effective data reduction and presentation [45]. Yet, driving simulators only present an artificial setting which alone cannot represent real-world conditions. For instance, even the most sophisticated driving simulators have limitations in simulating lengthwise and lateral accelerations (e.g., VTI driving simulator: [46], and Daimler-Benz driving simulator: [47]), and only a fraction of the complex transportation system can be accurately represented at this time. The discrepancies between simulated and actual driving environments may affect participants' driving behavior and performance, suggesting that performance metrics obtained in a simulator may not align with those recorded during real driving scenarios. Consequently, the evaluation of driving simulators becomes crucial to ensure that they yield transferable, reliable, and valid results across both environments [45].

2.3 Advanced Driving Assistance System

In general terms, advanced driving assistance system (ADAS) represents many different technologies, from some minor ones that provide information, guidance, and alerts to those systems that actually take supportive or active controlling action over the vehicle, peaking in fully automated driving solutions [48]. The main four kinds of ADAS are based on. The first class includes systems that are intended to enhance aspects of the driving task through the provision of information; such systems are collectively known as In-Vehicle Information Systems (IVIS). Examples include navigation aids, and systems that provide information on traffic and road conditions, for instance, TrafficMaster and RDS-TMC receivers [49]. The second class includes those that issue a warning or feedback to reduce driver errors or violations. The warnings could be in the form of auditory, visual, or haptic signals using either force feedback or vibration. Systems in the third category perform vehicle control without removing driver intervention and thus provide only limited driver override [49]. Examples include adaptive cruise control, stop-and-go systems, and various forms of intelligent speed adaptation. The fourth category is automated driving-better known as "autonomous driving"-in which the driver is completely removed from the control process and cannot override the actions of the system. It goes without saying that the implementation of different systems in the vehicle

modifies the role of the driver inasmuch as some aspects of the driving task are modified, while others are introduced or taken away. With more interventionist systems, the role of the driver is increasingly transferred from manual operation to one that is supervisory in nature [49].

2.4 Development in the Unity Game Engine

The driving simulator prototype is implemented and managed in the Unity 3D Game Engine. This section will preview the main concepts of this engine to understand further technical aspects in Chapter 5. Unity Technologies has developed a 3D game engine that operates across various platforms. This engine has consistently evolved to accommodate a wide range of devices, including gaming consoles, desktop computers, and mobile devices. Additionally, it offers support for virtual reality environments [50]. Unity has gained significant traction in the realm of game development for both iOS and Android platforms. Notable titles such as Beat Saber, Pokémon Go, Cuphead, Call of Duty, and Monument Valley have been created utilizing the Unity engine [51]. It allows the development of both 2D and 3D games; these are used to provide an interactive feel with simulation [52]. Users have a friendly interface, which accommodates beginners, and its user interface resembles those of Mac OS X Finder and Windows Explorer. Unity 3D provides collaboration of developers, artists, designers among others. The sectors of its application include leading Video Gaming, movies, automotive, engineering, construction, etc. Additionally, Unity has also been utilized for creating applications for the United States Armed Forces [53]. Unity game engine uses C# as one of the main scripting languages [54]. It has been designed as a cross-platform engine where it provides a user-friendly editor allowing the user to select assets and objects by simply choosing drag-and-drop. The Unity editor is compatible with various operating systems, including Linux, MacOS, and Windows. Furthermore, the Unity engine supports over 19 platforms, encompassing desktop systems such as Linux, Mac, and Windows, a range of gaming consoles including Nintendo Switch, PlayStation 4 and 5, Xbox Series, and Stadia Cloud Gaming Service, as well as mobile platforms like TvOS, iOS, and Android. Additionally, it accommodates virtual reality platforms, which include Google ARCore, PlayStation VR, Windows Hololens (Mixed Reality), Oculus, Unity XR SDK, Steam VR, Magic Leap, and Google Cardboard [55]. The Unity editor is composed of several sub-windows, among which the hierarchy, game view, project browser, inspector, and scene view are frequently utilized, as illustrated in Figure 3.

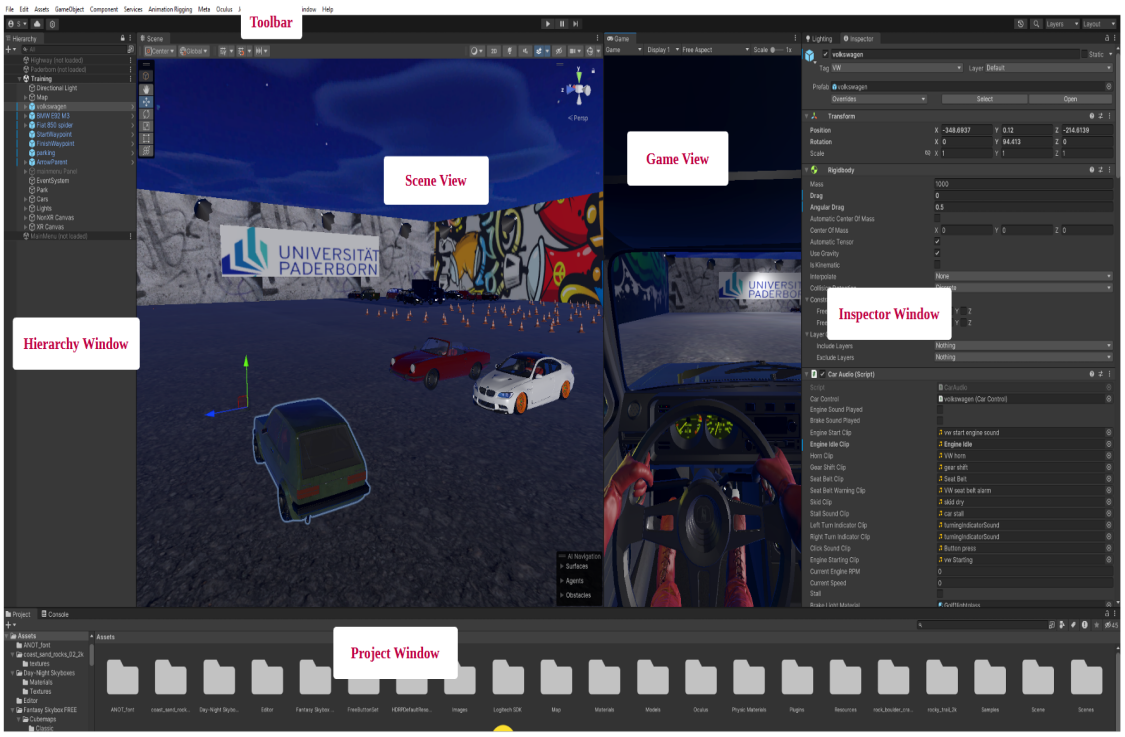


Figure 3: Interfaces of Unity editor.

2.5 Modelling in the Blender 3D Modelling Software

Blender is a free, open-source program intended for 3D (three-dimensional) graphics. Main features include infrastructure modeling, texturing, physics simulation, particle effects, rendering, animation, imaging, shading, compositing, and developing innovative real-time 3D game and animated film development [56]. In addition, it is loaded with learning resources like tutorial videos and documentation that can lead to learning about this software. Professionals and 3D artists recognize Blender as one of the most efficient, feature-packed, free, and open-source interventions for 3D computer graphics [57]. The version of Blender used to design the 3D environment to be used in this prototype is version 4.1. The Blender software makes it possible to model high-resolution, production-quality 3D models, animations, and graphics. The software is maintained by the Blender Foundation, which supports a substantial user community and records over 3.4 million downloads each year [58]. The development of Blender was initiated by the NeoGeo animation studio, with January 2, 1994, marked as its anniversary based on the timestamps of the source files. As an open-source software, Blender is free and can be downloaded by anyone for unlimited business uses from this website [57]. In fact, Blender can easily create animations similar to those from such great studios as Pixar and DreamWorks. Some of their feature films include Kung-Fu Panda, Toy Story, The Incredibles, Finding Nemo, Shrek, and Veggie Tales among many others [56]. In Blender, the user interface is based on several key components. Figure 4 presents a typical view,

which consists of the following elements:
3D viewport: This provides the view area of interaction with a three-dimensional scene, where modeling, animation, and texture painting are enabled.
Tool panel: The panel contains a series of utilities, which allow users to select and edit objects or meshes.
Properties Window: This is the information window where it displays, by default, the attributes of an object and allows its modification.
Animation Timeline: Presents the user with a timeline of what's happening in the scene about the animation, such as: the current frame, keyframes of the active object, start and end frames of the animation range, and markers from the user.
Scene Collection: This is about describing the object chosen and those present within the scene, enumerating all the components to a better management of the elements.
Header: At the right top or right bottom of the editor, a set of views may be obtained. Such views are Wire-frame View, Shading View, Solid View, and Render View.
Blender Modes: Nine different types of modes used in object editing, namely: Object Mode, Edit Mode, Sculpt Mode, Vertex Paint Mode, Texture Paint Mode, Weight Paint Mode, and Pose Mode.

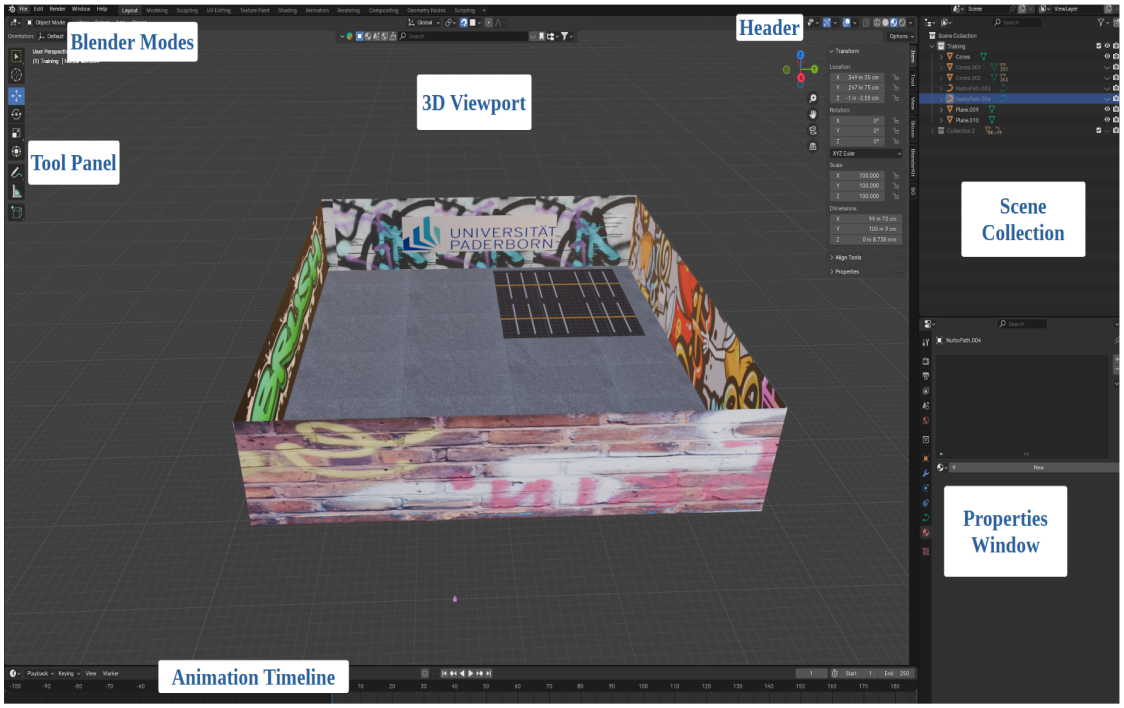


Figure 4: Interfaces of Blender 3D Modelling Software.

2.6 Motion Sickness within the VR Environment

Simulation sickness, or SSN, is a variant of motion sickness peculiar to stationary environments [16]. For instance, US Navy pilots were reportedly in considerable distress during the early sessions of simulator training when the first static helicopter simulator was in development in the 1950s. It was assumed that the absence of motion clues was responsible for these distresses [59]. SSN can become visually provoked even in the complete absence of real motion and can develop at any time during the exposure, which can result in prolonged aftermath effects that can become dangerous while driving a car [60]. Symptoms of SSN are very similar to those of MS, but there are some unique symptoms, like eye fatigue. These symptoms could be grouped into three major categories: nausea, which includes stomach awareness, nausea, vomiting, burping, and increased salivation; oculomotor symptoms, including eye strain, headaches, difficulty in focusing, and blurred vision; and disorientation, typified by general discomfort, vertigo, and dizziness [61]. Currently, head-mounted displays use advanced tracking technologies along with high refresh rates to minimize virtual reality sickness for normal users. This is through the correspondence of the perception of the user's vision and their vestibular input, so that any head movements virtually are matched exactly by actual head movements in real life [62]. Nevertheless, this solution does not cater to users who prefer to navigate the virtual environment using a controller or gamepad, often due to limited physical space or a preference for remaining stationary. In this regard, locomotion remains poorly solved for VR. Most open-space VR experiences leverage a range of locomotion strategies using controllers with joysticks or trackpads. Among the most common methods implemented into applications are teleportation, ratchet-based rotation, blurring, and FoV restrictors. Both teleportation and ratcheting have been identified as efficient techniques for enabling locomotion while reducing VR sickness because they allow the user's virtual avatar to instantly relocate or reorient without continuous movement [62]. These techniques minimize the amount of optical flow that can cause sensory conflict. Limiting the amount of optical flow may sacrifice, however, some degree of immersion and result in spatial disorientation. Because of this, many VR experiences have decided to use teleportation for primary locomotion in many open-space gaming scenarios by making smart navigation choices that avoid associated trade-offs [62].

2.7 Usability Evaluation in VR

Usability is a complex construct that can be analyzed from multiple viewpoints [63]. The concept of usability is widely interpreted and holds different meanings for various individuals. Some individuals associate usability with ease of use or user-friendliness, evaluating it primarily from the perspective of interface effectiveness. This perspective is justified, as usability is grounded in the principles of human-computer interaction. Numerous studies on usability concentrate on the design of interfaces [64]. For example, [64] notes that "the distinction between interface effectiveness and usability is ambiguous." Usability can also encompass notions of usefulness and usability. [65] made this observation, indicating that usability in-

cludes functions such as "Can I turn it on?" or "Can I access that function?" In contrast, usefulness pertains to questions like "Did it genuinely assist me?" or "Was the effort justified?" [66] differentiates between usability (ease of operation) and usefulness (filling a specific purpose), remarking that these two concepts are often intertwined in evaluative contexts. Usability comprises several characteristics. The International Standards Organization [67] defines usability as "the degree to which a product can be utilized by designated users to accomplish specified objectives with effectiveness, efficiency, and satisfaction within a defined context of use. [68] identifies five key attributes of usability: learnability, efficiency, memorability, a low error rate or ease of error recovery, and user satisfaction. [69] echo this sentiment, asserting that usability should be functionally accurate, efficient, easy to learn and remember, tolerant of errors, and subjectively enjoyable. Furthermore, [70] identifies four components of usability: usefulness, effectiveness (ease of use), learnability, and user attitude (likability). [71] categorize usability into initial performance, long-term performance, learnability, advanced feature usage, first impressions, and long-term user satisfaction. Usability engineering methodologies can be categorized into two primary types: user-oriented and expert-oriented methods. User-oriented methods involve potential software users executing specific tasks while being monitored by usability specialists. Conversely, expert-oriented methods entail usability professionals carrying out a series of predetermined actions aimed at formally evaluating particular aspects of usability [72]. Both categories necessitate significant manual effort. Furthermore, although there is extensive experience with these methodologies in the context of two-dimensional interfaces, there has been relatively little research assessing their effectiveness in virtual reality environments [73]. The approach delineated in this work aligns with user-centered methodologies, as it relies on users engaging with virtual reality (VR). However, it forks from traditional methods by allowing users to interact with the software in a manner that best suits their individual requirements, rather than adhering to predetermined tasks. Moreover, since users are not made aware of being monitored, they do not experience the pressure of a testing environment. Additionally, our approach eliminates the necessity for a human evaluator to oversee the user, as the monitoring process is entirely automated through the recording of user scores and interactions with the prototype. The subsequent analysis of these observations is also automated, negating the need for manual evaluation by an observer. The ISO 9241 standard established in 1998 [74], along with supplementary information from ISO/IEC 25010, defines usability as the degree to which specified users can effectively and efficiently accomplish designated objectives while experiencing satisfaction within a specific context of use. In this case, usability in general has many criteria, we will evaluate some of them in this approach which are: Effectiveness, Efficiency, User Satisfaction, and Learnability. First, Effectiveness refers to the degree of precision and thoroughness with which designated users are able to accomplish their objectives [67]. Efficiency pertains to the speed at which users can execute tasks after they have become familiar with the design [75]. User Satisfaction reflects the overall enjoyment experienced by users while engaging with the design [75]. Learnability refers to the degree to which a user can successfully complete fundamental tasks during their initial interaction with the design [75].

3 Related Work

To determine the usability criteria for our prototype that would enhance the existing progress of research and to evaluate the research questions that have been addressed within the domain of driving simulators, particularly in relation to virtual reality (VR), a comprehensive literature review was conducted.This review aimed to uncover a research gap that this thesis seeks to address. Consequently, this chapter will provide a detailed overview of prior studies in the area of XR-based driving simulators.

3.1 Review Research Questions

This section will discuss the review research questions and mention them. The difference between these review research questions and the main research question (RQ), discussed in Section 1.2, is that review research questions (RRQ) guide a related literature review or systematic review. They focus on summarizing and synthesizing existing research rather than collecting new data. On the other hand, research questions (RQ) are specific questions that a study aims to answer through user studies using a prototype or a survey. They guide the methodology and data collection process.

RRQ1: Which features from our feature list have been included in earlier related work?

RRQ2: Which research questions have guided the evaluation of usability in driving simulators, and what gaps remain for future exploration?

For this literature review, we will use the list of features, which is presented in Section 1.2, to better distinguish between the related research.

3.2 Previous Work

Before presenting related works, It is important to outline the procedure taken to locate and make decisions on these studies before mentioning the specific publications that have been linked. The steps entail ensuring a wide and relevant review of the current body of literature:

Literature Search: We have searched many academic resources, including Google Scholar, ACM Digital Library, Paderborn University Library, and IEEE Xplore. Some of the keywords that are used are the following: "VR driving simulation,"

"virtual reality in driving," "automated driving VR," and "AR driver-assistance systems". It included the inclusion on the study criteria that address VR driving simulation and system integration. The search is limited to data mined from a conference paper, peer-reviewed articles, and publication from trusted sites only.

Inclusion criteria: Include studies identified on VR driving simulation and system integration. Relevant peer-reviewed articles, conference papers, and credible sources have been considered.

Screening Process: The identified papers were scanned in terms of relevance based on their abstracts and conclusions. Only the documents that contained a technological discussion of user experience and integration of VR or AR in driving simulation had been selected as priorities.

Review and Analysis: The review and analysis of the selected papers have been done in such a way that from those selected works, key insights and findings pertinent to this research are looked for. In this respect, only the most relevant and impactful studies were selected for inclusion in the related works section.

This systematic process yielded the following related works, which are discussed here:

3.2.1 VR-OOM: Virtual Reality On-road driving Simulation

VR-OOM is an on-road driving simulator using VR technology. It takes on-road driving simulators into another level by integrating virtual reality interfaces and environments into a physical real vehicle. That is why It is considered more present and immersive than a conventional lab-based driving simulator, which also allows more flexibility than a traditional on-road driving experience. It was a very important break in driving simulation research. To explain the VR-OOM implementation idea, a system diagram (see Figure 5) was added to this paper. We see the driver sitting in a physical car and wearing a VR HMD, which has a leap motion sensor to register the driver's motions. behind him or her, we see a trained driving teacher (here, a driving wizard), who controls and models the VR scene. This VR environment includes the actual physical car and a virtual steering wheel as 3D models in an immersive VR environment. The participant sees himself or herself sitting on the car seat and holding the steering wheel with his virtual hands, which match his actual hands in the VR-controlled scene. Part of the participant's senses are natural and the other part is virtual. Visual senses are considered virtual and haptic and audible senses are natural [76]. The VR-OOM makes use of a well-known consumer VR headset well-matched with Unity3D, along with the OSVR Headset or the Oculus DK2. The visible gadget of VR-OOM operates on an HP zBook 17 HSTNNC76C PC that is connected to the VR Headset sensors and a tablet are used by the wizard driving force to look at the digital world. Leap motion controller is a sensor tool that captures hand and finger movements as input and is attached to the front of the VR headset to permit the VR-OOM machine to monitor the player's

hand movements. This sensor is attached to the VR-OOM computer through USB. The facts from the Leap Motion are hired to replicate the player's hand movements without delay in VR [76]. Finally, this work will be analyzed according to the different requirements. First, the impassiveness of VR-OOM is very low because the user will have to lift the headset to sustain a connection to the actual world, which will decrease the immersion of the driving simulator. This could be fixed if the environment of the Simulator just depends on the virtual environment. Next, the VR simulation system that has been used in VR-OOM has improved the driving experience using many sensors, such as the LEAP motion controller. That has also helped in capturing driver's behavior and interactions. Unfortunately, an assistance system was not implemented there to guide the user to overcome the errors during the use of a driving simulator such as ADAS. This work also included plans to deal with situations that occurred when the participants feel unwell. This paper [76] presents that the VR-OOM is a low-cost simulator that allows participants to enhance their driving experience with less money than in traditional driving schools.

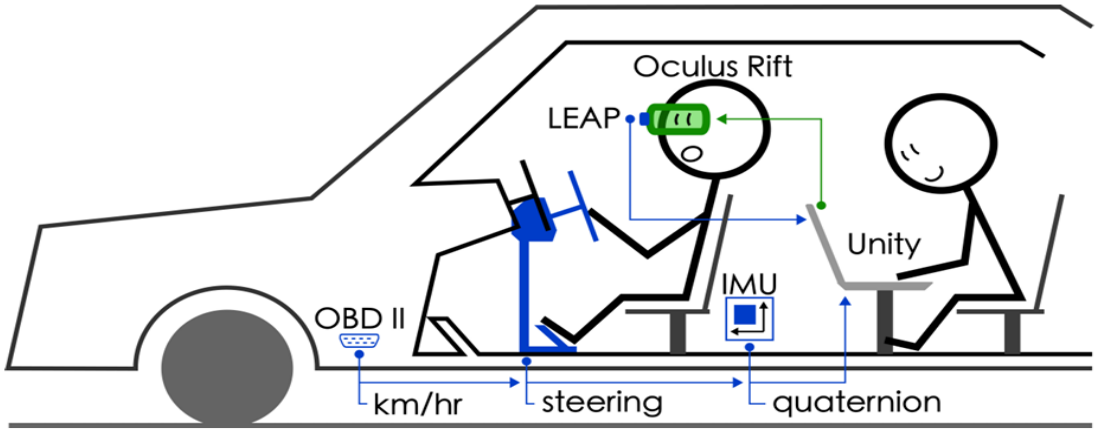


Figure 5: VR-OOM System Diagram (excluding the wizard driver).

3.2.2 DReyeVR: Democratizing Virtual Reality Driving Simulation for Behavioral and Interaction Research

DReyeVR is an open-source driving simulation in a virtual environment. It was developed to capture the driver's behavior and study the driver's interaction during the use of a driving simulator. As the name says, it uses mainly eye-tracking technology to capture where the driver looks during the experiment. DReyeVR implemented using unreal game engine and CARLA, an open-source urban conventional driving development [24]. CARLA does not support virtual reality and uses flat monitors, which reduce the eye and head tracking range [9]. The main difference between CARLA and DReyeVR is that DReyeVR involves immersive components and can be used to track driver's behavior and eye movements. DReyeVR costs about 5000 USD, which is cheaper than many traditional driving simulators used in driving schools [24]. The main goal of DReyeVR is to analyze and study the human

interaction of car drivers with the surrounding environment. DReyeVR has many features to achieve this goal, such as ego car view, which allows drivers to have the feeling of driving a real car. The second feature is using audio sources such as car engine traffic sound effects, adding more realism to this experiment. Finally, it uses an eye-tracking and capturing system. For this purpose, it uses HTC Vive Pro as an HMD (head-mounted display), which has eye and head tracking sensors and is supported by SteamVR [24]. It also includes two valve base stations for tracking, and two controllers. To improve the immersion, DReyeVR also uses a GTR arcade seat, as shown in Figure 6. Analyzing the different requirements, CARLA, which is used in DReyeVR, is an audio-less driving simulation without any sound effects that will decrease the immersion and presence felt, but in DReyeVR to improve these features, many sound speakers and cues have been added to it. As discussed above, this driving simulator used many features in the VR environment to improve driving experience, like ego vehicle control, sound designs, and eye tracking systems, etc. which are also very useful in capturing the driver's behavior and interactions. A practical assistance system to correct driver's errors is missing in this work, which can be improved in future works.

Figure 6: DReyeVR in first person perspective during a driving session with red crosshair representing the driver's eye position.

3.2.3 Automated driving using Virtual Reality

This study has two phases : training and a driving test, similar to a traditional driving schools. The training part is designed to let participants learn the main principles of level-3 Automated Driving System (ADAS)-equipped cars and to be able to recognize and interact with the Human-Machine Interface (HMI) in the vehicle. The second part is the driving test, which serves as a practical application of the theoretical aspects learned during the training phase [77]. The level-3 ADS contains three mods: manual, automatic, and take-over request. The HMI has different interfaces for each mode and the goal of the training part is to introduce these HMIs to the participants. The work presents also three different training scenarios: the traditional user manual, a fixed-base driving simulator using a real

steering wheel, pedals, and a wide screen, and finally a light VR system (LVR), containing an HMD with a racing steering wheel [77]. The driving test happens after the training part to test the participant's learning progress in a practical context. The hardware components used in this part, as shown (in Figure 7), are a real front body part of a vehicle and a panoramic surrounding display. The display has 170-degree FOV. The simulator also has four cameras to capture the environment and other important data, such as the speed and acceleration of the car [77]. Like in previous work, the different challenges will be analyzed for this work. Immersion is very low in the first phase, which is used to train participants to use the simulator, but in the driving test phase, the immersion is improved using a 170-degree FOV surrounding display and a real front body part of a vehicle. The training phase is very important to improve the driving experience of the participants. The simulator has many cameras integrated to capture the surrounding environment, but a real capture of the participants' driving behavior and interactions was not implemented. Also, an assistance system to improve the driving experience and overcome driver errors was missing. Considering the equipment used in this simulator, such as the car body, surrounding wide displays, and cameras, it is not the best option for cost-effective simulators.

Figure 7: The test drive simulator: (a) the real cabin with the 170° panoramic display and (b) a view of the cabin interior [77].

3.2.4 Augmented Reality-Based Advanced Driver-Assistance System for Connected Vehicles

In this work, an Augmented-Reality based ADAS(Advanced Driver Assistance System) was developed using connected vehicle technology [78]. These vehicles have two types of drivers: human drivers with AR-based Human-Machine-Interface, or automated controllers. The main architecture of the ADAS (as shown in Figure 8) contains: CAN BUS(a hardware component, that calculates speed, acceleration, and other relevant data of the ego vehicle), a localization component(which gives

information about the exact position of the ego vehicle), and finally the perception module(which provides a clear representation of the surrounding environment to the vehicle, these information contains hazard situations, traffic conditions, and surrounding vehicles). All these three components send their signals to the processor. which processes these signals and sends them to the cyber world [78].

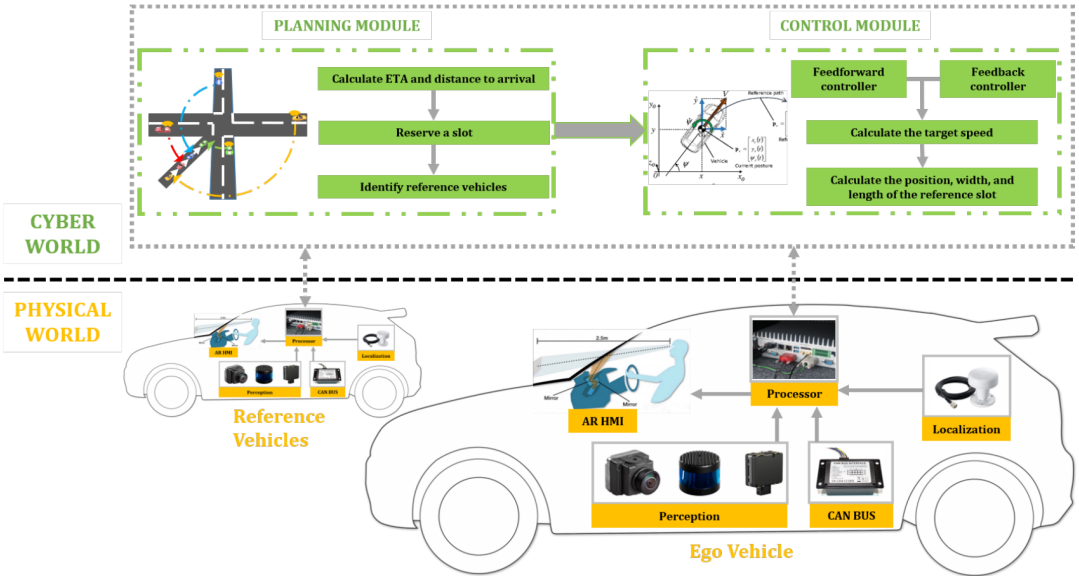


Figure 8: System architecture of the proposed AR-based ADAS for connected vehicles.

The processor also receives signals from the cyber world, containing orders and guidelines to the AR-HMI to guide the driver during his driving task [78]. A very important use case of this work uses AR-based ADAS to help reduce fatal accidents in unsignalized intersections (street intersections with no traffic lights), by sending signals to the connected vehicles and showing them in AR-HMI, which slots are available, reserved, or unavailable [78]. This is presented in figure 9. We can see three connected vehicles driving towards the unsignalized intersection, coming from different directions. The ego vehicle gets assigned to a specific slot according to its position, according to AR-HMI. This slot reservation will be shown as a red unavailable slot to the vehicles that have a driving conflict with the ego vehicle, so they can avoid accidents by staying in the green available slots.
Discussing the different challenges of driving simulators, this simulator lacks immersion and presence compared to the other works discussed before. Immersion was not one of the main goals of this work, but it was to overcome driving errors and hazards and enhance an ADAS system to lead drivers through real roads and improve their driving experience. The System architecture, which is shown in figure 8, improved the driving experience and the ADAS system is very important in overcoming driving hazards. Cost-effectiveness and motion sickness were not discussed explicitly in the original paper [78].

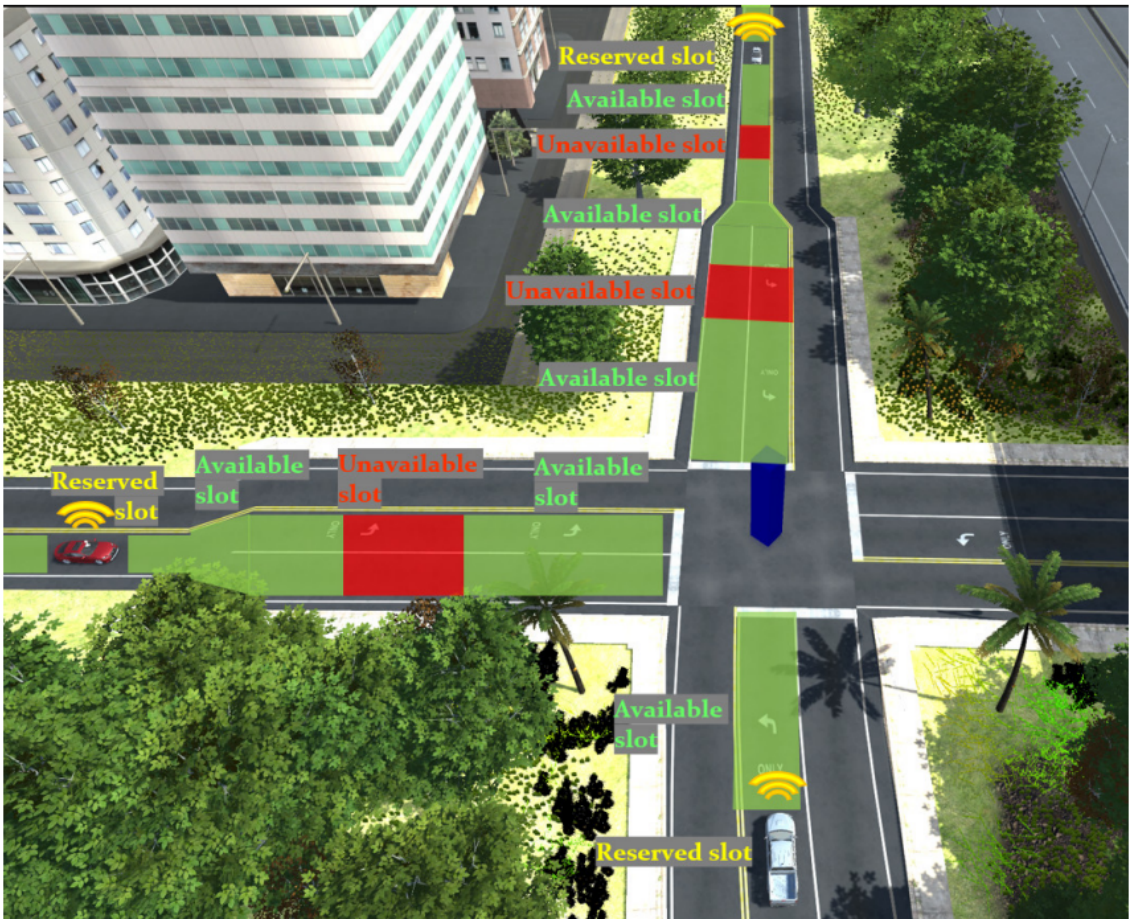


Figure 9: Illustration of the slot reservation concept for connected vehicles crossing an unsignalized intersection.

3.3 Results

This section will outline the key findings of the previous literature discussed before. Each contribution included addresses a specific subset of the features utilized to analyze the relevant literature. These features include immersive mixed reality support, simulation of realistic vehicle driving, driving assistance system integration, detection of driving hazards, and development of low-cost simulator. The first Subsection 3.3.1 will discuss immersive mixed reality support in all four related literature discussed in section 3.2. Next, Subsection 3.3.2 will present realistic Vehicle Driving features developed in each related work. Furthermore, Subsection 3.3.3 show if each work integrated a driving advanced assistance system to help drivers improving their driving skills and provide driving errors. Then, Subsection3.3.4 discuss in each work if it is possible to detect driving hazards and prevent it. Finally, Subsection 3.3.5 shows the financial aspect of each related literature.

3.3.1 Immersive Mixed Reality Support

Support for Head-Mounted Displays

VR-OOM [76] integrates the role of head-mounted displays (HMDs) in facilitating an immersive virtual reality experience. The authors highlight the significance of HMDs in delivering a realistic and captivating environment, implementing this technology to augment user engagement. In DReyeVR [24] the HTC Vive Pro Eye was utilized as the head-mounted display (HMD), featuring support for SteamVR, integrated eye-tracking capabilities, and an accessible eye-tracking software development kit (SDK). Third work [77] presents three different training scenarios: the traditional user manual, a fixed-base driving simulator using a real steering wheel, pedals, and a widescreen, and finally a light VR system (LVR), containing an HMD with a racing steering wheel. The AR-ADAS [78] uses AR-HMI to guide the driver during his driving task. The augmented reality human-machine interface (AR HMI) is projected onto an image plane, such as the windshield, utilizing a projector unit. To accurately overlay a slot onto the road surface within the driver's field of view, a front-view camera is required to capture the road's geometric features.

Head Movement with Six Degrees of Freedom

VR-OOM [76] accommodates head movement with six degrees of freedom, enabling users to explore their surroundings with ease. This capability is fully realized and is essential for fostering a natural and immersive driving experience. DReyeVR [24] uses HTC Vive Pro Eye as an HMD, which has six DOF [79]. In this paper [77] The capability of head movement encompassing six degrees of freedom (6DoF) is essential in this scenario, as it facilitates a training experience that is both more immersive and realistic. Finally, [78] uses AR to visualize guidance information for drivers, and the 6DoF head movement feature allows for precise tracking of the driver's head position and orientation.

Papers	Support for HMDs	Six DoF Head Movement
VR-OOM [76]	✓	✓
DReyeVR [24]	✓	✓
Automated Driving [77]	✓	✓
AR-ADAS [78]	✓	✓

Legend

✓ - Supported
✗ - Not Supported

Table 3.1: Comparison of Related Work Support for HMDs and 6DoF Head Movement.

3.3.2 Driving Scenarios and Challenges

Parking Training

VR-OOM [76] did not explicitly mention parking training. DReyeVR [24] also has not discussed this feature. This study [77] has not implemented any parking training in all modes. The AR-ADAS [78] did not discuss parking training.

Highway Driving

VR-OOM [76] did not discuss highway driving. In DReyeVR [24] highway driving was not presented. In this study [77], the target vehicle offered two driving modes for highway use: Manual Driving and Conditionally Automated Driving. The AR-ADAS [78] also did not implement highway driving.

City Driving

VR-OOM [76] did not explicitly discuss city driving. The main scenario in [24] is the city driving scenario, it incorporates appropriate sounds into different areas of the CARLA maps: for instance, regions rich in vegetation feature the cheerful chirping of birds, while urban and underground locations are filled with the sounds of wind and bustling traffic. This work [77] doesn't explicitly refer to city driving, but it highlights the crucial need for training drivers to react correctly to take-over requests when the automated system approaches its limits. The AR-ADAS [78] did not present any city driving mode.

Night Driving

VR-OOM [76] just had day driving and did not implemet a night driving mode. DReyeVR [24] did not developed a night driving either. [77] did not implement a night driving mode. The AR-ADAS [78] did not discuss a night driving mode.

Papers	Parking Training	Highway Driving	City Driving	Night Driving
VR-OOM [76]	✗	✗	✗	✗
DReyeVR [24]	✗	✗	✓	✗
Automated Driving [77]	✗	✓	✗	✗
AR-ADAS [78]	✗	✗	✗	✗

Legend
✓ - Supported
✗ - Not Supported

Table 3.2: Comparison of Related Work Support for Driving Scenarios and Challenges.

3.3.3 Driving Assistance System Integration

VR-OOM [76] briefly outlines the incorporation of voice guidance within the driving assistance system. This feature is designed to assist users throughout the simulation, providing real-time feedback and instructions. The simulator in VR-OOM [76], integrates visual components such as heads-up displays (HUD's) and other informational aids, is integrated into the simulation. These components are implemented to furnish users with essential information, thereby enhancing the overall driving experience. DReyeVR [24] is equipped with eye-tracking technology that enables the observation of the driver's gaze and focus, which is essential for analyzing driver behavior and their interaction with driving assistance systems. It also provides researchers with the ability to create tailored routes and traffic scenarios, facilitating the examination of various driving assistance systems in diverse conditions. It features comprehensive experimental logging and replay functionalities, which allow researchers to scrutinize driver behavior and assess the efficacy of driving assistance systems. This work [77] highlights the significance of educating drivers within a controlled environment through virtual reality (VR), which aids in acclimating them to automated vehicles and their operational systems. The AR-ADAS [78] employs augmented reality to present navigational information that is collaboratively computed by a network of interconnected vehicles. It utilizes an unsignalized intersection as a practical example, enabling drivers to traverse the intersection with augmented reality assistance without the necessity of coming to a complete halt.

Papers	Driving Assistance System Integration
VR-OOM [76]	✓
DReyeVR [24]	✓
Automated Driving [77]	✓
AR-ADAS [78]	✓

Legend
✓ - Supported
✗ - Not Supported

Table 3.3: Comparison of Related Work Support for Driving Assistance System Integration.

3.3.4 Detection of Driving Hazards

Violations of Traffic Signals

VR-OOM [76] does not specifically address the identification of traffic signal violations. DReyeVR [24] Offers a way to access semantic details about the simulated

environment, along with a versatile interface that allows for the creation of tailored traffic scenarios. This includes detailed control over other vehicles, pedestrians, and traffic signals, making it an invaluable resource for researchers. This work [77] does not provide any assistance to detect traffic signal violations. The AR-ADAS [78] focuses on a particular scenario involving unsignalized intersections, where connected vehicles, including connected and automated vehicles (CAVs), can work together to navigate through intersections without coming to a complete stop. This means that the AR-ADAS does not provide any assistance to detect traffic signal violations.

Detection of Lane Changes

This feature is not implemented in VR-OOM [76]. DReyeVR [24] did not implement a mechanism to detect lane changes. as shown in [77] simulator, it has many modes and one of them is the light virtual reality mode (LVR), where the trainee was instructed to utilize the steering wheel to execute a lane change maneuver while keeping the automated driving system active. Because of this they implemented a lane change detection system. As shown in Figure 9 the AR-ADAS [78] uses augmented reality to guide drivers through unsignalized intersections, providing them with information about available slots and reserved slots. For this purpose, it developed a lane change detection system that assigns slots to vehicles based on their position and direction, ensuring that they can navigate the intersection safely.

Adherence to Speed Limits

the OBD-II (On-board diagnostics) port in VR-OOM [76] serves as a polled serial interface commonly utilized for vehicle diagnostics and emissions testing. To link the VR-OOM laptop to the car. The VR-OOM software sends a request for the vehicle speed via this port, and a Node.js® script then relays that information over a separate UDP port to Unity. So the software can track the car movements and speed, however, it does not provide any assistance to the driver to adhere to speed limits. In DReyeVR [24] a heads-up display has been incorporated into the vehicle's dashboard, featuring a digital speedometer, indicators for drive and reverse, as well as turn signals to enhance vehicular monitoring. however, it does not provide any assistance to the driver to adhere to speed limits. In [77] the Human-Machine Interface in the vehicle featured a head-up display (HUD) along with a button located on the steering wheel. The HUD provides details such as the current speed, speed limit, distance traveled, and the vehicle's current status. The AR-ADAS [78] does not provide any assistance to adhere to speed limits.

Prevention of Accidents

The simulator in VR-OOM [76] is equipped with mechanisms to avert accidents and enhance user safety. This includes the provision of real-time feedback and alerts, which are designed to assist users in steering clear of hazardous situations and to encourage safe driving behaviors. DReyeVR [24] does not provide any assistance to prevent accidents. This work [77] does not provide any assistance to prevent accidents. The AR-ADAS [78] does not provide any assistance to prevent accidents.

Papers	Traffic Signals	Lane Changes	Speed Limits	Accidents
VR-OOM [76]	✗	✗	✗	✗
DReyeVR [24]	✓	✗	✓	✗
Automated Driving [77]	✗	✓	✓	✗
AR-ADAS [78]	✗	✓	✗	✗

Legend
✓ - Supported
✗ - Not Supported

Table 3.4: Comparison of Related Work Support for Detection of Driving Hazards.

3.3.5 Development of Low-Cost Simulator

VR-OOM [76] leverages affordable entertainment and gaming equipment to establish a cost-effective driving simulation experience. DReyeVR [24] uses a virtual reality (VR) simulator, which provides a much more affordable option compared to CAVE visual systems, costing just a few thousand dollars instead of the hundreds of thousands typically required, all while delivering an immersive 360-degree field of view. In this research [77], three distinct training systems were employed to educate participants on the operation of level-3 automated driving systems. a User Manual (UM) shown on a laptop, which is cost-effective. In addition to the UM, a Fixed-Base driving simulator (FB) featuring an authentic cockpit and controls, including pedals and a steering wheel, was used, which is not really very cheap. Finally, a light virtual reality (LVR) setup that included a Head-Mounted Display (HMD) and a gaming racing wheel, which also considered a cost-effective solution. in conclusion we can not say that [77] work is a low-cost simulator in total. The AR-ADAS [78] does not provide any information about the cost of the simulator, but it uses a projector unit to overlay the AR-HMI onto the windshield, which can be considered a cost-effective solution.

Papers	Development of Low-Cost Simulator
VR-OOM [76]	✓
DReyeVR [24]	✓
Automated Driving [77]	✗
AR-ADAS [78]	✓

Legend
✓ - Supported
✗ - Not Supported

Table 3.5: Comparison of Related Work Support for Development of Low-Cost Simulator.

3.4 Discussion

This part will begin by summarizing the results of the literature review, followed by an interpretation in Section 3.4.1. Then, in Section 3.4.2, we will explore potential threats to the validity of these findings.

3.4.1 Findings

Table 3.6 provides a comprehensive summary of the contributions we've previously explored, highlighting the features supported by their prototypes as well as not supported features.

Features	**VR-OOM** [76]	**DREYEVR** [24]	**Automated Driving** [77]	**AR-ADAS** [78]
Support for HMDs	✓	✓	✓	✓
Six DoF Head Movement	✓	✓	✓	✓
Parking Training	✗	✗	✗	✗
Highway Driving	✗	✗	✓	✗
City Driving	✗	✓	✗	✗
Night Driving	✗	✗	✗	✗
Driving Assistance System Integration	✓	✓	✓	✓
Traffic Signals Detection	✗	✓	✗	✗
Lane Changes Detection	✗	✗	✓	✓
Speed Limits Detection	✗	✓	✓	✗
Accidents Prevention	✗	✗	✗	✗
Low-Cost Simulator	✓	✓	✗	✓

Legend
✓ - Supported
✗ - Not Supported

Table 3.6: Comparison of Related Work Support for Various Features.

The comparative study of the four research prototypes (VR-OOM, DREYEVR, Automated Driving, AR-ADAS) has yielded important insights into the present landscape and future possibilities of driving assistance systems within driving simulators. The results from the feature analysis highlight several significant trends and areas that could benefit from enhancement. To begin with, the consistent integration of Support for HMDs and Six DoF Head Movement across all previous works emphasizes the necessity of immersive and realistic simulation experiences. These features are important for the realization of a realistic virtual space and realism is an essential part of any form of training or research. Assistance Systems

integrated into each prototype make it important that the provision of assistance to users during the simulation is availed. This would enable a user to move with ease in the virtual space and get assistance where needed, hence making it more user-friendly. Upon analysis, specific prototypes also bear unique characteristics. For example, Automated Driving covers unique usage scenarios of Highway Driving and Lane Changes, which shows a special interest in complex driving scenarios. On the other hand, DREYEVR is highly distinctive in the inclusion of City Driving, Traffic Signals, and Speed Limits, showing a comprehensive approach in the urban driving simulation. These highlight the variation in goals and target scenarios for each prototype. However, none of the prototypes provided features such as Parking Training, Night Driving, and Accidents, indicating specific areas that must be modified to achieve full utility in the future. The driving simulators, therefore, could be much more realistic and much more for the user to experience if those additions could be made. The cost aspect also appears from the analysis, say, for example, the Low-Cost Simulator feature appearing in three out of the four prototypes. This focus on affordability is of essence to drive wider usage and access to driving simulators. Based on the details just acquired, we can consider our first research review question **RRQ1** answered. Next, we will answer the **RRQ2** for each related work.**VROOM** [76] emphasizes the role of driving simulators in developing and assessing how humans interact with vehicles and their interfaces. Future studies might concentrate on analyzing the usability and effectiveness of these interactions across various driving situations. It suggests that in-lab driving simulators significantly diminish their capacity to create a feeling of immersion and presence when the user is not fully involved in the driving experience. It highlights how crucial inertial and vestibular cues are for detecting distance and steering effectively, and it proposes that upcoming studies might investigate ways to enhance the simulation of these cues in driving simulators.**DReyeVR** [24] discussed mainly two research questions: In what ways do eye tracking and head pose tracking work together to improve the assessment of driver attention and mental workload, and how well do directional road signs in the environment assist with navigation and influence where drivers look. Although the text discusses the use of virtual reality and external devices, it overlooks other innovative interaction techniques such as haptic feedback, which could be implemented in futuristic prototypes. **Automated Driving** [77] looks into how effective Virtual Reality (VR) training is when compared to conventional approaches like user manuals and fixed-base simulators, specifically in improving driver take-over performance in conditionally automated vehicles. The study also delves into user preferences and their perceptions of the usability of VR training for automated driving. Key findings reveal that VR training significantly enhances take-over performance and is favored by users over traditional methods. For future research, it is important to investigate the long-term impacts of VR training on driver performance. Additionally, exploring different VR scenarios and their effects on training effectiveness is essential. Usability studies should also aim to include a broader range of driver demographics to enhance the applicability of the results. **AR-ADAS** [78] focus on this research question: how An augmented reality (AR)-enhanced advanced driver-assistance system (ADAS) have the potential to significantly improve travel times and reduce energy usage in connected vehicles?

This research also explores how users interact with AR guidance in simulated driving environments. The results indicate that the AR-based ADAS can effectively boost travel efficiency and lower energy consumption. However, there are areas that need further investigation. It's essential to validate the AR-based ADAS in real-world driving situations to ensure the findings from simulations hold true. Additionally, long-term studies are needed to gauge how users perceive and trust AR systems. Lastly, examining how AR-based ADAS integrates with other in-vehicle technologies will help assess its overall usability impact.

3.4.2 Threats to validity

The main constraint of this structured literature review was the limited timeframe available for its preparation and implementation. As this is a bachelor's thesis, the literature review functions primarily as a contextual framework rather than the focal point of the research. As a result, the number of contributions included in this review was restricted. This constraint is evident in the relatively small selection of sources utilized for our study, which predominantly concentrated on papers pertaining to assistance systems integrated with driving mixed reality simulators. The narrow scope of our source selection suggests that there may be significant contributions that were not considered, which could have been relevant to this thesis. Although we endeavored to address this limitation by incorporating sources that cover a wide array of scientific literature, it is possible that some pertinent studies were inadvertently excluded.

3.5 Summary

Future studies should be directed at improving the simulation of inertial and vestibular signals, its practicality and effectiveness of interaction on a variety of driving scenarios, and exploring novel interaction methods, including haptic feedback. Besides, it is of great importance to conduct longitudinal studies in order to measure the impact of virTual reality training about driver performance and authenticate augmented reality-based advanced driver-assistance systems in real-world driving contexts. Broadening usability studies to encompass a wider array of driver demographics will significantly increase the relevance of the findings.
In other words, immersive and the existing prototypes have a reasonably good bottom, but quite a fair amount of improvement is needed in terms of variety and realism of the scenarios. In fact, future research and development should aim at adding vital features such as Parking Training, Night Driving, and Accident simulations to the driving experience. Moreover, cost-effectiveness considerations will widen access to such advanced simulators.

4 Conception

This chapter focuses on the conceptual framework developed to address the problem identified in Section 1.2, detailing the rationale for the principal design decisions undertaken. Section 4.1 will present an overview of the different elements constituting this solution and specify which of these elements will be further discussed in the following sections.

4.1 Solution Overview

Section 1.2 has identified a driving simulator prototype that applies the integration of virtual reality and driving assistance systems. It shall be designed in such a way as to improve driving performance and further alleviate the need for a driving instructor. It focuses, in particular, on two objectives: providing a VR driving simulator able to offer a better solution compared to traditional non-VR simulators and the use of an integrated driving assistance system to guide drivers along the road and prevent many possible accidents. It would also try to illustrate the feasibility and efficiency of a virtual reality application in such a potential field, possibly enjoying superiorities compared to the classic non-VR driving simulators mainly in respect to the goal of creating a low-cost driving simulator.

This section will provide a summary of the various components that comprise this conceptual solution. The subsequent sections will delve into the specifics of each element, detailing the design choices and implementation strategies employed. The subsequent sections will concentrate on individual components, explaining the key design choices that were necessary in relation to the challenges and research inquiries outlined in Section 1.2. Figure 10 illustrates the overall architecture of the proposed solution.

The prototype of the simulator operates in two distinct modes: Virtual Reality (VR) and non-VR. Initially, the player selects a device to engage with the prototype, opting for either a VR headset or a PC monitor. These devices correspond to the two operational modes of the simulator. Upon selection, the chosen device presents a virtual environment where all interactions and events occur. The player utilizes a driving setup, which includes a steering wheel, pedals, and a gear shifter, to navigate the main menu scene. For this scene, the player gets to set a few settings for: vehicle type, time of day, mode of play, and map. Options are three different cars, three different maps, and two times of day.

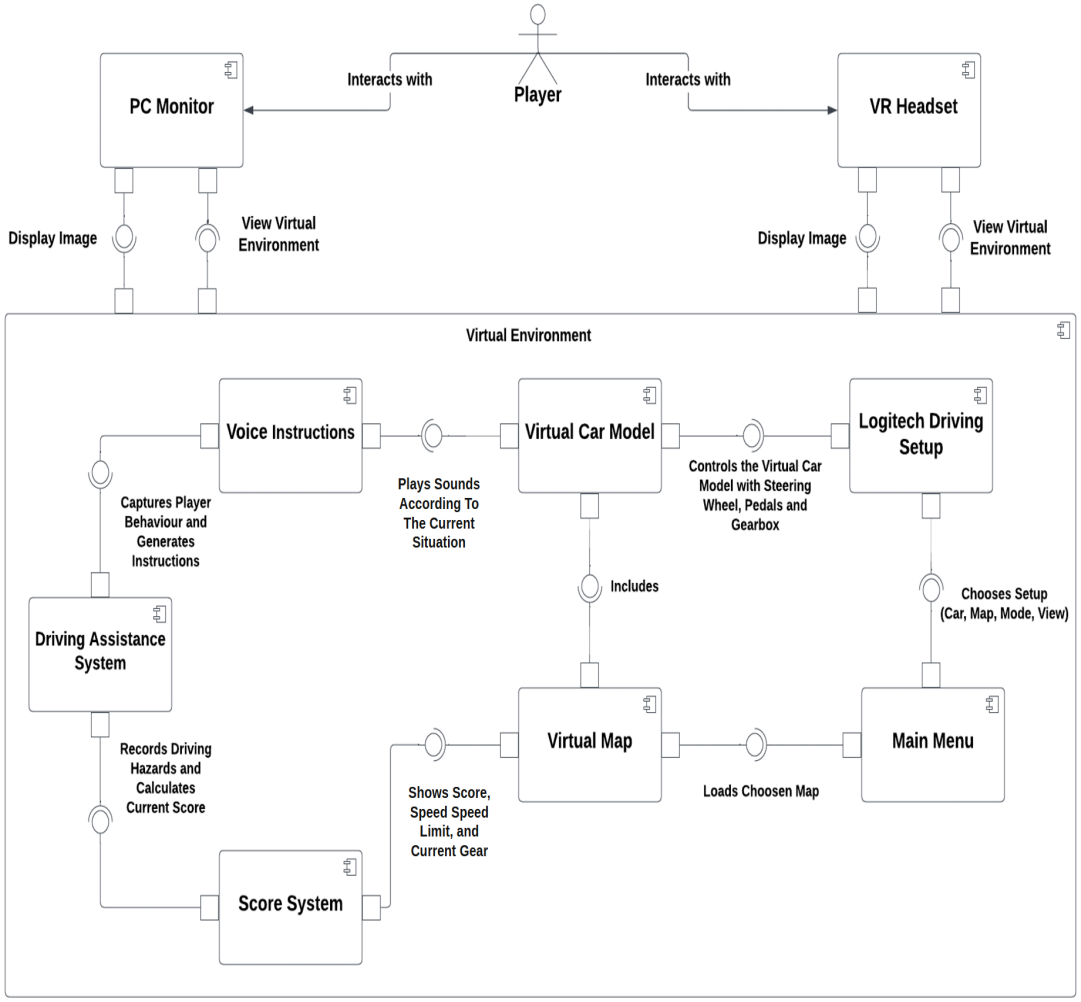


Figure 10: Conceptual Solution Overview.

Once all selections are entered, the player clicks on the play button to start the experience, loading a virtual map based on parameters with a selected vehicle. Afterward, the player can steer the car through this environment by using a driving setup. Scores are also recorded during the driving. An up front tally is given upon the camera viewing, which the player can monitor during the session. The driving assistance system monitors the driver's performance and modifies the driving score; negative scores are added if mistakes or not following the voice instructions of the system are done.

The proceses that are executed within this solution follow a specific pattern, which is done by the user of the driving simulator prototype, like controlling the car using the driving setup, score calculation logic and the driving assistance system. The following Figure 1 illustrates the general steps taken by users to complete the driving simulator tasks.

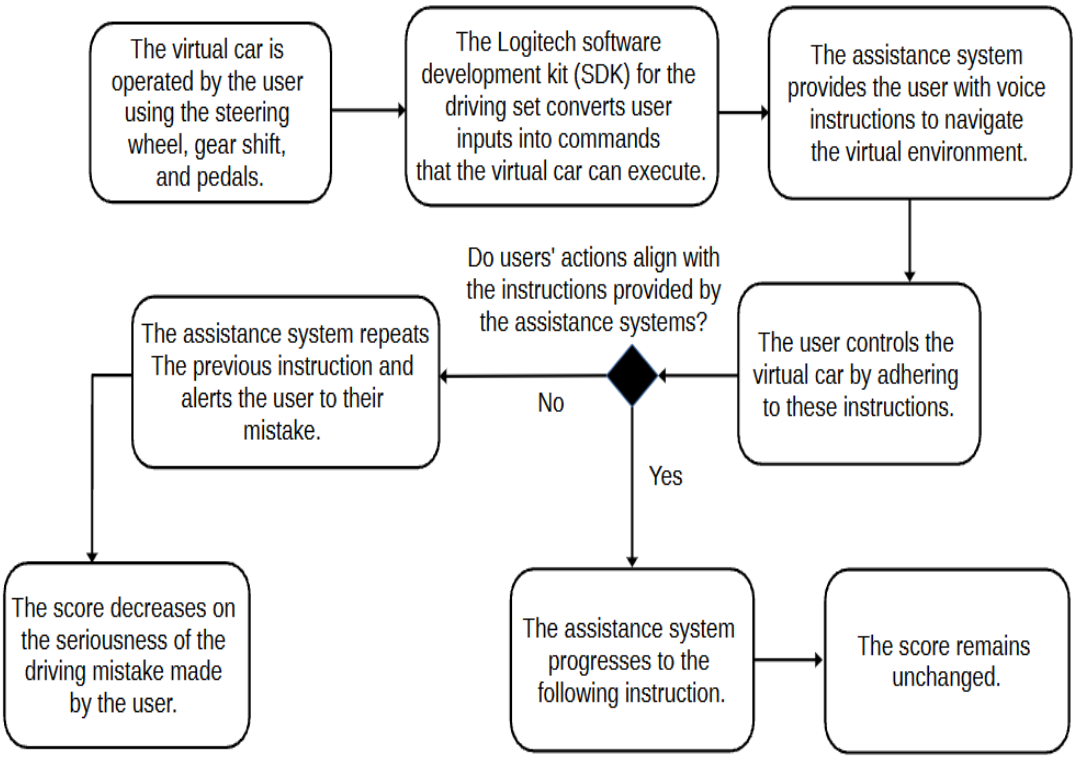


Figure 11: All Processes executed inside the solution.

The framework of the processes presented in Figure 11 allows for an exhaustive look at the steps followed by users in accomplishing all tasks in driving simulators. The first thing a user is supposed to do is controlling a virtual car through the driving setup: the steering wheel, the pedals, and the gear shifter. The Logitech software development kit (SDK), which is provided on their website [80] is used to convert user inputs into machine language, which can be executed by the virtual

car in the Unity game engine. The assistance system provides voice instructions to teach the user how to drive and guide the user along the virtual map. The user needs to drive the virtual car following the exact instruction given by the assistance system. Then, the simulator system compares the user inputs with the assistance system instructions and calculates the score based on the user performance. If the user makes any mistake, the assistance system will warn the user, and the score will be reduced. Otherwise, in case the user had actually followed the instructions, the score would remain the same and the aid system would move to the next instruction.

4.2 Virtual Environment

The Virtual Environment in Figure 10 is the space that contains all the software components. It is the space where the user can interact with the driving simulator prototype. As stated, the virtual environment must be as realistic as possible to provide users with a persuasive and engaging experience. We shall go into some technical details about how this works in Section 5.3 of the Implementation chapter. For now, it's important to know that users can navigate the Virtual Environment with the VR headset or view it on the PC monitor and control the virtual car with the driving setup. These will be elaborated on in the next section, which informs about the different modes and settings available to the user. The three environment modes are further defined in Subsection 4.2.1 as the training area, the city of Paderborn, and the highway. Subsection 4.2.2 refers to several daytime settings: day and night. In Subsection 4.2.3, it is discussed how the Non-VR and VR modes differ.

4.2.1 Environment Modes

The virtual environment is divided into three modes: Training Area, Paderborn City, and Highway. The modes can be challenging and offer different levels of complexity to cater to almost all players. The Training Area is ideal for users to hone their driving skills in a safer, less stressful environment, as this area serves as the newbie playground. Urban environments and challenging road layouts of the German city of Paderborn City. Highway mode conducts German autobahn testing, which can be used to simulate high-speed driving situations. The two additional modes are intended to cater to users at any skill level, whether they be novices or experienced drivers.

Training Area

The Training Area is an ultimate clear, clean canvas for users to experience tuning in with the driving simulator mechanic and control-wise. It has a simple track design with less traffic and obstacles so that users can try their hands at fundamental driving techniques such as steering, accelerating, and braking. For new drivers, it serves as a gentle lead-in to help build their ability and confidence before progressing onto tougher situations. In addition, this mode also brings in others is then equipped with a range of assistance technologies to help the user learn and adapt themselves to

these tools under low strain through voice instructions via an advanced assistance system. These features are designed to enrich the learning journey by providing immediate feedback and assistance. In this mode, users can also learn how to park in difficult parking areas, which is a crucial skill for any driver. Also will be able to learn how to start the car engine, start driving, and brake the car. functions like releasing the clutch slowly to move the car without stalling the engine, and how to change gears will also be learned in this mode. in Figure 12 we can see all the processes executed inside the Training Area.

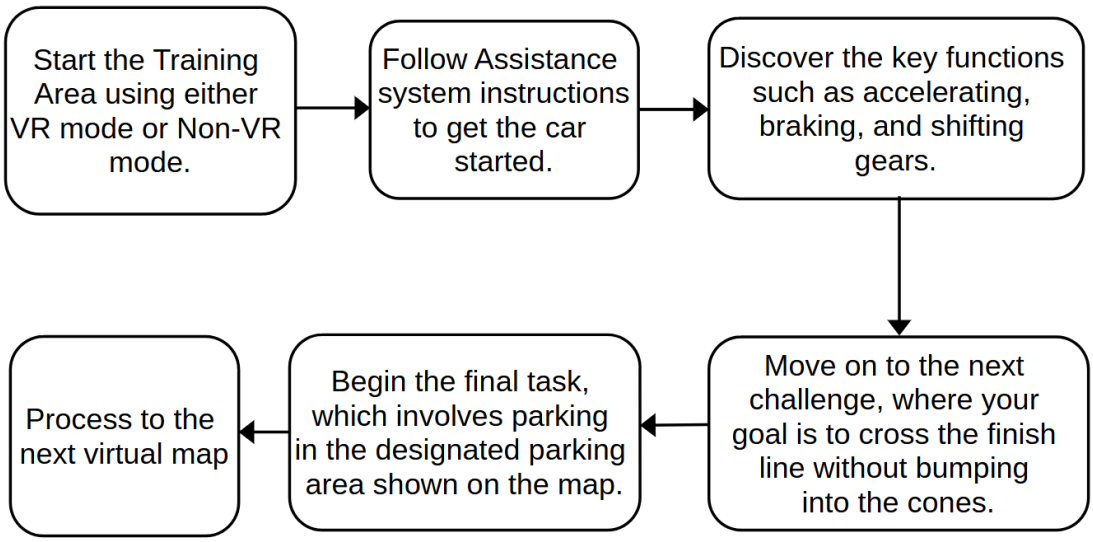


Figure 12: All Processes executed inside the Training Area.

Paderborn City

The actual value of this map is because it is an archetype of the real Paderborn City in Germany. We managed to design this map as close as possible to the real Paderborn City, putting all effort into making it quite realistic and bringing out tiny details. The Paderborn City environment has been designed to represent, for the most part, a western quarter of the city with a small part of the center of the city. It contains richly detailed features of an urban environment, such as traffic lights, crosswalks, and roadways with varied widths-from small lanes up to large boulevards. Traffic provides the user with the challenges of city driving: traffic management, following signs and signals, use of pedestrians. The Paderborn City environment serves to extend this into a more realistic simulation of actual city-driving scenarios that a driver would face in real German cities. Advanced driver-assistance systems in this mode further include lane change detection, collision avoidance, detection of other traffic participants including pedestrians, prevention of speed limit violations, and the monitoring of traffic light compliance. These technologies are designed to make life safer and actually inform people as to what a modern vehicle does in an urban environment. In this setting, users will explore both the advantages and challenges of ADAS within a safe simulated environment. The

Paderborn City environment aspires to fully immerse drivers in the experience of navigating a German city, enriching their understanding of urban driving dynamics and the importance of contemporary assistance technologies. In Figure 13 we can see all the processes executed inside the Paderborn City.

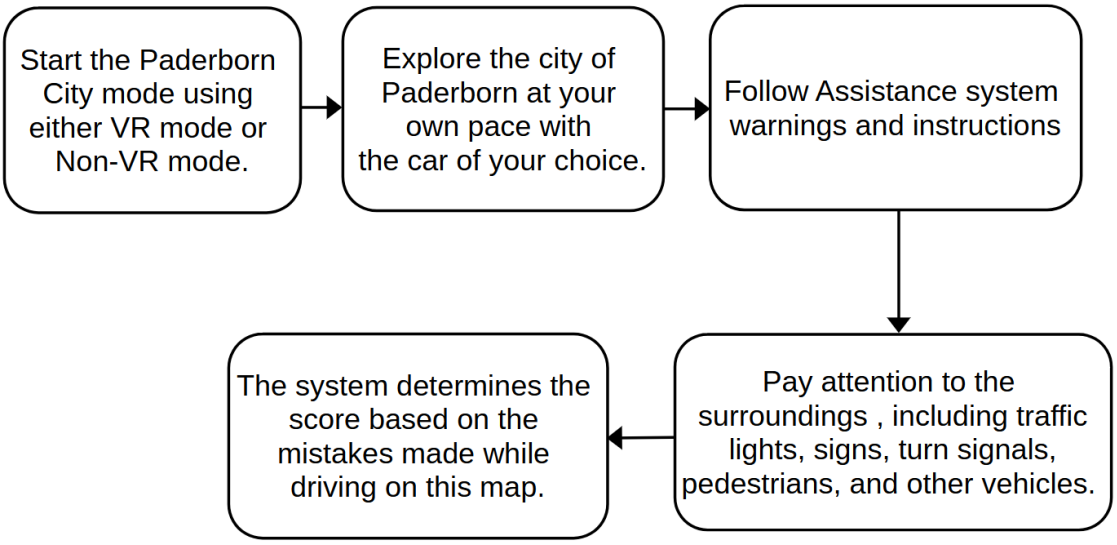


Figure 13: All Processes executed inside the Paderborn City.

Highway

This is a realistic simulation of driving in high speed on a multi-lane highway, prepared with due care to simulate the features of German autobahns. It has long roads with a lot of variation in traffic so that one can manage lane changes, overtaking, and maintaining safety distance at higher speeds. It tries to model the experience of driving on highways in Germany: the driver is very involved with situational awareness and making fast decisions. It also includes taking highway entrance and exit ramps, which is a basic skill that any driver should possess. It also detects the speed and warns the user if the speed limit is exceeded. The Highway environment also presents various challenges, such as unexpected lane closures and fluctuating traffic patterns, to evaluate the user's ability to adapt and respond to changing driving conditions. This map is also provided with the necessary traffic signs, which are usually used on German highways. Minimal environmental elements were also added beside the highway and traffic signs, like the University of Paderborn building and trees to give the users the feeling of moving forward. By capturing the distinctive elements of German highways, this environment seeks to provide drivers with the sensation and experience of driving on an autobahn, deepening their comprehension of high-speed driving dynamics and the importance of modern assistance systems. In Figure 14 we can see all the processes executed inside the Highway Mode.

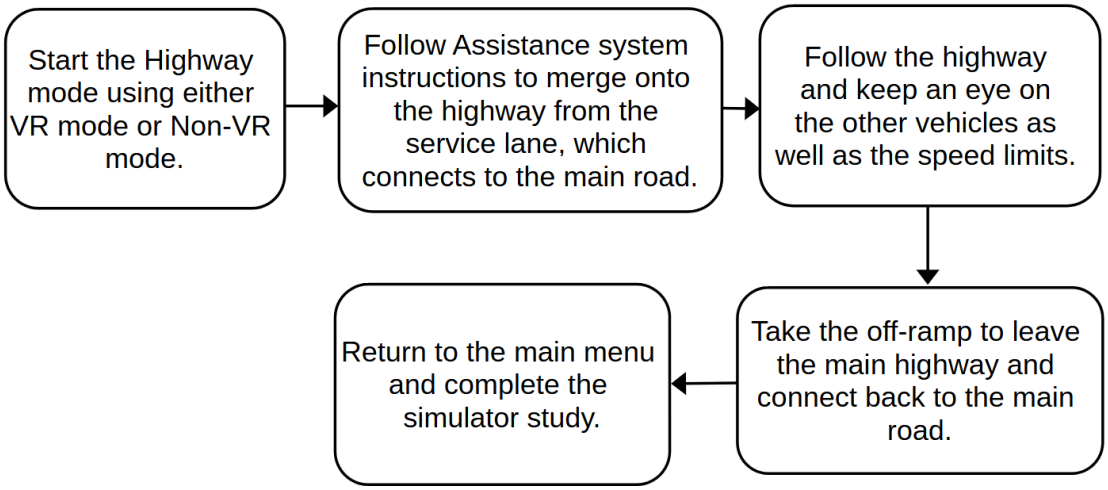


Figure 14: All Processes executed inside the Highway Mode.

4.2.2 Environment Daytime

The virtual environment in our prototype has two daytime scenarios: Day and Night. These two daytimes are designed to deliver a realistic and immersive driving experience, so close to the real life challenges and opportunities. The Day mode offers a bright and clear sight, which is the easiest mode to drive in. That increases visibility and requires less concentration, which is very convenient to practice the driving skill for novice drivers or driving beginners. Dark mode is more challenging because there are more difficult driving situations than in day mode. Drivers have to use the car features like the headlights and the horn while driving in dark conditions. The surroundings will also be equipped with some helpful elements, such as street lights, lighting signs, and building lights, all of which will assist the driver in driving through dark conditions. This model is intended to engage the driver with driving in crucial conditions and enhance their driving experience. The Virtual Environment will provide both integrated Day and Night scenarios suitable for drivers of different levels-from beginners to highly experienced ones. The Day scenario allows the novice driver to practice in a secure and controlled environment, while Night is targeted at those who want more demanding driving conditions. These two options make the Virtual Environment work towards a complete spectrum of driving training, relevant to users' preferences and competencies.

4.2.3 Non-VR vs VR mode

As shown in Figure 10, first the user has to choose between two modes, non-VR and VR. These two modes are developed to answer the research question about the difference between the conventional non-VR driving simulators and the VR one. By allowing the users to test both modes in each map they can decide which mode is more suitable for them. The non-VR mode is designed for users who do not like to use modern VR headsets or have motion sickness problems. This mode is also

designed for users who prefer to use a simple interface to control the virtual car. The VR mode is designed for users who like to have a more immersive and realistic driving experience. The choice between non-VR and VR modalities depends on personal preference, available technology, and the level of immersion desired. In such a way, Virtual Environment allows both users with different capabilities and preferences to take part in a driving simulation experience.

Non-VR Mode:

In the Non-VR version, users interact with the virtual world through a PC-screen type of interface. This is friendly and accessible to every user; thus, it is apt for a variety of users who cannot afford or are not in possession of VR equipment. Users can control the simulation using the driving setup provided to them. Comparatively, non-VR mode offers another representation of the driving scenario and allows users to focus on the acquisition and refinement of driving skills without the use of special equipment. Besides this, this modality is less stressful for the user because it does not require the use of VR headsets or large physical exertion. This is the non-VR mode for those who like the classic, straightforward driving simulation for skills enhancement and practice. In fact, this will also be very useful in cases where the user does not have any virtual reality equipment or prefers working with an interface that is simple and not very tricky to use.

VR Mode:

On the other hand, the VR mode provides users with an immersive and more realistic driving experience using modern technologies and equipment. This mode is designed for users who prefer a more engaging and interactive experience. The VR mode provides a 360-degree view of the virtual environment, which is important to copy real-life situations and give users the feeling of being inside a real car. This immersive experience significantly enhances the user's spatial awareness and depth perception, leading to a superior driving simulation. VR headsets are usually provided with motion-tracking cameras and sensors, which allow users to move their heads and bodies to interact with the virtual environment. This mode can substantially improve the learning process, as driving skills are practiced within a more realistic and captivating context, We will test this in the evaluation chapter 6. The VR mode is designed to provide users with a more immersive and engaging driving experience. By leveraging the latest VR technologies, users can explore the virtual environment in a more realistic and interactive manner.

4.3 Driving Assistance System

One of the most important features of the prototype in the driving simulator is indeed the driving assistance system; this is crucial because it provides enormous guidance, feedback, and instruction to users during any given driving time. The driving assistance system is designed in such a way as to enhance driving performance by users and to keep them from accidents; thus, it provides a full package

for learning. It will be integrated into the Virtual Environment and will be able to support users online in real time while navigating the virtual map. It will explain the driving assistance system in detail, such as the feature and functional description. Subsection 4.3.1 will provide an overview of the system architecture, while Subsection 4.3.2 will detail the individual elements that comprise the system.

4.3.1 System Architecture

The architecture of the driving assistance system is built on the premise of providing more user-friendly experiences and intuitiveness, thereby making each interaction natural and helpful. In this system, three components play the most important roles: the Sensor Module, the Decision-Making Module, and the Feedback Module. They interact with each other for data collection, interpretation of user behavior, and feedback with advice on a real-time basis. The driving assistance system integrates these components and therefore allows one to learn in tune and have more interactive skills in driving that are going to make them a better driver while ensuring responsible driving. Each of the subsections after this shall explain the components in detail, their functions, and how they relate to each other within the system. This was a design product of extensive research and analysis regarding existing driving assistance systems and principles of human-computer interaction. Although the structure for this system was developed independently, it nonetheless reflects a variety of studies and frameworks that so far have been advanced in the field.

Sensor Module:

Serving as the system's sensory apparatus, this module continuously collects data from the virtual environment, including user inputs, vehicle positioning, speed, and other essential metrics. The design choice for accurate and real-time data collection is supported by research in driving simulation systems, such as the work by Gietelink et al. [81]. It acts as the foundational element that keeps the system updated on all occurrences within the simulation. Some examples of the data collected by the Sensor Module include the user's steering angle, throttle and brake inputs, turning radius, physical collider interactions, current gear, clutch release rate, turning signals, and vehicle speed. By monitoring these variables, the Sensor Module provides the Decision-Making Module with the necessary information to evaluate user performance and provide relevant feedback. This component is essential for ensuring that the assistance provided is accurate and timely.

Decision-Making Module:

This part serves as the brain of the system. It processes the information gathered by the Sensor Module, analyzing how the user drives and performs. Using sophisticated algorithms and proven standards, as outlined in Pomerleau (1989) in "ALVINN: An Autonomous Land Vehicle in a Neural Network" [82], it determines the best actions to support the user. Whether it's suggesting a change in speed or warning the user about a possible danger, this module ensures that the help offered is timely

and relevant. By utilizing the data from the Sensor Module, the Decision-Making Module provides personalized feedback and advice that improve the user's driving experience. This component is essential for making sure the driving assistance system runs smoothly and effectively.

Feedback Module:

This module serves as the system's communicative interface. It relays the decisions made by the Decision-Making Module to the user through verbal instructions, warnings, and notifications. The design of this module is influenced by research emphasizing the importance of clear and actionable feedback in driving simulations, such as the review paper "Advanced Driver Assistance System (ADAS): A Review of Systems and Technologies" by Amditis et al. (2010) [83]. The objective is to deliver clear and actionable feedback that aids the user in navigating the virtual environment with safety and efficiency. By integrating these components harmoniously, the driving assistance system fosters a cohesive and interactive learning environment that enhances driving proficiency and encourages safe driving practices. The aim of the Feedback Module is to ensure that users are informed in due time and get relevant advice on what decisions they need to make for improvement in driving skills. This module will be very handy in developing good interaction between the system and user for the betterment of the experience of the driver. The module allows logging whereby it keeps track of users' performance over some time. This feature enables the users to review their progress and observe patterns in their driving habits. The Feedback Module is really crucial for the improvement of overall user experience and safe driving behavior with continuing constructive feedback.
Figure 15 provides a detailed view of the driving assistance system architecture In The driving simulator. This illustration showcases the interconnected nature of the system's components, highlighting the flow of data and feedback between the Sensor Module, Decision-Making Module, and Feedback Module. By integrating these elements, the driving assistance system offers users a comprehensive and interactive learning experience that enhances driving performance and safety. It is provided as a UML component diagram that shows the components of the system and their relationships.

Score System:

The Score System is a crucial part of the driving assistance system, providing users with a clear and concise overview of their performance. It makes it easy for users to get a general view of their performance. The system measures the user's score by his driving behavior, such as speed and accuracy, and according to traffic rules. The score would be real so users can monitor their progress and find what needs to be improved. It should be user-friendly, appealing, and easy to use; it should deliver clarity to its users so that they track their performances with ease and work for continuous improvement. This would, in turn, encourage the users to drive better in order to improve their scores. The score system is used for calculating the final score of the user at the end of the simulation. It constitutes the very foundation

of this score, since the performance of the user will be based on following the traffic rules, maintaining speed limits, and proper driving behavior throughout the session. Scoring is excellent, as it summarizes users' performance in a nutshell, which encourages self-assessment and improvement for better learning.

4.3.2 System Elements

The driving assistance system includes a variety of important components that collaborate to offer users guidance, feedback, and instruction during their driving journey. These components aim to improve driving skills, reduce the risk of accidents, and provide a thorough learning experience. In this section, we will explore the specific parts of the driving assistance system, explaining their functions and contributions to the overall system. The driving assistance system consists of several crucial elements, each contributing significantly to its overall effectiveness and performance.

User Interface (UI):

The UI is crafted to be intuitive and accessible, facilitating seamless interaction between users and the system. It features visual displays, control panels, and various interactive elements that deliver real-time feedback and guidance. It also includes the scoring system, which provides users with a clear and concise overview of their performance. The UI is designed to be user-friendly and engaging, ensuring that users can easily navigate the system and access the information they need to improve their driving skills.

Algorithms and Rules:

The Decision-Making Module utilizes a collection of algorithms and established rules to interpret the data gathered by the sensors. These algorithms are specifically designed to evaluate the user's driving performance and provide suitable feedback and guidance. More details about the algorithms and rules used in the Decision-Making Module will be discussed in Implementation Chapter 5.

Communication System:

This component guarantees that the system's feedback and instructions are conveyed to the user in an effective manner. It includes auditory alerts, signals, and other communication methods that assist the user in comprehending and responding to the system's guidance. The Communication System is designed to be clear, concise, and timely, ensuring that users can easily understand and act upon the information provided.

Data Storage and Processing:

The system incorporates elements for the storage and processing of data collected during the simulation. This functionality enables a comprehensive analysis of the

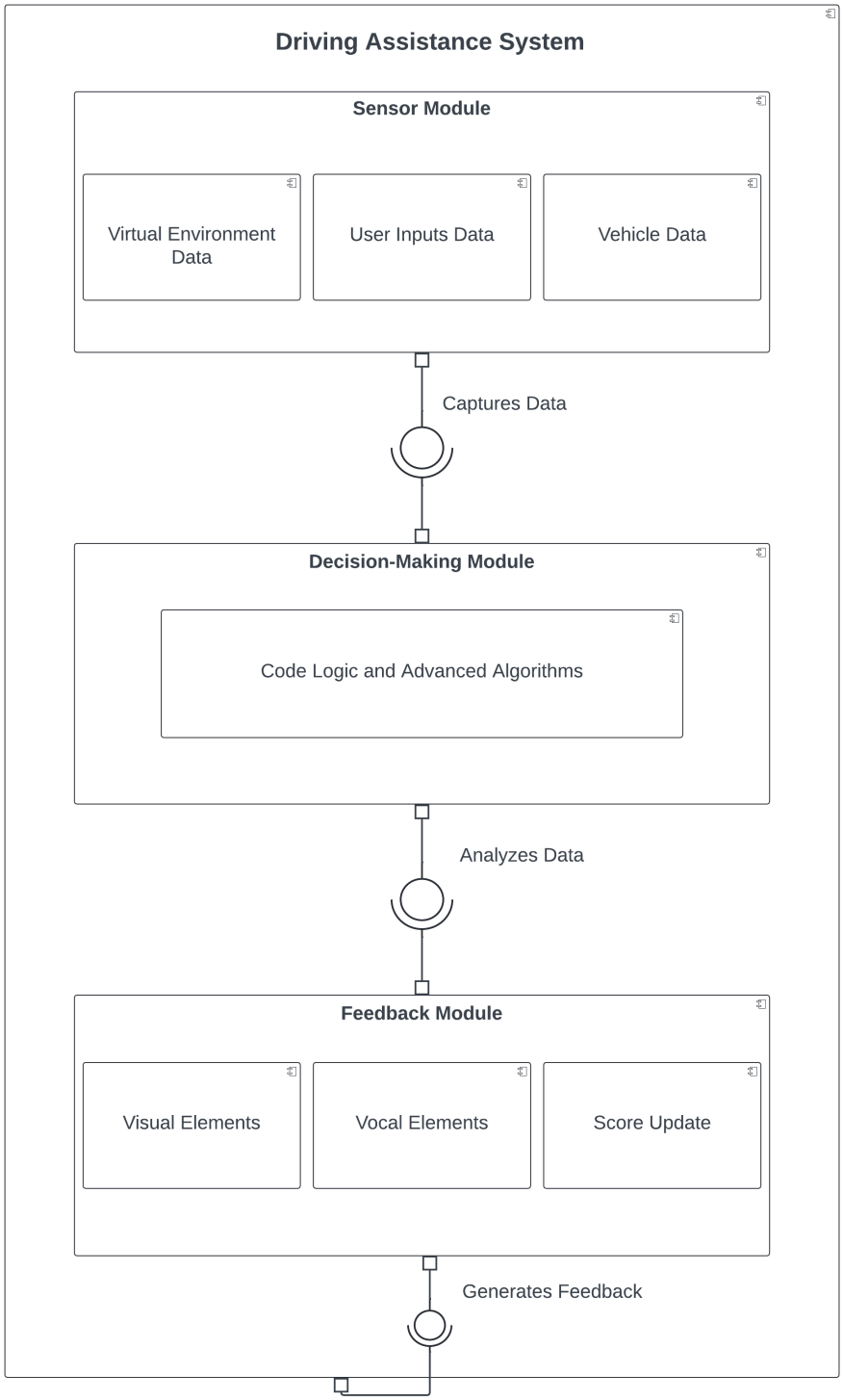


Figure 15: Detailed view of the driving assistance system Architecture In The Driving Simulator.

user's performance, leading to the generation of tailored feedback and recommendations. The Data Storage and Processing component is crucial for tracking user progress, identifying areas for improvement, and enhancing the overall learning experience.

4.4 Realistic Vehicle Functions

4.4.1 Car Engine Start

The Car Engine Start feature is a crucial part of the driving simulator prototype, letting users fire up the vehicle's engine and gear up for their driving adventure. This function replicates the actual process of starting a car, walking users through the necessary steps to get the engine running. By incorporating this feature, the driving simulator enhances the realism and excitement of the experience, making it feel more authentic. The Car Engine Start function is designed to be user-friendly and simple, so users can easily follow along and jump into their driving journey. In this section, we will dive into the specifics of the Car Engine Start function, outlining the steps needed to get the vehicle ready for the road.

4.4.2 Gear Shifting

The gear shifting feature is a vital component of the driving simulator prototype. It provides users with the ability to change gears and manage the vehicle's speed and acceleration, similar to the operation of a manual transmission. Users must interact with the gear shifter to influence their driving speed. The addition of this feature greatly enriches the realism and engagement of the driving experience, thereby enhancing the overall authenticity and ensuring that users remain actively involved. The design emphasizes simplicity and responsiveness, enabling users to make smooth adjustments to their speed and acceleration while exploring the virtual environment. This section will delve into the functionality of the Gear Shifting feature, outlining the necessary steps for gear changes and vehicle control.

4.4.3 Clutch Control

The Clutch Control feature is essential in the driving simulator prototype. It gives users the power to engage and disengage the clutch, which affects how the vehicle moves and its speed. This function is similar to a manual car's clutch, requiring users to coordinate the clutch pedal with the gear shifter for smooth transitions. By integrating this feature, the simulator creates a more realistic and challenging driving experience, enhancing its authenticity and complexity. The Clutch Control is built to be responsive and precise, enabling users to manage the clutch pedal effectively for gear shifts and vehicle dynamics. This section will delve into the Clutch Control function, detailing how to operate the clutch smoothly and change gears.

4.4.4 Accelerating and Braking

The Accelerating and Braking function is essential in the driving simulator prototype. It gives users the ability to control the speed and movement of the vehicle. This function replicates the actions of accelerating and braking, asking users to interact with the accelerator and brake pedals. With this feature, the simulator creates a realistic driving experience that enhances user engagement. It is designed to be intuitive and responsive, allowing users to navigate the virtual environment safely. This section will describe how to accelerate and brake smoothly.

4.4.5 Turning Signals

The Turning Signals function is a key part of the driving simulator prototype, allowing users to show when they plan to turn or change lanes. It mimics the real-life action of activating turn signals in a car, so users have to interact with the controls to let other drivers know their next move. This feature makes the simulator feel more realistic and interactive, boosting the overall safety and authenticity of the experience. It is designed to be straightforward and easy to use, so users can quickly activate the signals and effectively communicate their intentions. This section will go into detail about how to use the Turning Signals, including the steps to activate them and signal turns and lane changes accurately.

4.4.6 Lights Control

An essential part of the driving simulator prototype is the lights control mechanism, which gives users control over the illumination and visibility of the car. It simulates turning on headlights and other lights, encouraging users to engage with the controls to improve visibility under different driving circumstances. By including this function, the simulator greatly raises the level of engagement and realism, improving authenticity and safety. Users may effortlessly change the vehicle's illumination while navigating the simulated area thanks to the lights control's responsive and easy design. An explanation of the Lights Control system's operation will be given in this part, along with instructions on how to activate the lights and improve visibility in various driving situations.

4.4.7 Force Feedback

The Force Feedback function is a critical component of the driving simulator prototype, allowing users to have a realistic and immersive driving experience. This function simulates the actual feel of driving a car by providing tactile feedback via the steering wheel and pedals. By including this functionality, the driving simulator prototype delivers a vivid and fascinating experience, considerably increasing immersion and realism in the simulation. The Force Feedback feature is designed to be both sensitive and dynamic, allowing drivers to perceive road roughness, vehicle behavior, and driving conditions via the steering wheel and pedals. In this section, we will look at the specifics of the Force Feedback feature, focusing on the tactile sensations and feedback that users receive when driving.

4.4.8 Side and Rear Mirrors

The Front and Rear Mirrors feature is an important aspect of the driving simulator prototype since it allows users to keep an eye on their surroundings and be aware of them while driving. This function simulates the process of checking the front and back mirrors in a real car, motivating users to utilize the mirror controls to adjust their view and watch traffic. By including this function, the driving simulator prototype gives a more realistic and comprehensive driving experience, improving overall safety and alertness during the simulation. The Front and Rear Mirrors feature is designed to be simple and intuitive, allowing users to easily adjust the mirrors for optimal monitoring of their surroundings. This section explains the features of the front and rear mirrors functions, including the steps needed to check the mirrors and maintain awareness while driving.

5 Implementation

This chapter will present the technical realization of the developed prototype of a driving simulator with detailed design and the development of all components discussed in Chapter 4. Section 5.1 will hence disclose the main design choices and related strategies. All the following sessions will go in deep about every single most important component, explaining the technical path of every element and the difficulties emerging during the development. Section 5.2 will dedicate the VR headset used in the prototype of the driving simulator, while Section 5.3 describes the driving kit. Section 5.4 will focus on the virtual environment, detailing the different modes and settings available to the user. Section 5.5 will delve into the driving assistance system, explaining the system architecture and the individual elements that comprise the system. Section 5.6 will discuss the realistic vehicle functions, detailing the implementation of the car engine start, gear shifting, clutch control, accelerating and braking, turning signals, lights control, force feedback, and side and rear mirrors.

5.1 Implementation Overview

The prototype driving simulator was implemented through various design and development phases that resulted in an engaging simulation environment. This section describes the process of implementation, giving an overview of the important choices regarding design approaches. Development of the prototype driving simulator was performed in Unity, a game engine offering reliable and powerful possibilities for the creation of immersive and interactive simulations. This covers designing and developing various components comprising the virtual environment, driving assistance system, and even down to realistic vehicle functions; each was designed to be quality-oriented, highly realistic concerning driving simulation. The implementation process involved advanced technologies in terms of virtual reality and driving assistance systems to offer a state-of-the-art innovative simulation experience. These technologies combined in the prototype driving simulator realize natural elastically interactive driving that improves the user's driving skills and proper knowledge of driving. Each of the above elements will be detailed in the following sections from a technical point of view by presenting design choices and strategies adopted for their implementation, which would ensure comprehensive and interactive simulation experiences. Figure 16 presents the overview diagram once more, now updated to include the technologies utilized to develop the various components of the conceptual solution in our prototype implementation.

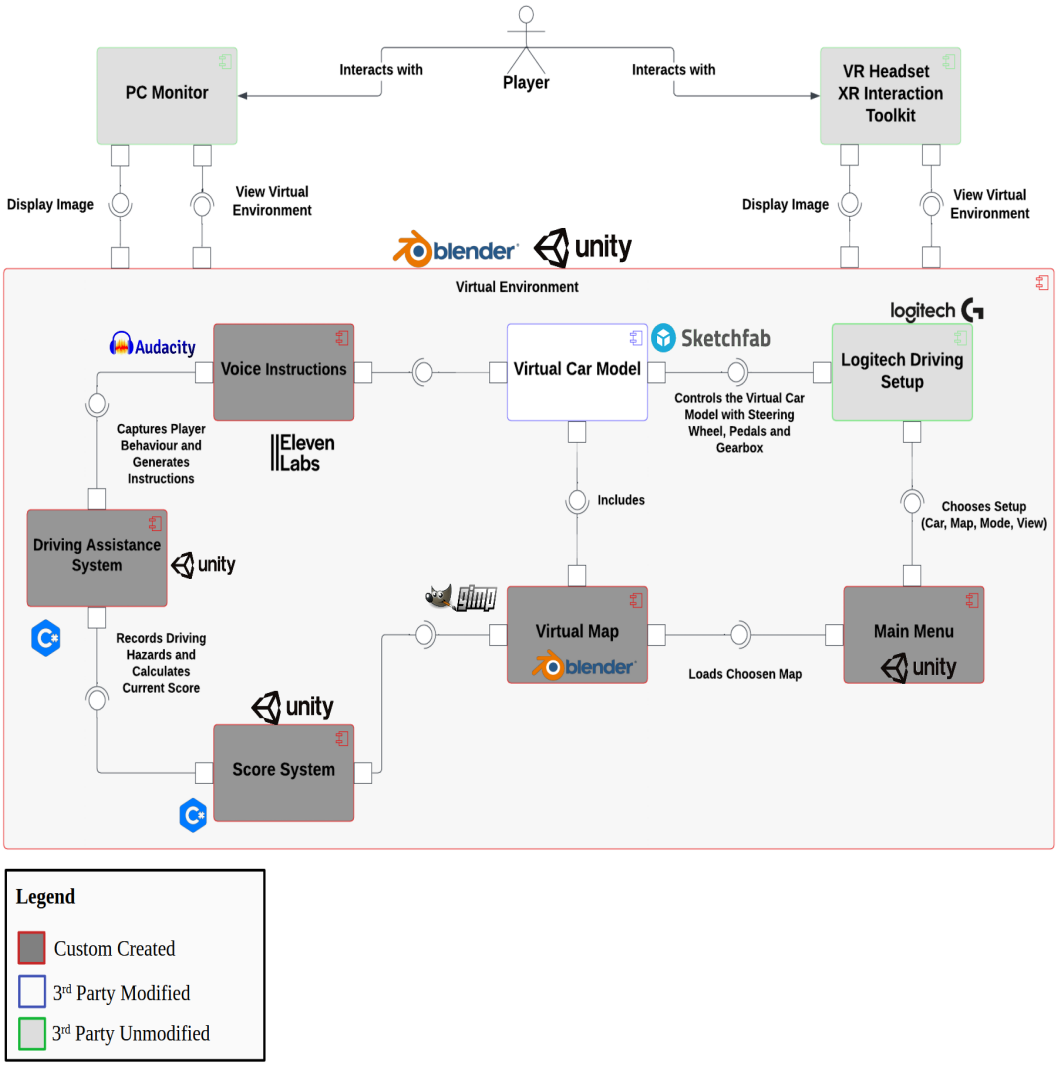


Figure 16: Implementation overview including the used technologies.

5.2 VR Headset

The VR headset is one of the most important features in the driving simulator prototype, as it provides users with an immersive and engaging experience. This section therefore provides information on the VR headset in terms of design choices and implementation strategies adopted. The VR headset that was used in this study includes the Meta Quest 2 and 3, since these have a high resolution and refresh rate that is very appropriate for a driving simulator. The Meta Quest headset is an independent VR headset with wireless and untethered capabilities, hence enabling free movement by the user and interaction with the virtual environment. The head-mounted display screen shows visuals clearly and in great detail. The headset also contains sensors and tracking technology that allow for accurate and precise

tracking of motion. This is designed to be comfortable and light, allowing users to wear the headset for extended periods without fatigue. The Meta Quest headset allows for a wide range of VR applications and games; as such, it was chosen to be the most suitable option for the driving simulator prototype [84]. The Meta Quest 3 headset costs about 530 USD, which is a reasonable price for a high-quality VR headset. The Meta Quest 3 headset is also compatible with the Unity game engine, which is used to develop the driving simulator prototype [22].

5.3 Driving Kit

The Logitech G923 Driving Force Racing Wheel and Pedals, with an additional gearbox, were the gaming driving settings used to simulate the functions in the real car cockpit. The driving kit works in providing a realistic and immersive experience while driving, manipulating the virtual car with much greater precision and accuracy. The Logitech G923 Racing Wheel features a leather-wrapped wheel with responsive force feedback-one can feel the road and make fine adjustments in steering. The wheel houses a series of buttons and controls that approximate what would come on an actual car, thereby affording the user a facile method of interacting with the virtual environment. Logitech G923 Pedals are responsive to provide a brake pedal, throttle pedal, and clutch pedal that will enable the user to accelerate, brake, and shift gears accurately. The pedals should be of a design that allows adjustability for customization to ensure comfort and an ergonomic driving position. The Logitech G923 Gear Shifter is another peripheral in the system, which provides a realistic way of changing gear for users. It features a six-speed H-pattern shifter, complete with a reverse gear for manual gear-shifting and speed or acceleration control of the vehicle. The Logitech G923 Driving Force Racing Wheel and Pedals with Gear Shifter are compatible with a wide array of driving simulators and games, hence an ideal choice for the prototype of the driving simulator [23]. We provided users before the user study with the necessary controls in order for them to operate the gaming steering wheel kit. Logitech G923 driving kit costs about 400 USD, which is a reasonable price to get a high-quality driving kit. The Logitech G923 driving kit also supports Unity game engine, which we are using for developing the driving simulator [23]. Figure 17 shows the controls of Logitech G923 Driving kit used in the driving simulator.

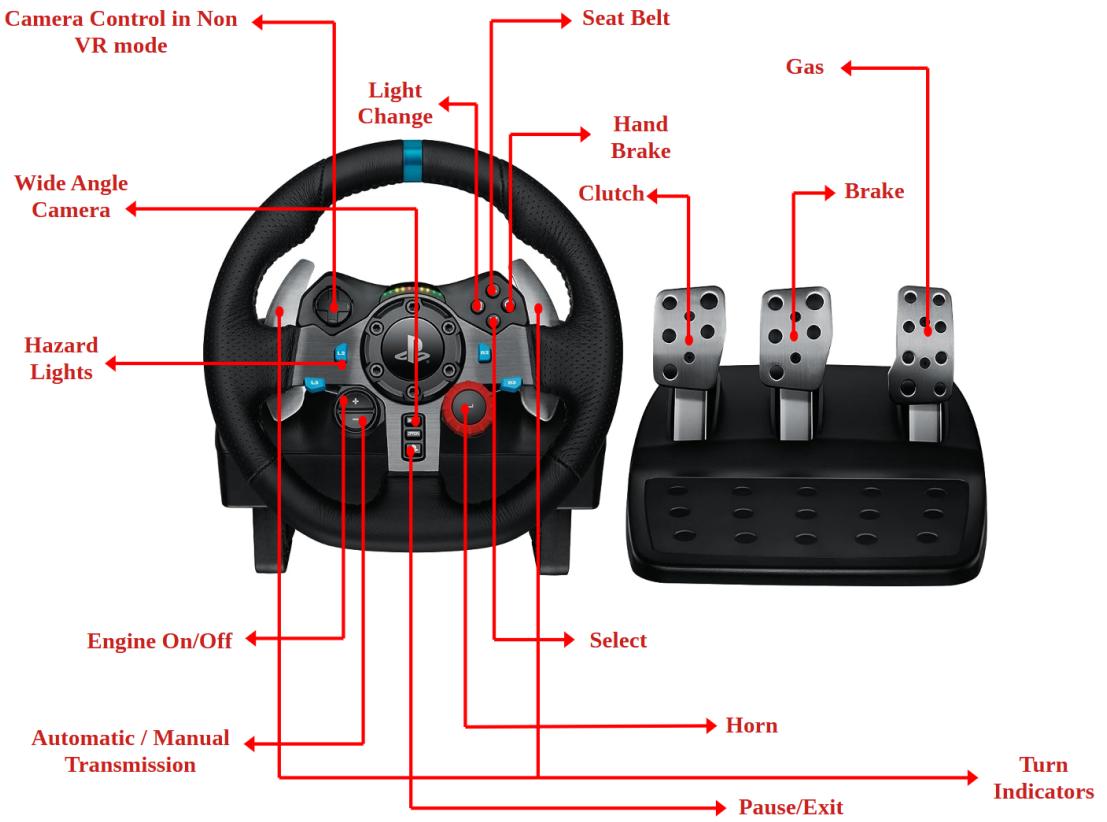


Figure 17: Logitech G923 Driving Kit Controls Used In The Prototype.

5.4 Virtual Environment

As discussed in Section 4.2 and shown in Figure 16, the virtual environment containing the Training Area, Highway, and Paderborn City is developed and designed mainly in Unity and Blender software. In Unity, the main function and the different 3D models have been integrated together. In Blender, most of the 3D models have been designed. Some of the 3D models have been downloaded from the internet (Sketchfab [85]) and modified to fit the virtual environment. The virtual environment is designed to be as realistic as possible, providing users with an immersive and engaging experience. The virtual environment is divided into three modes: Training Area, Paderborn City, and Highway. Next, we will provide an overview of the design and implementation of the virtual environment, detailing the various modes and settings available to users.

5.4.1 Training Area

The Training Area map is completely designed in Blender software [57], street cones models were designed to serve the first task which is driving through the path made by these cones to learn how to steer the car and drive in narrow streets without colliding with the cones, and a blue arrow was used to guide the driver where to

start driving in this task. Figure 18 shows the first task cones and guiding blue arrow. The next task is to finish the first task and reach the last cone line defined as a physical collider in Unity. The last task is about learning how to park between many parking cars without colliding with any of them. Figure 19 shows the parking lot is designed in Blender and the blue guiding arrow, which points to the parking area, where the driver has to park exactly. Before the first task, the user takes the initial training session, which is about learning how to start the car, accelerating, braking, and using different buttons of the driving setup. All these instructions are played in the form of audio instructios as a part of the advanced driving assistance system integrated into the driving prototype.

Figure 18: Training Area First Task.

Figure 19: Training Area Parking Task.

5.4.2 Paderborn City

The Paderborn City map in Blender was the most time-consuming task. It is a replica of the real Paderborn City in Germany. It is designed to be as authentic as possible. This object was hard to make because we used a part of the real Paderborn City in Germany from Google Maps [86] to give drivers the real feeling of driving in Paderborn City. Figure 22 shows the top view of the Paderborn City map created completely in Blender software for this study (top), and the real map of Paderborn City in Germany was taken from Google Maps (bottom). The Paderborn City environment primarily showcases the western quarter of the city along with a small section of the city center. As shown, there was no big difference between the real and the virtual map, which is a good sign that the virtual environment is designed to be as realistic as possible. To simulate exactly the same streets and roads in Paderborn City, we used Blender to design the 3D models in the map, like; streets, buildings, traffic signs, street lights, and parking lots, in a way that copies the exact shapes of real-life objects. Figure 20 shows some of the 3D models created in Blender for the purpose of simulating a realistic Paderborn City map. On the upper left side, we can see the streets with the textures, which are created in GIMP software [87]. On the right upper side, it shows a very famous building in Paderborn City, which is the Home Deluxe Arena football stadium, which is the main stadium for the Paderborn football team SC Paderborn 07 [88]. On the lower side we see some of the German traffic signs, designed especially for this study. Finally, on the right lower side, we can see the street lights, which are designed to be as realistic as possible and copy the exact shape of the German street lights. Besides the 3D design in Blender, there are many functions developed in VS Code [89] with C# programming language, like street lights color changing logic below 5.1. Other technical details about the development of the Paderborn City map is the use of the AI Navigation package in Unity to make it possible to move the traffic AI cars, bicycles, and pedestrians around the city to make the environment more realistic. The C# code, which is written for this purpose, is very long so everyone can check it out in the GitHub repository provided for the source code, which can be found in Appendix Section A.

5.4.3 Highway

The Highway map is also created and desiged in blender from scratch. The Highway map is designed to be a long straight road with three lanes in each direction. We also created landscape on the highways both sides to give the users the feeling of driving forward. The highway textures like the white markings defining different lanes borders are created in GIMP [87]. We also integrates the highway entrance and exit ramps. Figure 21 shows different scenes taken in the Highway map used in our driving simulator prototype.

```
public float transitionTime = 15f;
public void SetRedLight()
{
    StartCoroutine(TransitionToRed());
}
private IEnumerator TransitionToRed()
{
    SetYellowLight();
    yield return new WaitForSeconds(transitionTime);
    SetLights(true, false, false, false, true, false, false, true);
}
public void SetYellowLight()
{
    SetLights(false, true, false, true, false, true, true, false);
}
public void SetGreenLight()
{
    StartCoroutine(TransitionToGreen());
}
private IEnumerator TransitionToGreen()
{
    SetLights(true, true, false, false, true, false, true, false);
    yield return new WaitForSeconds(transitionTime);
    SetLights(false, false, true, true, false, true, false, false);
}
private void SetLights(bool red, bool yellow, bool green, bool pedestrianRed, bool
    pedestrianGreen, bool bikeRed, bool bikeYellow, bool bikeGreen)
{
    if (red) SetEmission(redLight, true); else SetEmission(redLight, false);
    if (green) SetEmission(greenLight, true); else SetEmission(greenLight, false);
    if (yellow) SetEmission(yellowLight, true); else SetEmission(yellowLight, false
        );
    pedestrianRedLight.color = pedestrianRed ? Color.white : Color.black;
    pedestrianGreenLight.color = pedestrianGreen ? Color.white : Color.black;
    if (bikeRed) SetEmission(bikeRedLight, true); else SetEmission(bikeRedLight,
        false);
    if (bikeYellow) SetEmission(bikeYellowLight, true); else SetEmission(
        bikeYellowLight, false);
    if (bikeGreen) SetEmission(bikeGreenLight, true); else SetEmission(
        bikeGreenLight, false);
}
```

Listing 5.1: C# Code For Changing Traffic Lights.

Figure 20: Some 3D Models created For The Paderborn City Map.

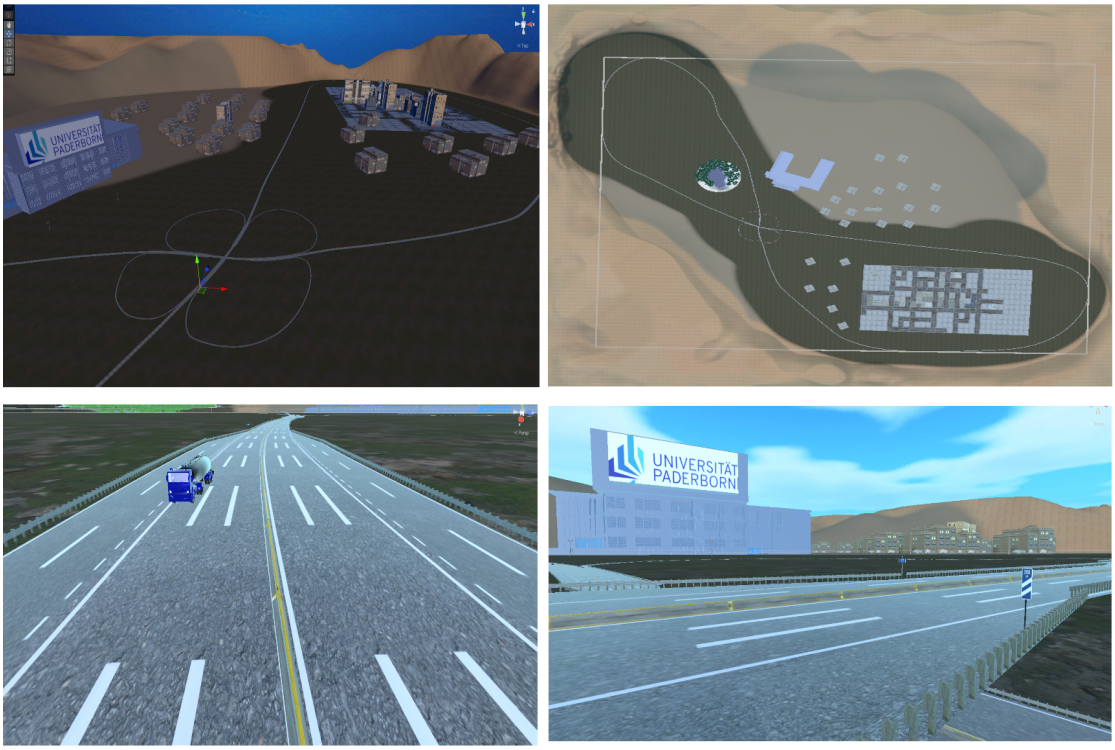


Figure 21: Scenes Taken In The Highway Map.

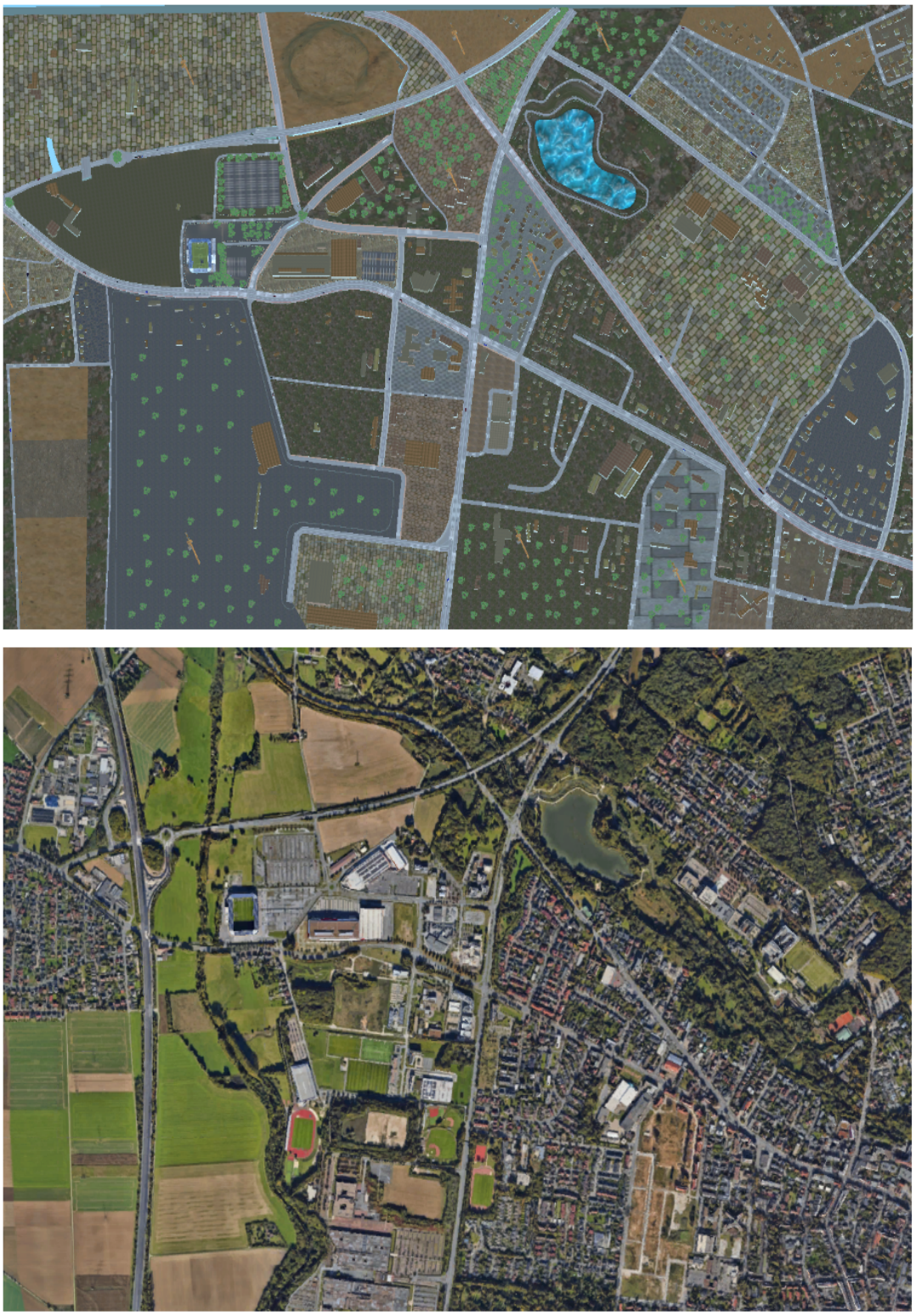


Figure 22: Top View of the Paderborn City Map created Completely in Blender software For This Study (Top), and The Real Map of Paderborn City in Germany Taken from Google-Maps (Bottom).

5.4.4 Daytime Settings

The Daytime settings are designed to provide users with the option to choose between day and night modes. To simulate these functions we used in Unity the skybox material function in the lightining window to change the skybox material to a night or day material. Additionally, we used the directional light function to change the light intensity and color to simulate the day and night modes. Figure 23 shows the light settings in Unity to simulate the day and night modes.

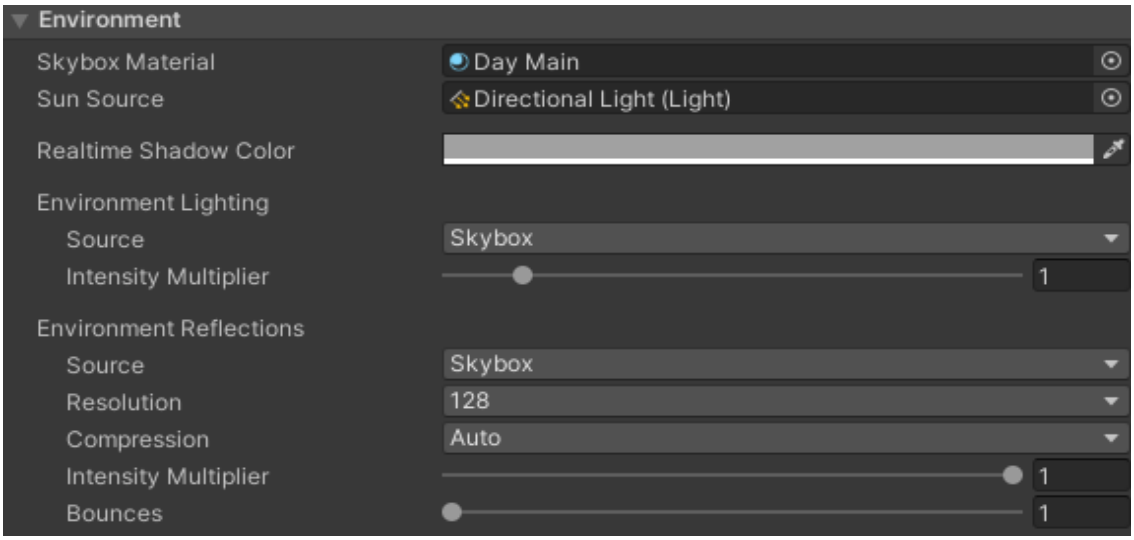


Figure 23: Unity Lightining Settings.

5.4.5 Non-VR VS VR Mode

These modes are an essential part of the driving simulator prototype, as they provide users with different ways to interact with the virtual environment and they help us answer one of the research questions we seek to answer in this study. Therefore, we will discuss some of the implementing aspects we used to develop these modes. First, in Unity we used different cameras for each mode, for VR mode we used the XR-Origin(XR-Rig), which is typically used in Unity to integrate the VR headset and show the user the environment in VR perspective. For the XR-Origin we used a 110 FOV which is perfect for VR view in Meta Quest. Then for the non-VR mode we used the main camera in Unity, which is used to show, the user the environment in a non-VR perspective. The main camera is set to 60 FOV, which is perfect for non-VR view. As mentioned earlier, the view mode can be changed in the Main Menu scene, where the user can choose between VR and non-VR views. Figure 28 shows the Main Menu scene where the user can choose between VR and non-VR view.

5.5 Driving Assistance System

There were different technologies used to develop the driving assistance system, which is an essential part of the driving simulator prototype. The driving assistance system includes many elements like the User Interface (UI), Algorithms and Rules, Communication System, and Data Storage and Processing. These elements require different technologies to be developed. We will present these different technologies with the different development techniques used to develop the driving assistance system in the following sections.

5.5.1 System Architecture

The driving assistance system architecture is designed to prioritize user experience and intuitiveness, ensuring that each interaction is both natural and beneficial. The architecture of the driving assistance system is designed to be modular and flexible, allowing for easy integration of new features and functionalities. The system architecture is divided into three primary components: the Sensor Module, the Decision-Making Module, and the Feedback Module.

Sensor Module

In Section 4.3.1, we discussed the Sensor Module as the assistance system sensory apparatus, which continuously collects data from the virtual environment, including user inputs, vehicle positioning, speed, and other essential metrics. In this section we will present the different implementation techniques used to develop the Sensor Module. The Sensor Module collects data from different resources as shown in Figure 15. First, the virtual environment data is collected from different maps which are presented in the previous sections. For example, the data, which is collected from the surrounding environment in the Training Area is the physical collision data when colliding with the cones, and other parking cars, and when reaching the invisible colliders of the task finish line. This function is implemented in Unity using the OnTriggerEnter and OnTriggerExit functions, which are built-in Unity functions that are called when a collision occurs. Figure 5.2 shows the C# code used to implement the Training Area collisions logic. The data collected from the virtual environment is then sent to the Decision-Making Module to be processed and analyzed. In the Paderborn City map, the data collected will vary more than the data collected in the Training Area. There data will be collected from the physical collider integrated in the environment such as; pedestrians, traffic cars, walls, street lights invisible stop lines, different lanes, intersections, and colliders provided to define different speed limits on different streets. This was very time-consuming, cause the map desiged for Paderborn City was very big and complex. Other data collected in the Paderborn City map is the data collected from the street lights script, which then will be stored in the data structure in the assistance system script, all these logic codes are provided in the source code in the A Section. On the Highway the data is mainly collected from the traffic AI cars, different lane colliders to detect changing lanes, and the invisible colliders to detect the highway entrance and exit ramps.

```
void OnTriggerEnter(Collider other)
    {
        if (other.CompareTag("Waypoint") && !isAtWaypoint)
        {
            isAtWaypoint = true;

            if (currentWaypointIndex < waypoints.Length)
            {
                currentWaypointIndex++;
                PositionArrow(waypoints[currentWaypointIndex]);
            }
        }
        if (other.CompareTag("Park") && hasPlayedParkingClip && !hasPlayedLastClip)
        {
            PlayFeedbackNormal(lastClip);
            hasPlayedLastClip = true;
        }
        if (other.CompareTag("Cone"))
        {
            // Check if enough time has passed since the last sound was played
            if (Time.time - lastPlayedTime >= cooldown)
            {
                PlayFeedbackNormal(coneClip);
                scoreManager.DecreaseScore("collision with cones");
                // Update the last played time to the current time
                lastPlayedTime = Time.time;
            }
        }
        if (other.CompareTag("Car"))
        {
            // Check if enough time has passed since the last sound was played
            if (Time.time - lastPlayedTime >= cooldown)
            {
                PlayFeedbackNormal(carColisionClip);
                scoreManager.DecreaseScore("collision with cars");
                lastPlayedTime = Time.time;
            }
        }
        if (other.CompareTag("ParkingSpace"))
        {
            CheckParkingSuccess();
        }
    }
    void OnTriggerExit(Collider other)
    {
        if (other.CompareTag("Waypoint"))
        {
            isAtWaypoint = false;
        }
    }
```

Listing 5.2: C# Code For Training Area Collisions.

Decision-Making Module

The Decision-Making Module is the core of the driving assistance system, responsible for processing the data collected by the Sensor Module and generating appropriate feedback and guidance for the user. The Decision-Making module is realized mainly in the C# code, there the different code logic will analyze different data collected in the Sensor module and generate the appropriate feedback and guidance for the user. The Decision-Making Module utilizes a collection of algorithms and

established rules to interpret the data gathered by the sensors. These algorithms are specifically designed to evaluate the user's driving performance and provide suitable feedback and guidance. More details about the algorithms and rules used in the Decision-Making Module will be discussed in Section 5.5.2. For this purpose, we developed an assistance system code for each one of the three maps, Training Area, Paderborn City, and Highway. The main functions used for the Decision-Making module in the assistance system codes are; CheckTrafficLight, HandleCollisions, HandleSpeedingWarnings, OnTriggerEnter, OnTriggerExit, and many other functions more about these functions can be found in the source code.

Feedback Module

The feedback module is the last module in the assistance system. After collecting the data in the sensor module and analyzing it in the decision-making module, the resulting data will be sent to the Feedback module to generate the feedback and guidance for the user, which is the most important function in our study. For this, we developed many audio sources in C# code, which will be played at the right time to give the user the right feedback. These audio sources are created in the ElevenLab AI voice generator [90], which is a very good tool to generate the right voice for the right feedback and make it very close to the real human voice, which gives users the feeling of hearing a real voice while driving. Some of the feedback texts are; "You are driving too fast, please slow down", "You are driving too slow, please speed up", "You changed the lane without indicating, please be careful next time", "You just crossed the red light", "You made an accident with another car (or pedestrian), please drive more carefully", "The car stalled cause you released the clutch so quickly while starting the car", and many other feedback texts, which help the user to drive more carefully and learn how to drive the car in the right way. The Feedback Module is designed to be clear, concise, and timely, ensuring that users can easily understand and act upon the information provided.

Score System:

As discussed in Section 4.3.1, the scoring system is one of the most crucial aspects of the driving assistance system, given that it shows the visual of driving performance and progress to the user. Our driving prototype's scoring system would be based on motivating users for advancement in driving performance and calculating the number of errors taken by the user throughout the driving session. In Unity, the scoring is done by using UI canvas and UI text elements. These elements are very helpful in designing an interactive and appealing-looking score display. The score system panel shows the current score, current speed, current gear, and the speed limit. The current score starts from 20000 points and decreases according to the severity of the error made by the user. These errors are; collision with pedestrians and other cars, crossing the red light, changing the lane without indicating, driving too fast, driving too slow, collision with cones in Training Area, and stalling the car. The car speed text reflects the current car speed while driving and show it to the user. The current gear text reflects the current gear the user is driving in, and the speed limit text detects the current speed limit where the user is currently driving.

To make the speed limit detection possible we used the Unity physical collision detection function OnTriggerEnter, which is called when the car collides with the invisible colliders, which are placed in the different streets. The speed limit is then stored in the data structure in the assistance system script and then sent to the UI canvas to be shown to the user. Figure 24 shows the score system panel used in the driving prototype. The score system uses a set of algorithms and rules to measure the performance of the user driving and base the score on predefined criteria. This starting score of 20,000 points is intended to give enough buffer time for the user to afford several mistakes before they can get to such a critical low score and therefore keep the user motivated. The magnitude of penalty values for different errors was prescribed according to the seriousness and hazardousness of every action. For example, a penalty of 500 points has been set for collision with pedestrians, while not giving a blinker receives a penalty of 50 points. The penalty structure is as follows: a cross over a red light will incur 200 points, collision with pedestrians attracts 500 points, collision with cars is 350 points, collision with cones attracts 500 points, failing to put on a blinker incurs 50 points, while speeding attracts 100 points and stalling the car attracts 300 points. These were set to represent the relative value of safe driving practices and to incentive users away from more hazardous behaviors. In Listing 5.3 we can see the C# code used to implement the logic of the score system.

Figure 24: Score Canvas In Unity Editor.

The score system uses a series of algorithms and rules to evaluate the user's driving performance and calculate the score based on predefined criteria. In Listing 5.3 we can see the C# code used to implement the score system logic.

5.5.2 System Elements

These are the main components: User Interface, Algorithms and Rules, Communication System, and Data Storage and Processing are major components that are key in the carrying out of all functions of the driving assistance system in providing users with valuable feedback and guidance. Each of them is fundamental, and very instrumental in the overall functionality and efficiency of the driving assistance system, whereby it would ensure that users receive the adequate support to improve driving skills and knowledge. We now proceed with presenting the details of each

```
    public class ScoreManager : MonoBehaviour
{
    public int score = 20000;
    public TextMeshProUGUI scoreText;
    void Update()
    {
        if (scoreText != null)
        {
            scoreText.text = "Score : " + score.ToString();
        }
    }
    public void AssignTextComponents(GameObject activeCanvas)
    {
        if (activeCanvas.activeInHierarchy)
        {
            scoreText = FindChildWithTag(activeCanvas.transform, "ScoreText").
                GetComponent<TextMeshProUGUI>();
        }
    }
    private Transform FindChildWithTag(Transform parent, string tag)
    {
        foreach (Transform child in parent)
        {
            if (child.CompareTag(tag))
            {
                return child;
            }
            Transform result = FindChildWithTag(child, tag);
            if (result != null)
            {
                return result;
            }
        }
        return null;
    }
    // Method to decrease score based on the type of fault
    public void DecreaseScore(string faultType)
    {
        switch (faultType)
        {
            case "redLight":
                score -= 200;
                break;
            case "collision with pedestrians":
                score -= 500;
                break;
            case "collision with cars":
                score -= 350;
                break;
            case "collision with cones":
                score -= 500;
                break;
            case "noBlinker":
                score -= 50;
                break;
            case "speeding":
                score -= 100;
                break;
            case "stalling":
                score -= 300;
                break;
            default:
                break;
        }
    }
}
```

Listing 5.3: C# Code For The Score System.

system element through describing its design and implementation in the prototype of the driving simulator.

User Interface (UI)

The first part of the driving assistance system is the User Interface, which should provide the user with a clear and intuitive way to interact with this system. The UI is implemented by using Unity's UI system, which, as can be understood from its name, enables developers to create interactive and eye-catching interfaces with ease. It includes, among others, buttons, text fields, and images that provide users with direct access to various features and functions provided by the driving assistance system. The UI is designed in a way to be user-friendly, easy to navigate, so that users could find the information in time and get their feedback right away. The UI also includes the scoring system, whereby the user gets a graphical view of his driving performance and progress. The scoring system encourages users to practice better driving skills for improvement; it also calculates the number of errors the user made during the driving session. The UI also includes an information panel in the left corner of the current view, which shows the current speed, the current gear, the current score, and the speed limit. Another element of the UI is the car dashboard, which is designed in Unity, especially for this study and added to the car model to let users see the car speed, the gear, the seat belt warning indicator, the current front light status, the turning signals indicator, and a background light to light the dashboard to help the user driving in the night mode. Figure 25 shows some of these UI elements used in the driving simulator prototype.

Figure 25: User Interface Elements Used In The Driving Simulator Prototype.

Algorithms and Rules

The Algorithms and Rules element of the driving assistance system is responsible for processing the data collected by the Sensor Module and generating appropriate feedback and guidance for the user. The Algorithms and Rules are designed to evaluate the user's driving performance and provide suitable feedback based on predefined criteria. These algorithms are developed mostly in C# language in Unity, and they are used to analyze the data collected from the virtual environment and the user's inputs. All these algorithms are provided in the source code in the A Section.

Communication System

The Communication System element of the driving assistance system is designed to facilitate communication between the different modules of the system. The Communication System ensures that data is transmitted accurately and efficiently between the Sensor Module, Decision-Making Module, and Feedback Module. The Communication System is also responsible for sending feedback and guidance to the user in a timely manner, ensuring that users receive the information they need to improve their driving skills. The Communication System is implemented in Unity using audio sources and text-to-speech technology from ElevenLab AI voice generator [90] and also using some of the functions in C# to manipulate and play the audio sources at the right time. Most of these audio manipulation functions can be found in the CarAudio script provided in the source code in the A Section.

Data Storage and Processing

For the data storage element, we used many data structures to store the data gathered in the sensor module and analyzed in decision-making module. The data structures used are; arrays, lists, and dictionaries. These data structures are used to store the data in a structured and organized manner, ensuring that it can be easily accessed and processed by the different modules of the driving assistance system. The Data Processing element is responsible for analyzing the data stored in the data structures and generating appropriate feedback and guidance for the user. The Data Processing element is implemented in C# language in Unity, and it is used to process the data collected from the virtual environment and the user's inputs. Figure 5.4 shows a code snippet captured from the CarControl script, which is the most important script used in the driving simulator prototype, it contains all the functions used to control the car and the driving assistance system. The snippet shows the different variables and data structures used to store the different data used to control the car and some of the driving assistance system functions.

5.6 Realistic Vehicle Functions

The CarControl script is developed in the C# language in Unity, which is used to control the 3D Virtual Car model. In this script, there is a package of functions controlling the car's engine start and gear shifting, clutch control, accelerating and

```
1     [SerializeField]private float speedDiff; // Speed difference
2     public int gear; // Gear input from the input_Manager script
3     public bool hasStartedDriving = false; // Has the car started driving
4     public bool stall = false; // Is the car stalled
5     public bool wasClutchPressed; // Was the clutch pressed
6     public float previousClutchInput; // Previous clutch input
7     public float clutchReleaseRate; // Clutch release rate
8     public int previousGear; // Previous gear
9     public const long BRAKE_FORCE_MULTIPLIER_HANDBRAKE = 100000000000000; // Brake
          force multiplier when the handbrake is applied
10    public const long BRAKE_FORCE_MULTIPLIER_NORMAL = 1000000000000; // Normal
          brake force multiplier
11    List<float> speedValues = new List<float>(); // Create a list to store the
          speed values
12    List<float> clutchValues = new List<float>(); // Create a list to store the
          clutch input values
13
14    [Space(5)]
15    // Car Settings
16    [Header("Car Settings")]
17    public float maxEngineTorque = 500f; // Maximum torque of the engine
18    public float idleRPM = 1; // Idle RPM of the engine
19    public float redlineRPM = 7000f; // Maximum RPM of the engine
20    public float brakeForce; // Brake force
21    public float downForce = 200; // Downforce to keep the car on the ground
22    public float oppositeForce = 70f;
23    public float rollOverSpeed = 10f; // value to roll over the car at the
          beginning
24    public float[] gearRatios; // Gear ratios
25    public float[] maxSpeed; // Speed limits for each gear in KPH
```

Listing 5.4: C# Code Snippet From The CarControl Script Used For Data Storage and Processing.

breaking, turning signals, lights control, force feedback, front and rear mirrors. These functions provide a more realistic and interactive driving experience to the user and enable him/her to practice how to drive an automobile in a controlled and safe environment. The CarControl script also has functions simulating car engine start, gear shifting, clutch control, accelerating and braking, turning signals, lights control, force feedback, and front and rear mirrors. These features are put together to provide users with a near-actual and interactive experience of driving, so that they learn how to drive an automobile well and in the safest possible environment. In Section A you can find the source code of the CarControl script, which contains all the functions used to control the car and the driving assistance system. Figure 26 reflects the realistic vehicle functions on the most preferred car model in our driving simulator prototype, which is the BMW E92 M3.

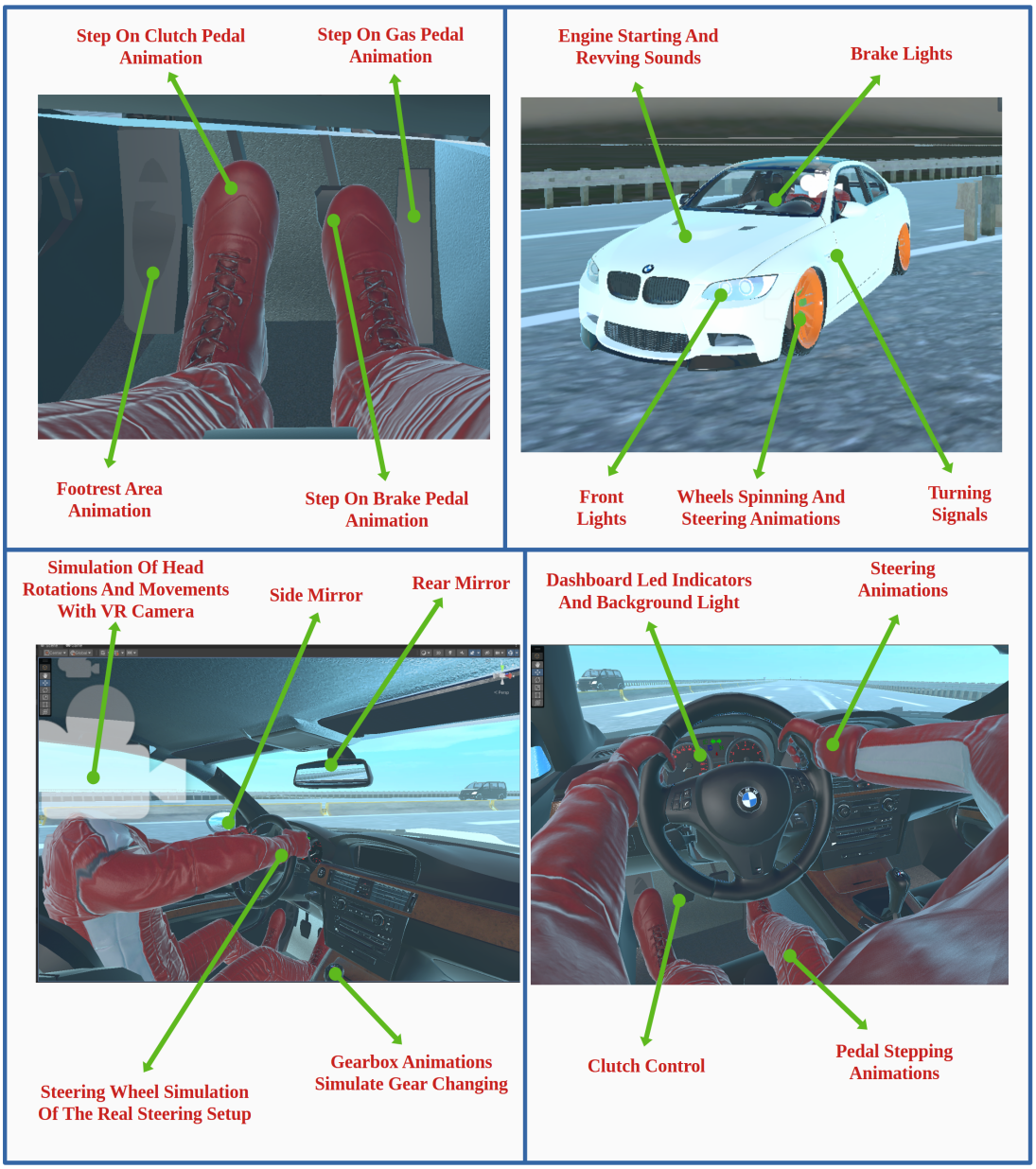


Figure 26: Different Car Functions And Animations.

6 Evaluation

This chapter focuses on the evaluation of the study made using the driving simulator prototype. Initially, Section 6.1 will provide an overview of a preliminary case study conducted prior to the user study to evaluate our collaborative application. Following this, Section 6.2 will outline the design and implementation of the user study. In Section 6.3, the findings from the study will be presented, and Section 6.4 will engage in a discussion of these findings in relation to the research questions posed in Section 1.2.

6.1 Case Study

First, we will assess our solution via a user study. We undertook a preliminary case study to examine the VR-based driving simulator. This section will provide a succinct overview of the case study's components. The objective was to test the effectiveness, efficiency, and user satisfaction, alongside other aspects of the advanced assistance system integrated into the VR-based driving simulator designed and developed for the purpose of this study. Figure 1 shows a general overview of the steps taken by users to complete the driving simulator tasks.

Users of the driving simulator prototype must complete specific tasks in order to respond to the associated questionnaire, these tasks include :

1. First, the user has to learn the controls to be able to operate the VR headset and the gaming steering wheel kit.
2. The user then starts the game in the main menu scene, where they have to choose different prototype settings, as shown in Figure 28.
    - 2.1 The user then chooses their favorite car from three 3D car models and presses select.
    - 2.2 Next, the user chooses the specific map according to the current questionnaire.
    - 2.3 Then, the user chooses the time of day, either day or night.
    - 2.4 After that, the user has to choose the view according to the current questionnaire, either VR view or non-VR view.
    - 2.5 Finally, the user chooses if they want to play with or without the assistance system.

3. After that, the game starts with the chosen settings, and for every mode and map, the user has to drive for about 15 minutes.

4. While the user is driving, the advanced assistance system will guide the user, warn them in dangerous situations, calculate the score, and teach the user how to drive the car.

5. The goal at the end is to answer the user study questionnaire and get the score resulting from each mode.

Figure 27 shows the driving simulator tasks that users have to complete in order to respond to the associated questionnaire.

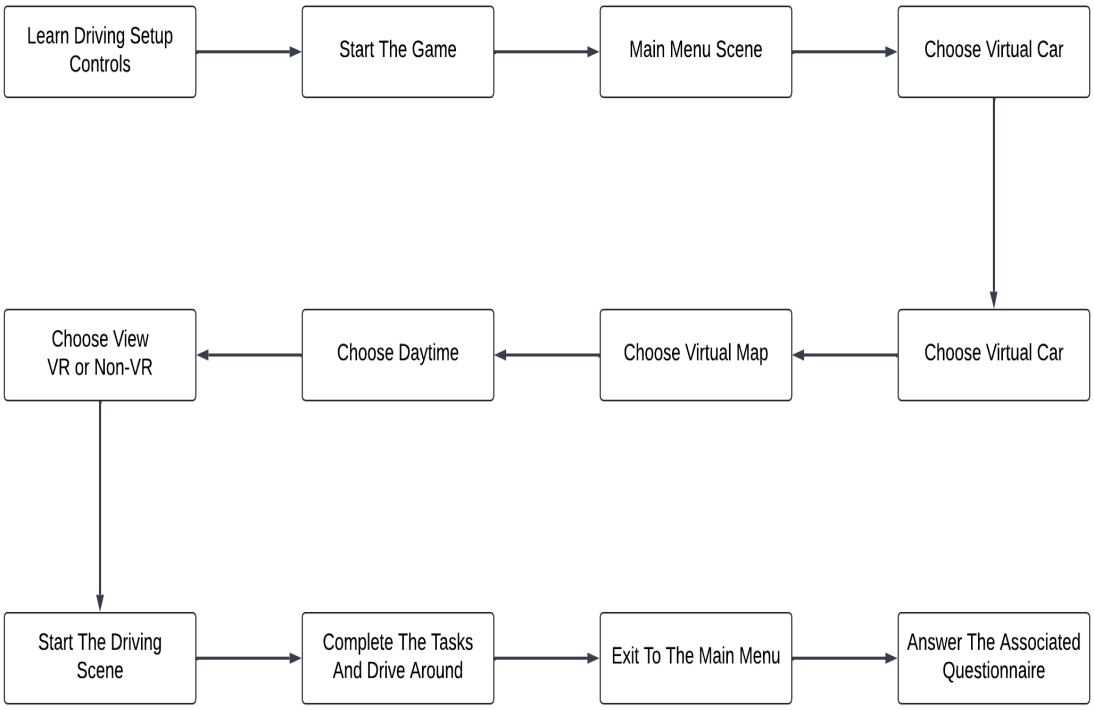


Figure 27: Driving Simulator Tasks.

6.2 Method

This section will detail the methodology employed in the design and implementation of the usability study. Subsection 6.2.1 will discuss the configuration and overarching design choices that were made. Subsection 6.2.2 will present the criteria used to assess usability evaluation within the study. Subsection 6.2.3 will provide a summary of the participants recruitment process, and finally, subsection 6.2.4 will delineate the procedural steps undertaken by each group of participants.

Figure 28: Driving Simulator Main Menu.

6.2.1 General Design

This work tend to answer the main two research questions, which has mainly been discussed in Section 1.2:

RQ1: How much can an assistance system in a VR-based driving simulator enhance driving performance and potentially replace the role of a driving instructor?

RQ2: How does the transition from non-VR to VR advanced driving assistance system affect the performance scores and the usability criteria in a simulated environment?

Considering the research questions of the study, it is crucial that the responses directly compare the VR-mode with the non-VR mode in each map of the developed driving simulator with the integrated advanced assistance system. The non-VR simulator mode used a computer monitor with a gaming steering wheel setup. The VR simulator mode used the VR headset with a gaming steering wheel setup. In conducting comparative usability assessments, two primary design paradigms are recognized: within-subjects and between-subjects designs. Within-subjects studies involve exposing each participant to both versions of a program, allowing them to complete tasks in each. In this context, participants would engage with both the VR and non-VR modes of the driving simulator across three different maps: the training area, the highway, and Paderborn city. This approach enables direct comparisons between the two modes; however, it necessitates a longer study duration, as each participant must complete tasks in both modes for all three maps.

Comparisons are made by posing identical questions to all participants and assessing whether their responses exhibit statistically significant differences based on the mode they experienced. The subject design requires a larger participant pool to achieve statistically meaningful results This design facilitates direct comparisons, enabling participants to articulate their preferences regarding specific aspects of the modes. Given that both modes are centered on driving simulation, we consistently evaluated participants who completed tasks in both modes across all three maps.

Table 6.1 shows the order in which the participants tested the prototype. As shown each participant has to choose the maps in this order: Training Area, Highway, then Paderborn City. And in each map first they have to choose the VR-mode then the non-VR mode to be able to answer all questionnaire.

Participant	**Map**	**Mode**
Participant_1	Training Area	VR
Participant_1	Training Area	non-VR
Participant_1	Highway	VR
Participant_1	Highway	non-VR
Participant_1	Paderborn City	VR
Participant_1	Paderborn City	non-VR
Participant_2	Training Area	VR
Participant_2	Training Area	non-VR
...	...	...

Table 6.1: The order in which Participants used the VR and the non-VR modes in different maps in this study and then answered the associated questions.

The driving simulator prototype under evaluation was developed as a PCVR application that can be run in a Windows 11 environment. Each participant in the study used a gaming driving setup Logitech G932 with a computer monitor and VR headset. As discussed in previous sections the participants had to choose one map and one mode VR or non-VR. In the VR mode each participant sits in front of the gaming driving setup and wears the VR headset, which by some participants the Meta Quest 3 and by others the Meta Quest 2. In non-VR mode the participants

take the VR headset off and sit in front of a computer monitor and gaming driving setup. The prototype has also instruction sounds which were used in the advanced driving system, for this purpose music box or box speakers were used. Next, we will clarify the exact steps the user in the driving prototype has to accomplish each task. First, the user will be told to choose the Training Area map, VR mode, day or night as the user prefers, his favorite car, and assistance system. Then, the user puts on the VR headset and grabs the steering wheel. After this, the user will hear voice instructions recorded to simulate the function of the assistance system, which will guide the user across the Training Area and teach the user how to drive a car, drive carefully around a path made of street cones, and park in parking lots. Next, the user goes back to the main menu and continues with the next task which is the Training Area in non-VR mode, as shown in table 6.1. In each mode, the user will get a score which will be twenty thousand in the beginning and decrease if the user makes some errors. Next, map will be the Highway in VR and then in non-VR mode, where the user has to drive on a highway that looks like the highways in Germany and has to follow the instructions from the assistance system. The last map task will be to make a free drive in Paderborn City, to test the skills learned in previous modes and tasks. After each mode, the user will answer questions about the previous mode. In the end, the user will answer general questions, the IPQ questionnaire, the SUS questionnaire, and the NASA TLX questionnaire, like it is shown in Subsection Usability Study Questionnaire in the Appendices Section.

6.2.2 Usability Measurements

This Section will discuss some usability evaluation criteria, used to evaluate the use cases specified in this work. This work will focus on some criteria, which have been briefly discussed in Section 2.7. After each mode in the application, a list of questions will be presented to the user and have to be answered. These questions are categorized according to usability criteria: effectiveness, efficiency, and finally user satisfaction. In Section Results 6.3 we will show evaluation charts ordered according to these usability criteria. To measure the effectiveness of the driving system, users have to answer some questions after each mode, like how is it easy to understand the tasks, does the assistance system improved their performance, does the assistance system gives them good feedback while driving, does the score improved when changing from VR to non-VR mode, and finally does the prototype help them to understand some German street rules. The efficiency was measured mainly by the time needed to complete some tasks, how easy was it to understand the assistance system instructions, and how easy was it to control the vehicle using the gaming driving setup. For user satisfaction, we used many questionnaire available online like the SUS questionnaire and the TLX NASA questionnaire, to evaluate the user reaction to the VR application and the whole driving tasks. Other general questions were presented to know which car model, mode, map, or transmission, the user preferred in his driving experience. In Figure 29 we provide a summary of the study tasks, the data collected during the user study, along with the measurements that were extracted from it, and finally which usability evaluation variables were measured in each task. As shown in Subsection 6.2.1 the user study is designed

so that the user will first drive in VR-mode then in non-VR mode in all three maps one by one and after each mode, the user answers the specific questionnaire. This allows us to measure the score of each scenario and compare VR and non-VR driving simulators in the same study which will save us a lot of time. The variables that will be measured after each mode separates are effectiveness and efficiency. User satisfaction and custom variables will be measured when all modes finish, cause it depends on the facts, the user learned during the whole experience. In the study, the participants have also to answer demographic questions like age, educational level, VR experience, driving simulator experience, driving frequency, and many others. We were also interested in some general questions, which may help other researchers in other driving studies. These questions include questions about the user's favorite car model, map, transmission mode, daytime, and VR or non-VR mode. In addition to these questions, users were asked to answer questions about graphics, assistance system scenarios, assistance system functions, and User interface elements. The full list of questions included in the final questionnaire is available in Appendix B, and a detailed analysis of each question will be conducted in Section 6.3 during the evaluation of the results.

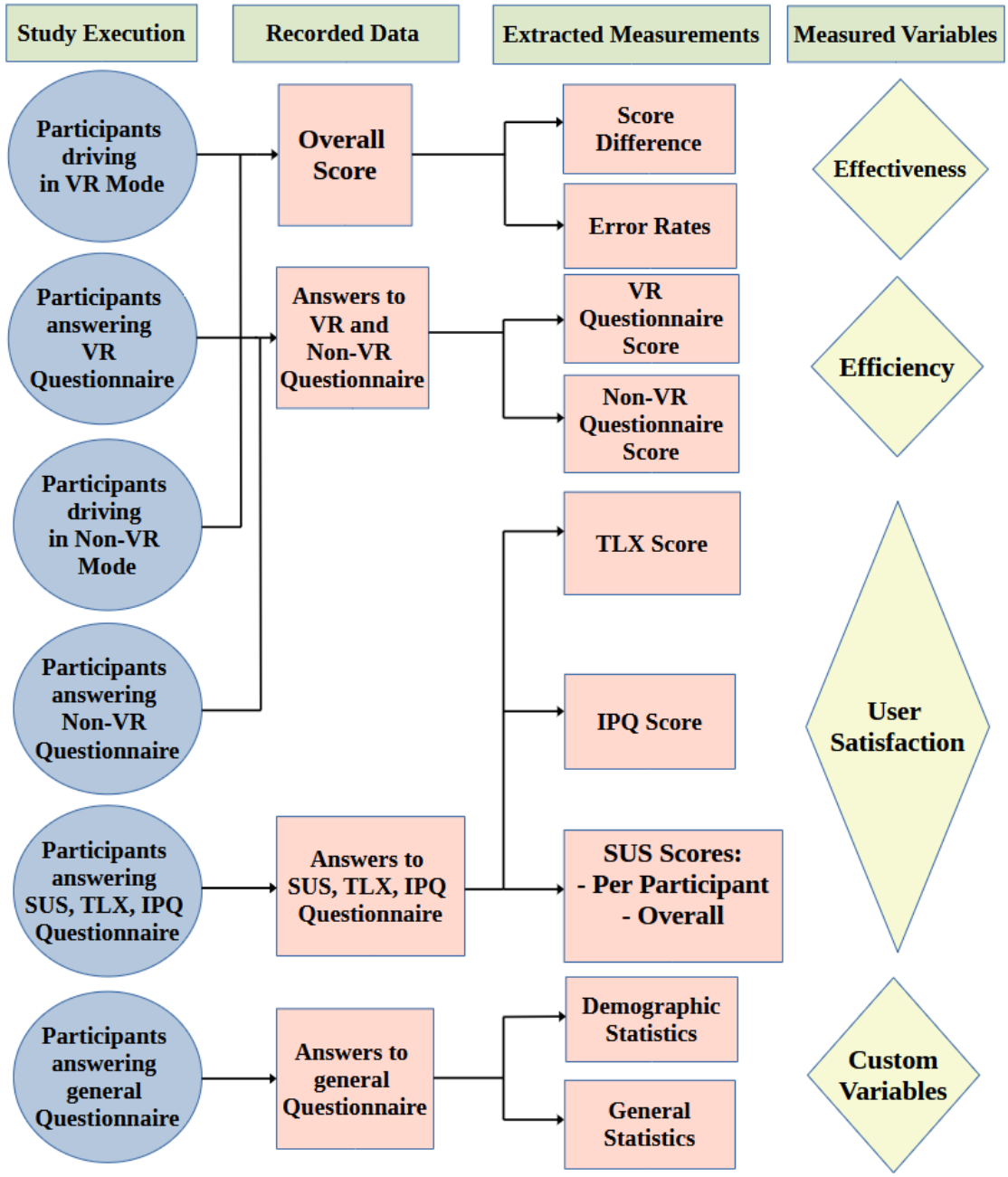


Figure 29: An overview of what was measured within the user study.

6.2.3 Participant Acquisition

The driving simulator is developed in a way that includes three maps with two different view modes for each map. The Training Area map includes a driving course to teach users without any driving experience how to drive a car and also to teach normal car drivers how to use the gaming driving setup and avoid committing deadly accidents on real roads. Therefore, participants do not have to have previous driving or VR experiences to be able to take the user study, which makes

it easier to acquire participants from different places with different ages. Our user study was mainly done in Stuttgart City, Germany. This choice made it very interesting, cause some participants had not visited Paderborn City before and were able to drive the virtual car in the application on the Paderborn City map, which gave them some information about the city. In total, 12 participants took part in the final study, they answered at the beginning many demographic questions about their personal information, the results are discussed in detail in Section 6.3.1. demographic questions included name, age, gender, educational level, VR experience, driving experience, driving frequency, and motion sickness likelihood.

6.2.4 Execution Procedure

The user study was ultimately carried out over the course of one week in August 2024 in Stuttgart City, Germany. The implementation process was designed to maintain uniformity across all participating groups. As previously discussed in Section 6.2.3 any participant without any proper driving experience was able to take the user study. This allowed us to accept participants of all ages. Initially, participants were welcomed and provided with an overview of the study's primary tasks and modes. Subsequently, the fundamental procedures for their involvement and the activities they would engage in during the study were outlined. Following this, the main controls of the VR headset and the gaming driving setup were presented to participants as an image, which they can always call and learn how to control the setup used in the study. Then, participants were told to choose their favorite car model to drive in. The mode and the map, which participants will drive the car, are chosen according to the current questionnaire and are received from the study supervisor. As discussed in Section 6.2 the first mode was the VR mode in the Training Area map. The purpose of this map is to teach novice drivers how to drive a car if they have not done it before, complete cone training sessions, and finally learn how to park in difficult parking lots. Next, participants test this also in non-VR mode, then move to the next map which is Highway in Germany. There participants will learn how to drive on a highway in Germany and which rules they have to pay attention to. Finally, participants can test what they learned during previous sessions in Paderborn City, Germany. After each mode and session participants have to answer the questionnaire made for this session and in the end, they will answer general questions about the assistance and driving systems.

6.3 Results

This Section will provide the results collected in different questionnaires across the user study. These results are collected from the 12 participants and presented in different charts. We performed in some cases statistical tests to evaluate the enhancement made from non-VR to VR mode. The demographic characteristics of participants, as derived from their responses in the initial segment of the questionnaire, will be detailed in Section 6.3.1. Following this, Sections 6.3.2 and 6.3.3 will address the assessed efficiency and effectiveness, respectively. In Section 6.3.4, the findings from the System Usability Scale (SUS) questionnaire will be presented.

Subsequently, Section 6.3.5 will focus on the responses to our general questionnaire, specifically designed to address additional issues in driving simulators. The open-ended questions included in this questionnaire, aimed at eliciting more nuanced feedback, will be discussed separately in Section 6.3.6.

6.3.1 Demographics

A total of 12 individuals, took part in the study. Following their involvement, they completed a demographic questionnaire, which will be elaborated upon in the subsequent sections.

Age:

Figure 30 shows the age distribution among all participants. Participants were between 17 and 65 with a median of 27 as shown in the box-plot chart. We can see that the age of the participants in this study was distributed over all generations, which gives a better overview of the effect of the assistance system on different age groups. Most of them were between the ages of 25 and 30, which can be understood cause youth will be more interested in driving simulator and testing driving gaming setups.

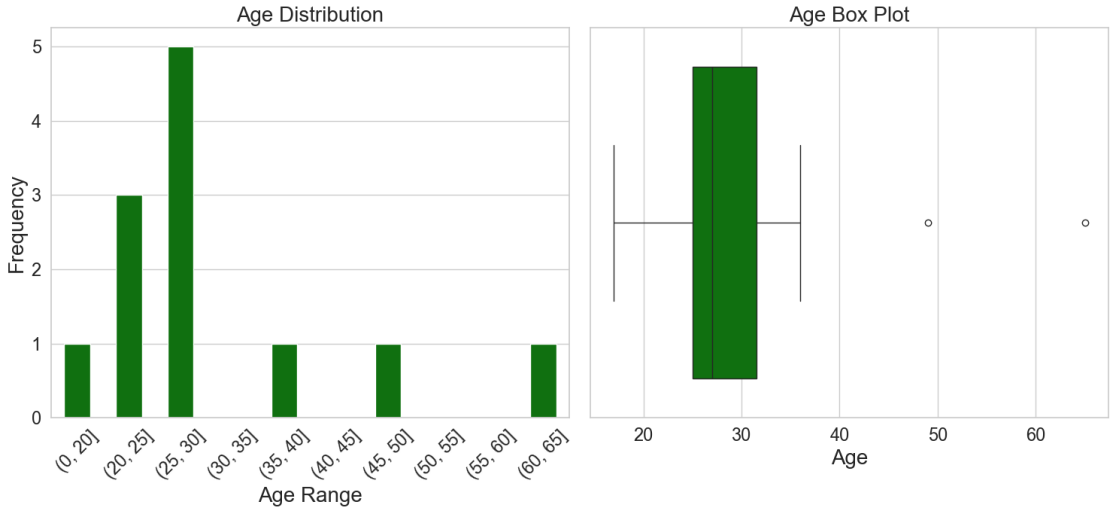


Figure 30: Age distributions among the participants.

Gender:

The 12 participants included 9 males and 3 females as shown in Figure 31. That means one quarter of all participants were females, which is considered acceptable. It is important to acknowledge that the options provided for gender were not exhaustive; we included a text field for respondents to specify the gender they want.

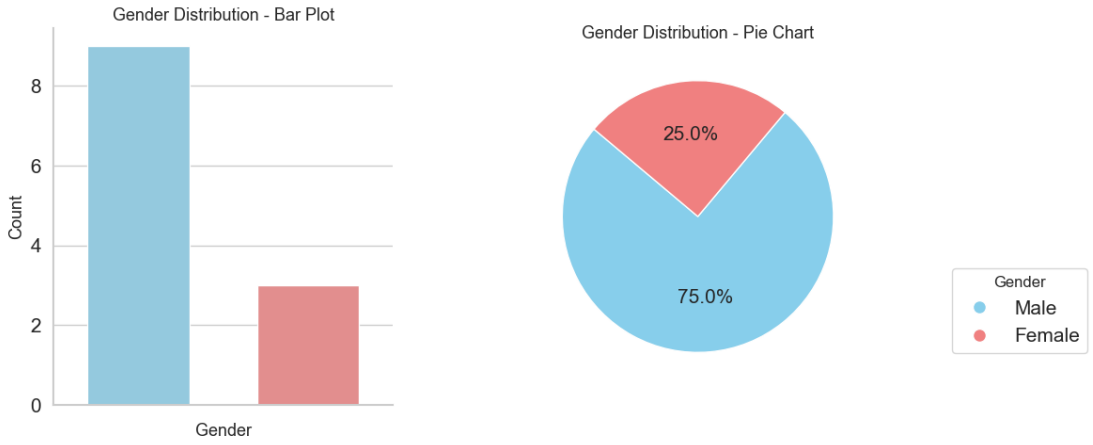


Figure 31: Gender distributions among the participants.

Educational Level:

Figure 32 illustrates the educational level of the participants through both a bar plot and a pie chart. The study encompasses individuals with a range of educational backgrounds, including high school graduates, bachelor's degree holders, master's degree recipients, trainees, retirees, employees undergoing training, and those who are self-employed. This diversity enables us to evaluate the driving simulator across a spectrum of societal members and analyze their responses to it.

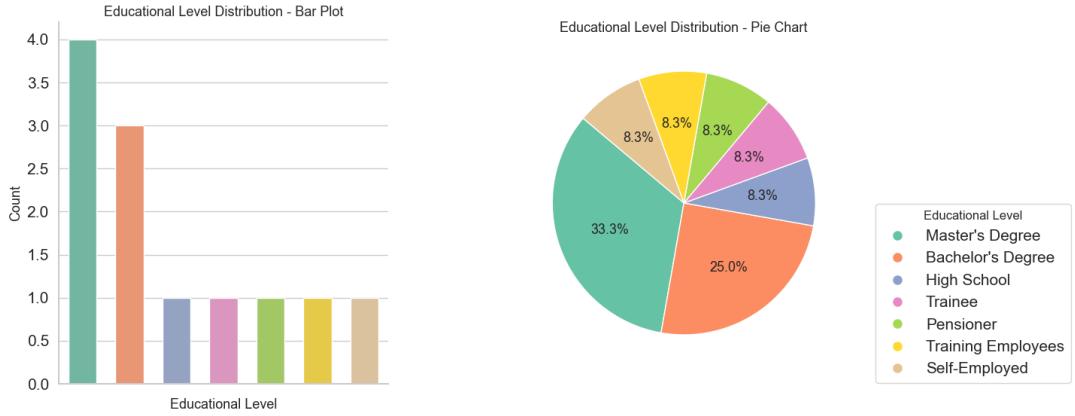


Figure 32: Participant's educational level.

Driving Experience:

Given that this research aims to evaluate the usability of a driving simulator in both virtual reality (VR) and non-VR modes, it was essential to inquire about the participants' familiarity with VR devices, their experience with driving simulators, their driving history and frequency, as well as the intensity of any motion sickness

they may experience in a VR setting. Figure 33 illustrates the driving experiences of the participants, categorizing their frequency of driving a real vehicle as daily, weekly, monthly, or rarely. Notably, one participant, aged 17, reported having no prior driving experience. Additionally, the data indicates that the majority of participants possess one to three years of driving experience, which is advantageous for the efficiency of the study.

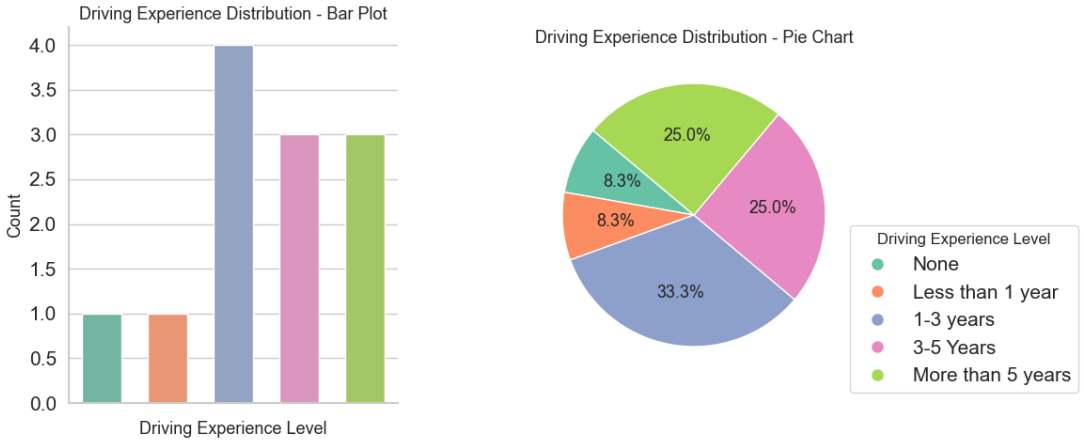


Figure 33: Participant's Driving Experience.

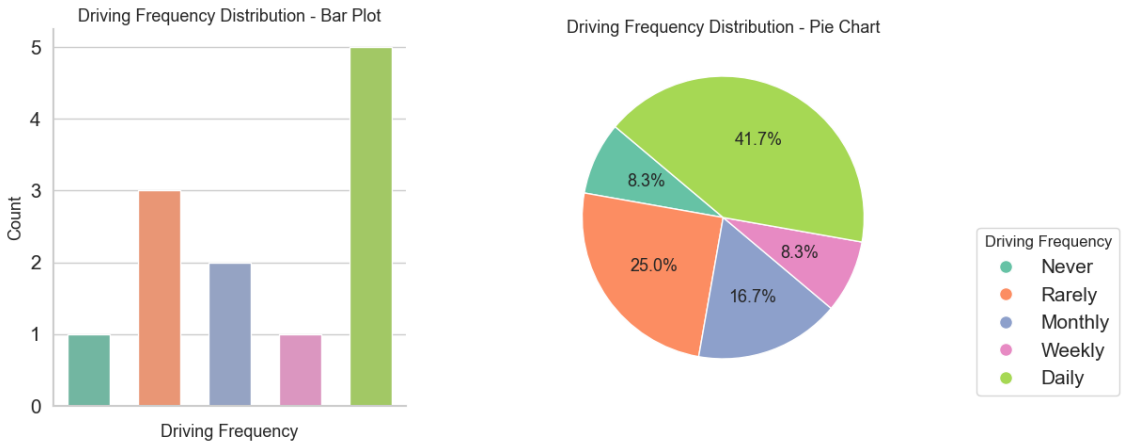


Figure 34: Participant's Driving Frequency.

VR and Driving Simulator Experience:

Figure 6.3.1 presents the participants' experiences with virtual reality (VR) and driving simulators. The findings reveal that while most participants lacked familiarity with driving simulators, a significant number had prior experience in a VR environment. Otherwise, the lack of experience with driving simulators, gave us the ability to evaluate how fast participants can learn how to use driving setup.

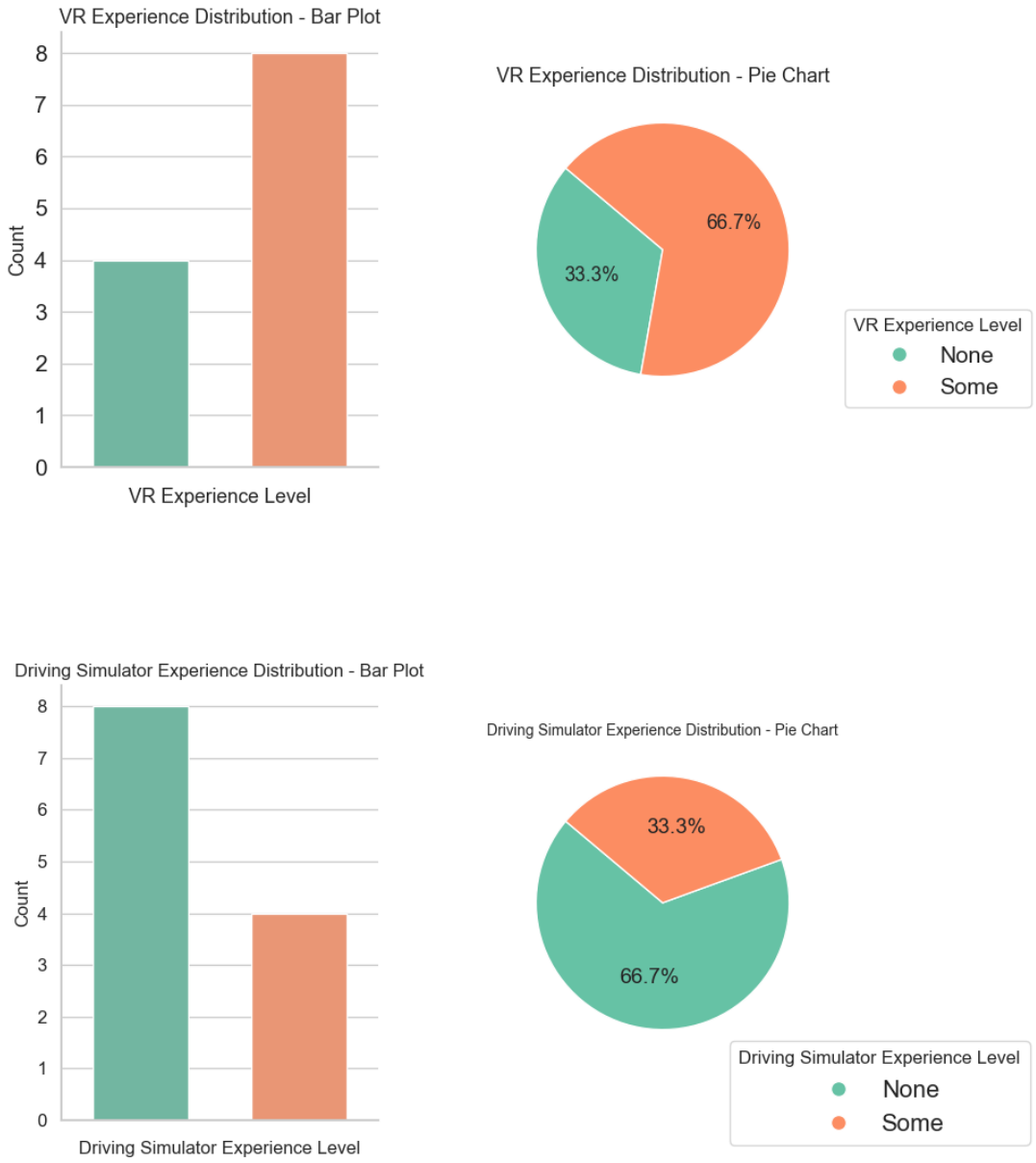


Figure 35: Participant's VR and Driving Simulator Experience.

6.3.2 Efficiency

As outlined in the general study design subsection 6.2.1, the efficiency of the usability evaluation was assessed for each participant following their experience in both modes across each map, as illustrated in table 6.1. Participants were required to complete an efficiency questionnaire after engaging with either the VR or non-VR mode. The questionnaire included inquiries regarding the ease of controlling the virtual 3D vehicle, the speed at which the assistance system was comprehended, and whether the allotted time was sufficient to complete all tasks. Consequently, we evaluated the efficiency for each map in both VR and non-VR modes, revealing

varied results across different maps. Participants rated their responses on a scale from 1 to 5, with 1 indicating poor performance and 5 indicating excellent performance. Figure 36 presents a comparison of VR and non-VR efficiency within the Training Area map. Figure 37 illustrates the efficiency comparison in the Highway map, while Figure 38 depicts the comparison in the Paderborn City map. Analyzing the responses to the first question across all maps, "Easy to control vehicle," it is evident that participants found it easier to control the virtual vehicle in VR mode compared to non-VR mode. This can be attributed to the 360-degree view provided by the VR camera, which enhances situational awareness. Conversely, it was observed that the assistance system was more comprehensible in non-VR mode across all maps, a finding that was unexpected. This discrepancy may be explained by the clarity of the audio output from the desktop non-VR mode speakers, which appeared to be more intelligible than that of the Meta headset. Additionally, participants reported having sufficient time to complete driving tasks in VR mode compared to non-VR mode. The box plots displayed on the right side of each figure indicate that the overall efficiency score was higher in VR mode than in non-VR mode (5 versus 4). This finding contributes to addressing the second research question, RQ2, which suggests that transitioning to VR mode enhances the efficiency of the usability evaluation. The data previously presented may exhibit a normal distribution around its mean, although this is not a requirement. To conduct a significance test, it is essential to ascertain whether the data follows a normal distribution, as this determines the appropriate statistical test to employ: the T-test for normally distributed interval data or the Wilcoxon Signed-Rank test for data that does not conform to a normal distribution [91]. The Shapiro-Wilk test serves as a robust method for assessing whether sample data adheres to a normal distribution [91]. The results of this test are as follows:

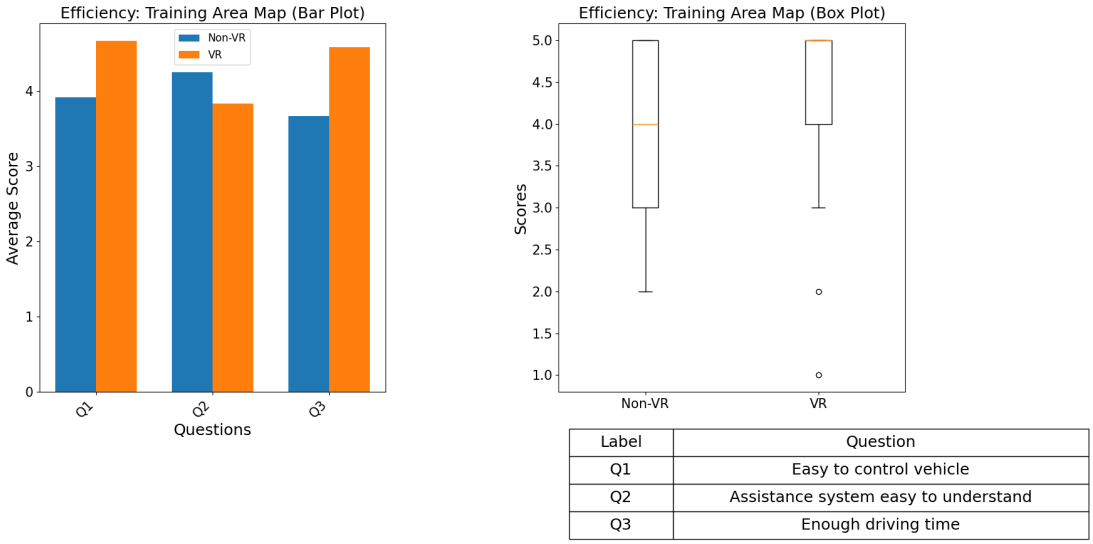
For the **Training Area Map:**
non-VR: p=0.000206
VR: p=2.285e-07
At a significance level of $\alpha = 0.05$, both non-VR and VR data are deemed not to follow a normal distribution.

For the **Highway Map:**
non-VR: p=3.439e-06
VR: p=7.929e-08
Similarly, at a significance level of $\alpha = 0.05$, both non-VR and VR data do not conform to a normal distribution.

For the **Paderborn City:**
non-VR: p=4.676e-05
VR: p=4.898e-08
Again, at a significance level of $\alpha = 0.05$, both non-VR and VR data are not considered normally distributed.

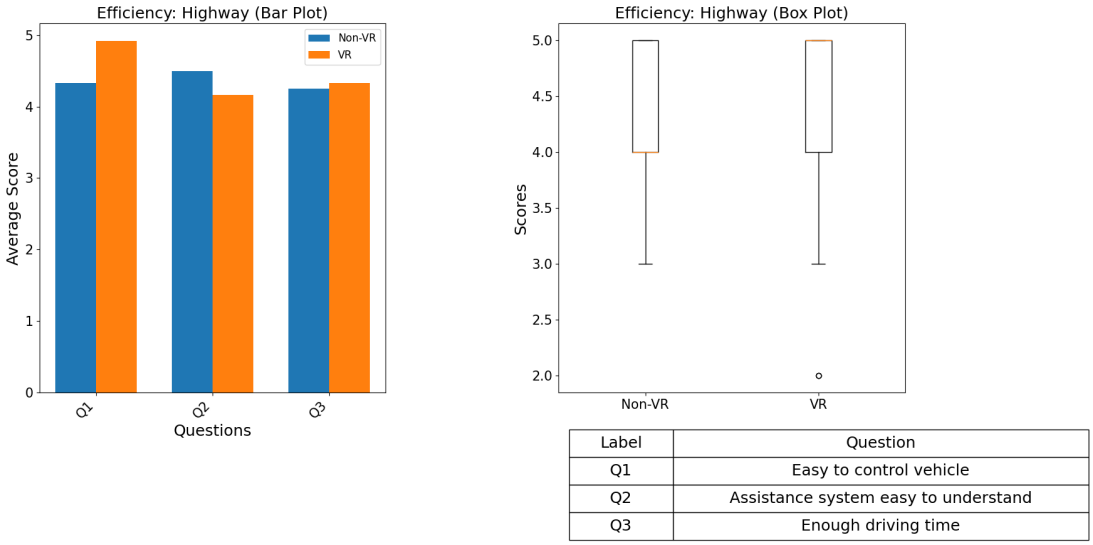
Consequently, the Wilcoxon Signed-Rank test was employed, as it does not pre-

suppose a normal distribution, and our findings indicate that neither sample can be confirmed as normally distributed. The results from the Wilcoxon Signed-Rank test are as follows: For the Training Area Map: Z=-4.517, p=0.017943, indicating statistically significant differences with $p < 0.05$. For the Highway: Z=-4.305, p=0.387371, indicating no statistically significant differences with $p >= 0.05$. For Paderborn City: Z=-4.344, p=0.000435, indicating statistically significant differences with $p < 0.05$. These findings suggest that for the Training Area Map and Paderborn City, the responses in the VR condition were statistically significantly different from those in the non-VR condition, whereas for the Highway, no statistically significant differences were observed.



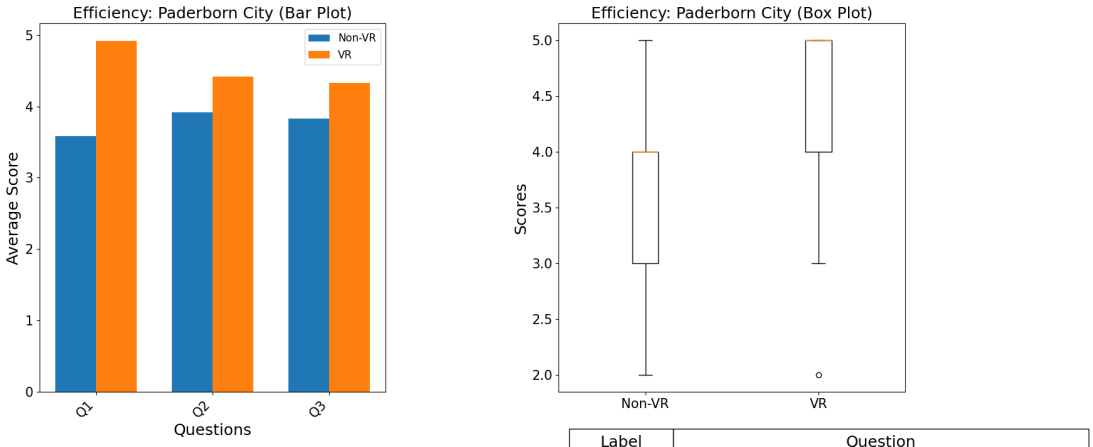
Label	Question
Q1	Easy to control vehicle
Q2	Assistance system easy to understand
Q3	Enough driving time

Figure 36: Efficiency questions comparison in Training Area map between VR (blue) and non-VR (orange).



Label	Question
Q1	Easy to control vehicle
Q2	Assistance system easy to understand
Q3	Enough driving time

Figure 37: Efficiency questions comparison in Highway map between VR (blue) and non-VR (orange).



Label	Question
Q1	Easy to control vehicle
Q2	Assistance system easy to understand
Q3	Enough driving time

Figure 38: Efficiency questions comparison in Paderborn City map between VR (blue) and non-VR (orange).

The efficiency in training provides insights into how well participants were able to complete the tasks within the given time. Higher efficiency means that participants completed the tasks faster and possibly with fewer errors. Measuring efficiency is important to understand how fast the learning process is. Especially when using VR and non-VR modes, it is crucial to know which mode helps participants complete tasks more efficiently. This can provide indications of which mode is better suited for training purposes. Several conclusions can be drawn from the efficiency

measurements: Participants found it easier to control the virtual vehicle in VR mode, which can be attributed to the improved situational awareness provided by the 360-degree view. The assistance system was more comprehensible in non-VR mode, possibly due to the clearer audio output. Participants had sufficient time to complete driving tasks in VR mode, indicating higher efficiency. Overall, efficiency was higher in VR mode than in non-VR mode, suggesting that the VR mode improves the user-friendliness and effectiveness of the training. These findings contribute to answering the second research question (RQ2) by showing that transitioning to VR mode enhances the efficiency of the usability evaluation.

6.3.3 Effectiveness

The effectiveness of our work was measured with the help of effectiveness questionnaires after each mode.
Similar to the data gathered for efficiency, we will analyze the data gathered and observed in Figures 39 40 41. We noticed that the effectiveness overall score in VR and non-VR in the Training Area was nearly the same (reached the average of 5). In the Highway map this was not the case, cause the VR mode score was slightly higher than the non-VR mode. In Paderborn City both VR and non-VR modes were equal (in average 4). The Shapiro-Wilk test showed the following results:

For the **Training Area Map:**
non-VR: p=6.944e-10
VR: p=3.231e-11
Both non-VR and VR data are not considered normally distributed.

For the **Highway Map:**
non-VR: p=3.127e-10
VR: p=3.366e-13
Both non-VR and VR data are not considered normally distributed.

For the **Paderborn City:**
non-VR: p=1.779e-07
VR: p=5.680e-09
Both non-VR and VR data are not considered normally distributed.

Consequently, the Wilcoxon Signed-Rank test was employed, as it does not presuppose a normal distribution. The results from the Wilcoxon Signed-Rank test are as follows: For the Training Area Map: Z=-6.372, p=0.052177, indicating no statistically significant differences with $p >= 0.05$. For the Highway: Z=-7.037, p=0.001342, indicating statistically significant differences with $p < 0.05$. For Paderborn City: Z=-6.212, p=0.096340, indicating no statistically significant differences with $p >= 0.05$. These findings suggest that for the Highway, the responses in the VR condition were statistically significantly different from those in the non-VR condition, whereas for the Training Area Map and Paderborn City, no statistically significant differences were observed.

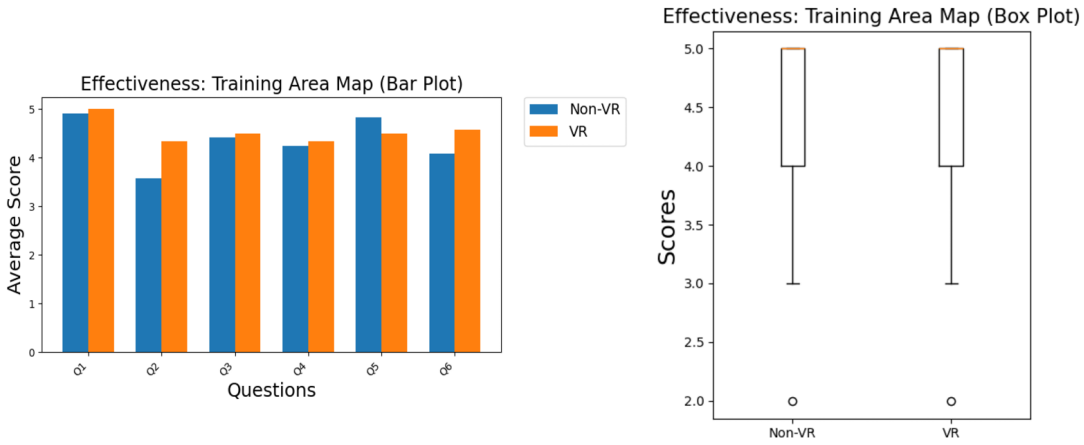


Figure 39: Effectiveness questions comparison in Training Area map between VR (blue) and non-VR (orange).

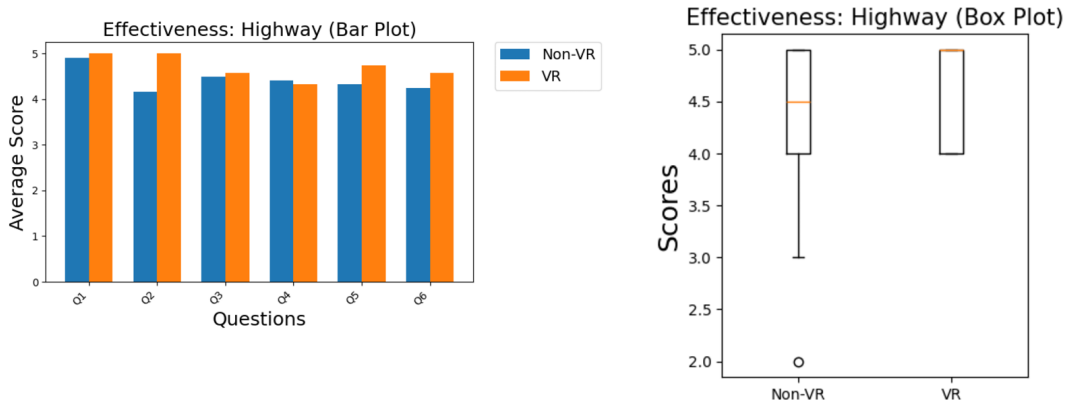


Figure 40: Effectiveness questions comparison in Highway map between VR (blue) and non-VR (orange).

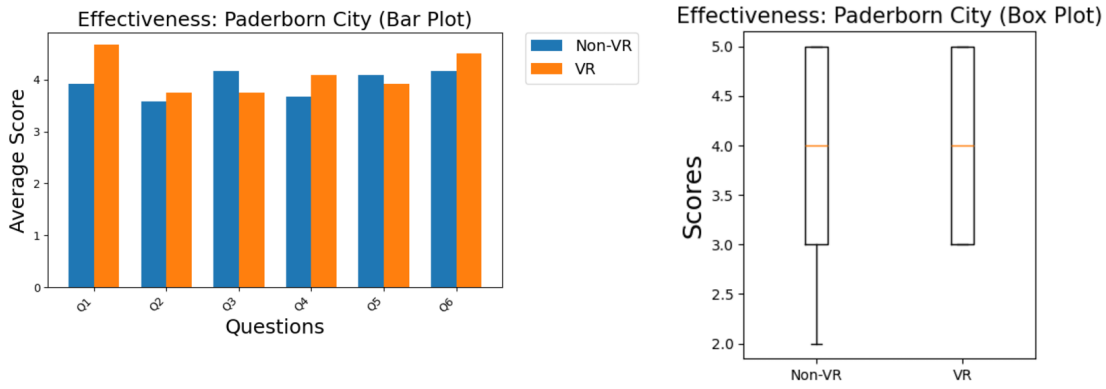


Figure 41: Effectiveness questions comparison in Paderborn City map between VR (blue) and non-VR (orange).

Label	Question
Q1	I found the tasks easy to understand
Q2	The system improved my driving
Q3	The system gave good feedback
Q4	The simulator helped with vehicle controls
Q5	I understand German street rules better
Q6	The system helped me drive more carefully

Table 6.2: Effectiveness Questions.

Additionally, the overall score calculated and captured during every session helped with the evaluation of the effectiveness, efficiency, and user satisfaction of our study. As shown in Figure 42 the overall score was clearly higher in VR mode than in non-VR mode in every map. This indicates that using the overall score system to calculate the error rate and the effectiveness of the assistance system was a very good idea, as we would not have been able to identify a clear significant difference between VR and non-VR effectiveness without this additional parameter, which is the overall score for each driving session.

Effectiveness: The Shapiro-Wilk test showed the following results:

For the **Training Area map:**
The results of the normality test revealed that the non-VR data did not follow a

normal distribution, in contrast to the VR data, which was normally distributed. As a result, the Wilcoxon signed-rank test was utilized to evaluate the presence of statistically significant differences between the scores of the non-VR and VR conditions. The analysis produced a p-value of 0.000488, indicating significant differences ($p < 0.05$). The Z-value obtained from the Wilcoxon test was -3.059. The mean score for the non-VR condition was 12,787.5, while the mean score for the VR condition was 16,791.67. These findings imply that the VR condition facilitated superior performance scores for the Training Area Map.

For the **Highway map:**
The normality test results indicated that the non-VR data exhibited a normal distribution, whereas the VR data did not. Consequently, the Wilcoxon signed-rank test was employed to analyze the differences between the non-VR and VR scores. The test yielded a p-value of 0.000488, signifying statistically significant differences ($p < 0.05$). The Z-value for the Wilcoxon test was -3.059. The average score for the non-VR condition was 14,441.67, compared to an average score of 17,708.33 for the VR condition. These results suggest that the VR condition resulted in enhanced performance scores for the Highway Map.

For the **Paderborn City map:**
The normality test results indicated that both the non-VR and VR data were normally distributed. Consequently, a paired t-test was performed to compare the scores between the non-VR and VR conditions. The test produced a p-value of 7.63e-06, indicating statistically significant differences ($p < 0.05$). The average score for the non-VR condition was 12,579.17, while the average score for the VR condition was 17,100. These results suggest that the VR condition led to improved performance scores for the Paderborn City Map.

Efficiency: The higher scores in VR mode across all maps suggest that users were able to complete tasks more efficiently in VR mode compared to non-VR mode. This is evidenced by the consistently higher performance scores, which indicate that users made fewer errors and completed tasks more quickly in VR mode. The statistical analyses confirmed that these differences were significant.

User Satisfaction: The significant differences in scores between VR and non-VR modes also imply better usability in VR mode. The higher scores in VR mode suggest that users found the VR interface more intuitive and easier to use, leading to better performance. The statistical analyses confirmed that these differences were significant.

Across all maps analyzed, the VR condition consistently yielded superior performance scores in comparison to the non-VR condition. The statistical analyses confirmed that these differences were significant, indicating that the VR condition not only improved effectiveness but also enhanced efficiency and usability.

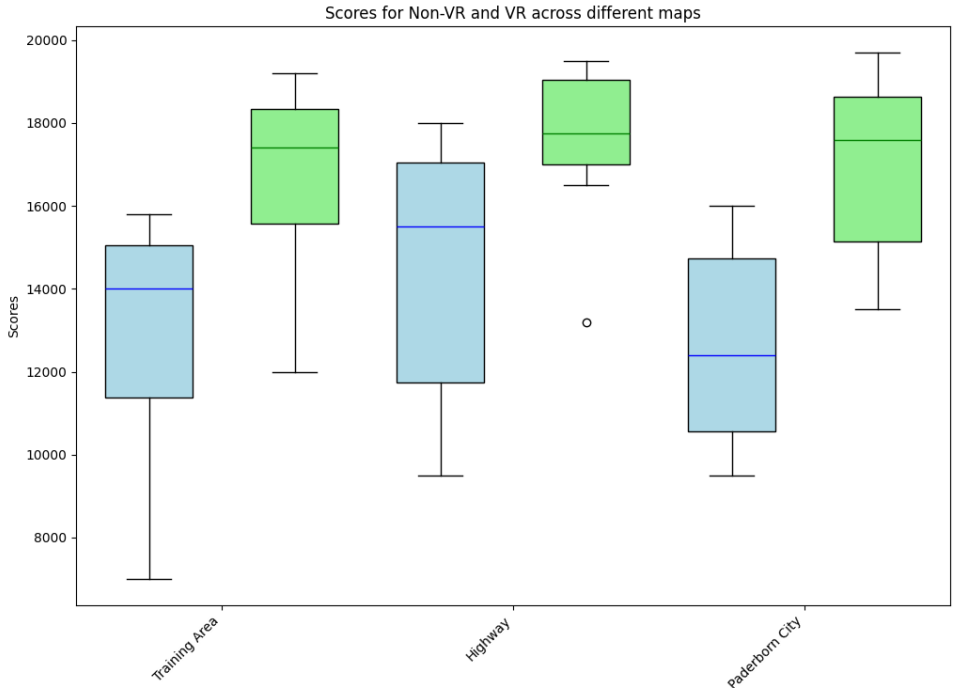


Figure 42: Overall Score distributions for VR (green) and non-VR (skyblue) across different maps.

6.3.4 User Satisfaction

User satisfaction with the driving simulator application was assessed through the administration of a 10-question System Usability Scale (SUS) questionnaire for each of the 12 participants. The results, including individual SUS scores for each participant and the overall average score, are presented in Figure 43. Participants provide their responses on a scale of 1 to 5 for each question. For questions with odd numbers, a higher score is indicative of a more favorable response, while for even-numbered questions, a lower score is preferable. To facilitate score calculation, these responses are converted to a 0-4 scale, where higher values consistently represent better outcomes. The results on this 0-4 scale are presented in the individual cells of Figure 43. Each participant is identified by their ID, from 1 to 12. Bangor et al. [92], have sought to enhance the System Usability Scale (SUS) by incorporating an adjective rating scale. This scale categorizes scores as follows: a score around 50 is designated as "OK," indicating a mediocre experience; "Good" corresponds to a score of 71; "Excellent" is assigned to a score of 85; and the highest rating, "Best Imaginable," is given to a perfect score of 100. Furthermore, they discovered a correlation between traditional letter grades and their adjective scale, where an "A" represents scores from 90 to 100, a "B" ranges from 80 to 89, a "C" spans 70 to 79, and so forth. According to this framework, our driving simulator application usability score would be classified as a "C" in letter grading and "OK" in the adjective scale. In conclusion, the VR prototype exhibits certain usability issues; however, it is largely

functional. The responses to the open-ended questions in the questionnaire will provide a more detailed understanding of these usability challenges and potential solutions (refer to Section 6.3.6).

Label	Question
Q1	I think that I would like to use this system frequently
Q2	I found the system unnecessarily complex
Q3	I thought the system was easy to use
Q4	I think that I would need the support of a technical person to be able to use this system
Q5	I found the various functions in this system were well integrated
Q6	I thought there was too much inconsistency in this system
Q7	I would imagine that most people would learn to use this system very quickly
Q8	I found the system very cumbersome to use
Q9	I felt very confident using the system
Q10	I needed to learn a lot of things before I could get going with this system

Table 6.3: SUS Questions.

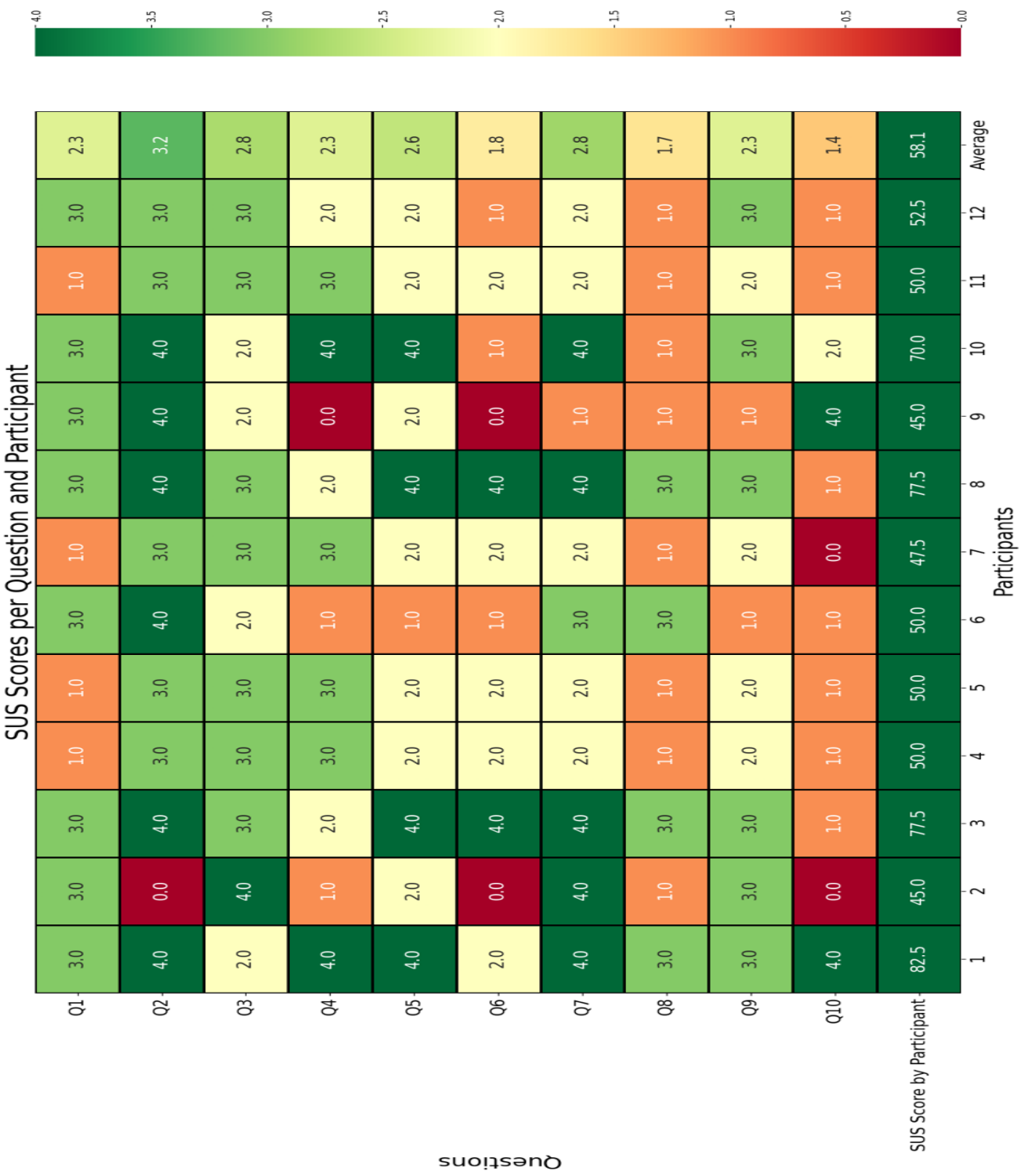
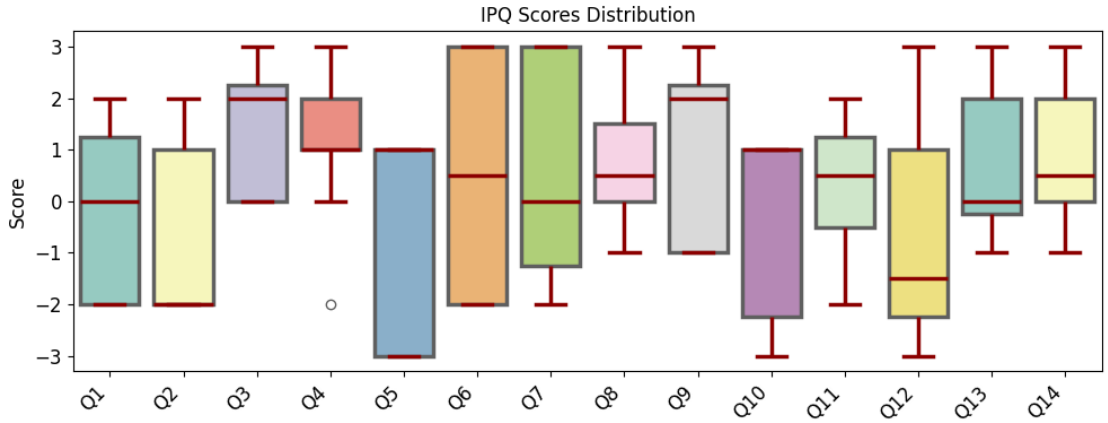


Figure 43: The overall SUS score of the driving simulator application and the scores per participant. Individual ratings for questions are on a 0-4 scale where higher is better.

6.3.5 User Immersion and Presence in Virtual Environments

The IPQ is a self-administered questionnaire that was first created by Schubert et al. in 2001. The initial iteration of this instrument was designed to assess several variables, including spatial presence, involvement, realness, as well as global presence [93]. In virtual reality The IPQ (Igroup Presence Questionnaire) is a tool used to measure the sense of presence experienced by users in virtual environments. It assesses different dimensions of presence: spatial presence, involvement, and experienced realism. This questionnaire is widely applied within research to evaluate the efficiency of Virtual Reality systems, as well as to understand how different factors influence the user's feeling of presence. For each question, there is usually a corresponding scoring criterion indicating whether the score is better if it is higher or lower, which is captured by the good bad mapping dictionary, where each question is mapped either to "high" or "low". Two types of charts were constructed to visualize the data: bar chart and box plot. The following is a bar chart displaying the average scores of each question: the color representation for each bar is followed by good bad mapping: Red: Indicates that a lower score is better Blue: Indicates that a higher score is better. This coloring helps in an immediate grasp of which questions are better with higher scores and which are better with lower scores. This box plot shows the distribution of scores for each of these questions. The coloring scheme in the box plot keeps the palette consistent, and focuses on the median and quarterlies of the data. The edges of the boxes are done in black color, while the lines within boxes are done in dark red for better readability.



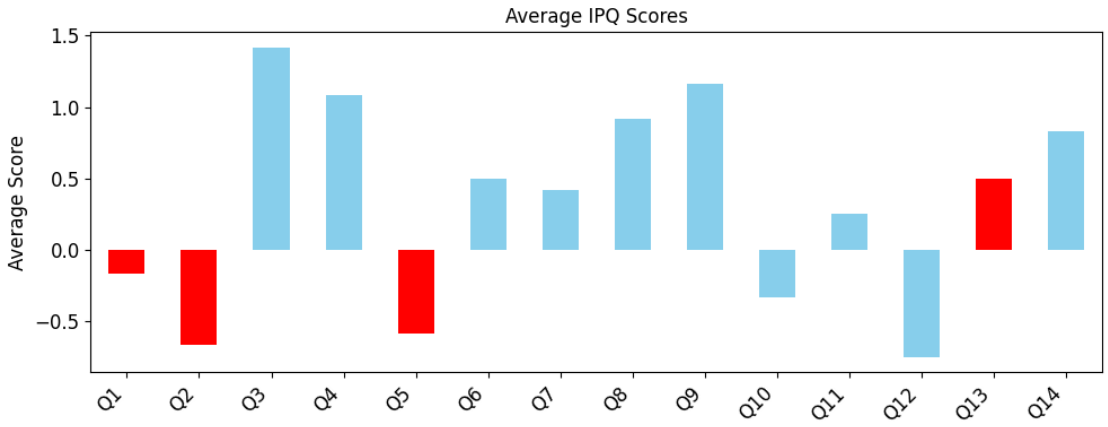


Figure 44: IPQ Score of the Driving Simulator Application.

Label	Question
Q1	How aware were you of the real world surrounding while navigating in the virtual world?
Q2	How real did the virtual world seem to you?
Q3	I had a sense of acting in the vir- tual space, rather than operating something from outside.
Q4	How much did your experience in the virtual environment seem consistent with your real world experience ?
Q5	How real did the virtual world seem to you?
Q6	I did not feel present in the vir- tual space.
Q7	I was not aware of my real envi- ronment.
Q8	In the computer generated world, I had a sense of "being there."
Q9	Somehow, I felt that the virtual world surrounded me.
Q10	I felt present in the virtual space.
Q11	I still paid attention to the real environment.
Q12	The virtual world seemed more realistic than the real world.
Q13	I felt like I was just perceiving pictures.
Q14	I was completely captivated by the virtual world.

Table 6.4: IPQ Questions.

6.3.6 User Workload in Virtual Environments

The NASA Task Load Index (NASA-TLX) is a widely used tool for assessing perceived workload. It measures six dimensions: Mental Demand, Physical Demand, Temporal Demand, Performance, Effort, and Frustration. The goal is to understand the workload experienced by individuals performing tasks.

The scores range from 1 (very low) to 20 (very high).
The bar chart in Figure 45, presents the average scores for each dimension of the NASA Task Load Index (NASA-TLX).
Mental Demand: The average score for Mental Demand is approximately 9.92, indicating a moderate level of mental effort required by the tasks.
Physical Demand: The average score for Physical Demand is 13.00, suggesting a relatively high level of physical effort.
Temporal Demand: The average score for Temporal Demand is 9.08, reflecting a moderate level of time pressure experienced by the participants.
Performance: The average score for Performance is 9.17, indicating that participants felt their performance was moderately satisfactory.
Effort: The average score for Effort is 12.50, suggesting that participants exerted a relatively high level of effort to accomplish the tasks.
Frustration: The average score for Frustration is 2.08, indicating a very low level of frustration experienced by the participants.

The box plot in Figure 45, illustrates the distribution of scores for each NASA-TLX dimension. The box plot provides a more detailed view of the data, including the median, inadequately range (IQR), and potential outliers.
Mental Demand: The median score is 10.0, which is very close to the average score, indicating a symmetrical distribution of scores around the median.
Physical Demand: The median score is 13.0, identical to the average score, suggesting a symmetrical distribution with no significant outliers.
Temporal Demand: The median score is 9.5, slightly higher than the average score, indicating a slight skew towards higher values.
Performance: The median score is 9.0, very close to the average score, suggesting a symmetrical distribution.
Effort: The median score is 13.0, slightly higher than the average score, indicating a slight skew towards higher values.
Frustration: The median score is 2.0, very close to the average score, indicating a symmetrical distribution with very low levels of frustration.

The analysis of the NASA-TLX scores reveals that Physical Demand and Effort are the most significant contributors to the overall workload, with moderate levels of Mental and Temporal Demand. The low Frustration scores suggest that participants did not experience significant frustration during the tasks. The close alignment of the median and average scores indicates a relatively symmetrical distribution of the data, providing a reliable assessment of the workload experienced by the participants.

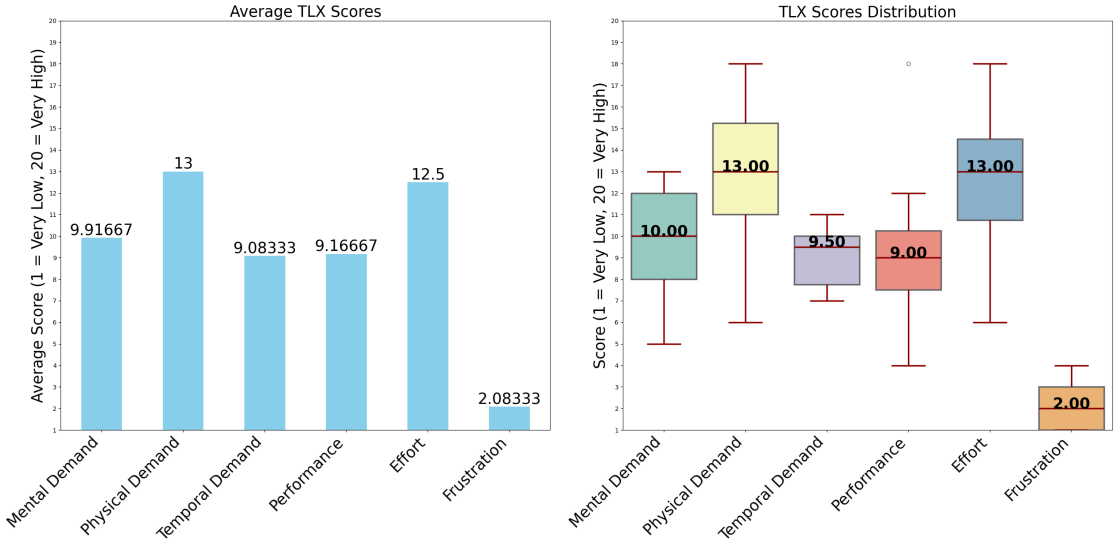


Figure 45: TLX Score of the Driving Simulator Application.

6.3.7 General Questionnaire

Motion Sickness:

The bar plot in Figure 46 illustrates the distribution of motion sickness levels among the participants. The data is categorized into four levels: "Not at all," "Slightly," "Moderately," and "Severely." The chart reveals that the majority of participants reported experiencing motion sickness "Slightly," with a count of 6. This is followed by "Not at all," "Moderately," and "Severely," each with a count of 2. The visualization effectively highlights that a significant portion of the participants experienced only mild symptoms of motion sickness.

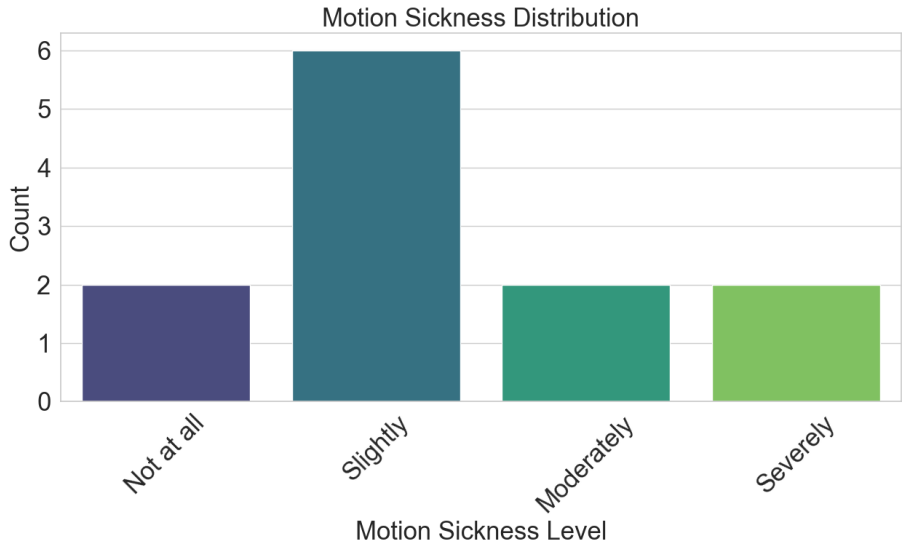


Figure 46: Motion Sickness Likelihood of the Driving Simulator Application.

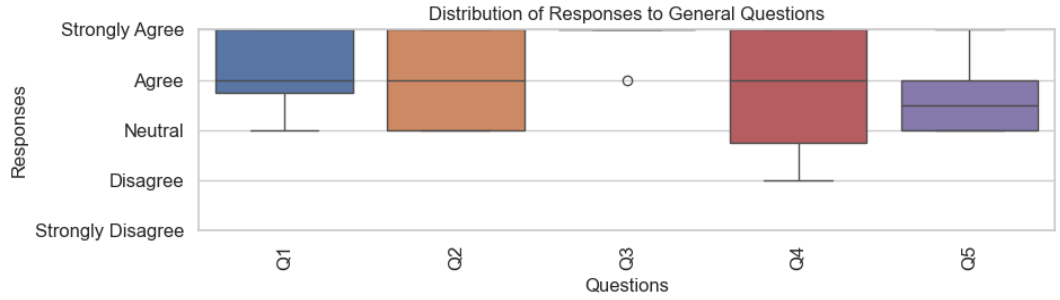
General Information:

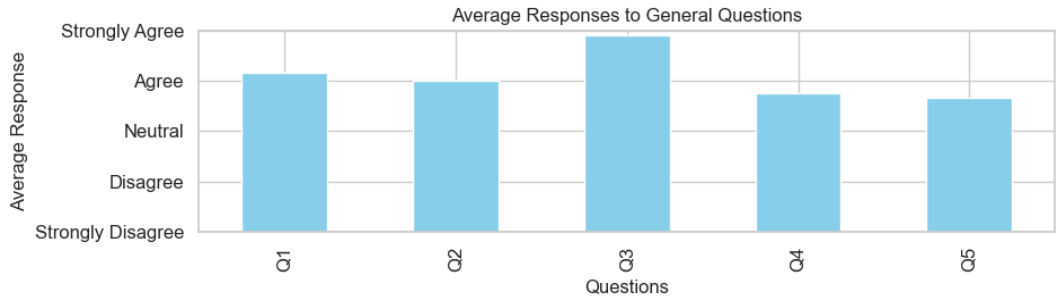
The visualizations in Figure 47 provide a comprehensive analysis of the responses to the general questions posed to the 12 participants. The data is represented through two distinct plots: a box plot and a bar chart.

The box plot on the left side of Figure 47 illustrates the distribution of responses for each of the five questions. The responses are measured on a Likert scale [94] ranging from 1 to 5, where 1 represents "Strongly Disagree" and 5 represents "Strongly Agree." The box plot reveals the median, quarterlies, and potential outliers for each question. It is evident that the responses are generally positive, with most medians falling between 4 ("Agree") and 5 ("Strongly Agree"). The inadequate ranges are relatively narrow, indicating a consensus among participants.

The bar chart on the right side of Figure 47 presents the average responses for each question. This chart provides a clear comparison of the mean scores, which are also measured on the same Likert scale [94]. The average responses for all questions hover around 4, suggesting that participants generally agree with the statements. The highest average response is observed for the question "Assistance system is an excellent tool," indicating strong approval. Conversely, the question "I suggest enhancing the graphics" has a slightly lower average, suggesting room for improvement in this area. Overall, the visualizations indicate a positive reception of the training scenarios and tools, with minor suggestions for graphical enhancements. The box plot and bar chart together provide a comprehensive view of the participants' feedback, highlighting areas of strength and potential improvement. The high median and mean responses for most questions suggest that the participants found the training scenarios realistic, the assistance system useful, and the

navigation easy, with some room for improvement in the graphics.





Question	Description
Q1	Realistic Training Scenarios
Q2	I suggest enhancing the graphics.
Q3	Assistance system is an excellent tool
Q4	Good example for driving in different daytimes
Q5	Easy Navigation

Figure 47: General Questions answered by the Participants of the Driving Simulator Application.

Personal Preferences:

The visualizations in Figure 48 provide a comprehensive analysis of the distribution of various categories related to the driving simulation data. The data is represented through five distinct pie charts, each offering insights into different aspects of the simulation: "Transmission Mode," "Map," "Daytime," "Car," and "Time." The pie charts reveal the following distributions:

1. Transmission Mode: The pie chart for "Transmission Mode" shows that the data is almost evenly split between "manual transmission" and "automatic transmission." Specifically, "manual transmission" accounts for approximately 50% of the instances, while "automatic transmission" also accounts for around 50%. This indicates a balanced preference of both transmission modes in the simulation.

2. Map: The "Map" pie chart illustrates the distribution of different driving

environments. "Training Area" and "Paderborn City" are the most frequently used maps, each accounting for around 33.3% of the instances. "Highway" is used slightly less frequently, making up approximately 25% of the instances. This distribution suggests a diverse range of driving scenarios, with a slight preference for urban and training environments.

3. Daytime: The "Daytime" pie chart shows a significant preference for daytime driving, with "Day" accounting for approximately 75% of the instances. "Night" driving is less common, making up around 25% of the instances. This indicates that most of the driving simulations were conducted during the day.

4. Car: The "Car" pie chart reveals the distribution of different car models used in the simulation. "BMW e92" is the most frequently used car, accounting for approximately 58.3% of the instances. "Fiat Spider" and "Volkswagen MK1" are used less frequently, each making up around 16.7% and 25% of the instances, respectively. This suggests a preference for the "BMW e92" model in the simulations.

5. Time: The "Time" pie chart shows the distribution of driving duration's. The most common duration is 90 minutes, accounting for approximately 58.3% of the instances. Other duration's include 60 minutes (25%), 100 minutes (8.3%), and 120 minutes (8.3%). This indicates that the majority of the driving sessions lasted for 90 minutes.

Overall, the visualizations provide a clear and detailed overview of the driving simulation data. The balanced use of transmission modes, diverse driving environments, preference for daytime driving, frequent use of the "BMW e92" model, and common driving duration of 90 minutes are all highlighted in the pie charts. These insights can help inform future simulations and analyses by identifying common patterns and preferences in the data.

Figure 48: General Information given by the Participants of the Driving Simulator Application.

6.3.8 Open Ended Questions

The research concluded with three open-ended questions designed to gain insights into the participants' interactions with the assistance system within the VR driving simulator. The first question sought to identify the positive and negative feedback that participants associated with their use of the driving simulator. The second question aimed to ascertain whether participants would consider utilizing the driving application in real-world situations. The third question is to collect additional comments to improve the driving simulator application in the future. This subsection examines the insights gained from these questions. The three questions were articulated as follows:

1. Could you provide us with any constructive or critical feedback regarding the driving simulator application?
2. Would you be inclined to utilize the driving simulator with the assistance system if you want to take driving lessons?
3. Do you have any further observations or remarks concerning the driving application?

The responses to the first question exhibit a range of both positive and negative feedback. Subsequently, we will outline the positive and negative comments gathered from each participant at the conclusion of the questionnaires. It is important to note that some participants did not provide any feedback, whether positive or negative.

Positive Feedback:

1. The representation of Paderborn city is highly realistic and stands out as my favorite aspect.
2. I enjoy driving on the highway to test the maximum speed capabilities of each virtual car model.
3. The presence of AI pedestrians enhances the overall realism of the experience.
4. The assistance system proves to be extremely beneficial when changing lanes without signaling, running red lights, and avoiding collisions.
5. The German street signs are visually appealing and possess realistic textures akin to those found in real life.
6. I appreciate that each car model features varying horsepower, distinguishing them from one another.
7. The AI-driven speech audio of the assistance system sounds remarkably human, contributing to an immersive experience.
8. The animations of the 3D virtual driver model accurately replicate the driver's movements, enhancing immersion.
9. The clutch function effectively simulates a real-life clutch, which is advantageous for novice drivers.
10. I find the left and right indicator lights to be well-developed.
11. I am fond of the design of the main menu and the ease of navigation between different user interface elements using the Pad function in the driving setup.

Negative Feedback:

1. I noticed that the 3D buildings lack a sense of realism.

2. The AI cars encounter issues that cause them to drive improperly.
3. Controlling the camera in Non-VR mode can be quite challenging.
4. The car's braking function is not very responsive and is slow, making it difficult for the car to decelerate effectively.
5. The cones in the Training Area map are positioned too closely together, making it tough to finish the second task.
6. Not all driving game setups are compatible with the application, such as the Logitech G25.
7. Sometimes, the car fails to decelerate when the gas pedal is released.
8. Some graphics could use improvement.
9. The highway area feels rather empty.

The results for the second open-ended question show that a strong preference was given to the driving simulator with the assistance system for driving instruction. Indeed, out of 12 there were 9 participants who showed a positive attitude, citing the benefits of a safe and controlled environment to practice in, and the constructive feedback provided by the assistance system. This strong preference shows that the driving simulator, in a combination with the assistance system, has great potential for driving education and building confidence among learners. Yet, it also has to be put into consideration that the preference of 3 participants was different, which may indicate that there is some variation in learning styles or further point to areas of improvement in the design and functionality of the simulator. Of course, the same factors are worthy of investigation in future research to further ensure that the simulator will accommodate the diverse needs of a wider array of learners.

The participants also gave further comments to answer the last open ended question and we will present these comments that we received:

- "You can add to The driving simulator's the ability to simulate different weather conditions, such as rain and fog."
- "The virtual driving instructor's tips and guidance are very helpful and make the learning process more engaging."
- "The loading times between different driving scenarios are quite long, which can be frustrating."
- "The tutorial mode could be more comprehensive, covering more advanced driving techniques."

6.4 Discussion

This section presents the findings from the usability study. The results provide insights relevant to the research questions RQ1 and RQ2, which will be addressed in Section 6.4.1. In order to conclude this study, Section 6.4.2 will identify potential threats to the validity of these results and examine the inherent limitations of the research.

6.4.1 Findings

The information provided in Section 6.3 enables us to derive specific conclusions regarding the primary research inquiries of this thesis, namely RQ1 and RQ2. These research questions were articulated at the outset of the development of this contribution.

RQ1: How much can an assistance system in a VR-based driving simulator enhance driving performance and potentially replace the role of a driving instructor?
RQ2: How does the transition from non-VR to VR advanced driving assistance system affect the performance scores and the usability criteria in a simulated environment?

We were expecting the answer to RQ1 to be 'yes', due to a variety of reasons discussed throughout this thesis, namely immersive capabilities of VR, such as 360-degree view and enriched situational awareness provided by the camera in VR. Besides, more realistic simulation of driving conditions is expected by a VR system to improve driving performance and probably replace a driving instructor. Accordingly, this expectation is best clearly supported by our data in the case of the Highway Map, where indeed the VR mode outperformed the non-VR mode with a statistically significant difference in effectiveness scores. In the case of the Training Area and Paderborn City maps, however, the score of effectiveness is almost identical for the two systems, showing no unique advantage of the VR system in these cases. Given former research and the fact that our participants were familiar with traditional computer programs, the answer to RQ2 had greater potential to add to the existing state of research. In this respect, we expected a temporary drop in task execution speed and an increase in task execution errors when switching to VR. The data from our study partially confirms this expectation. Simultaneously, participants felt it was easier to manage the virtual vehicle in VR mode of viewing probably because the field of view allows better situational awareness. On the other hand, the assistance system was easier to understand in non-VR mode because of the clarity of audio output through the desktop speakers. The overall efficiency score, however, was higher in VR mode with respect to all maps, which would go to say that switching to VR mode actually increases the efficiency of the usability assessment. These may be biased by the general familiarization of the subjects with PC applications as well as conventional driving simulators. Perhaps one becomes efficient in VR once one gets accustomed to the VR driving assistance system. Such acclimatization would reduce the initial disadvantage and perhaps even turn it off

completely.

6.4.2 Threats to Validity

The user study, along with the development of the conceptual solution and prototype, encountered specific limitations that may compromise the validity of the findings. This section will address these factors in detail. With only 12 participants, the sample size is quite small, which may limit the validity of the findings. A larger sample size gives data that is more robust, with fewer effects from outliers. While there is a wide range in age, the majority of participants cluster around the age bracket between 25 and 30 years. This might give a certain bias in the results and may not be representative of the effects of the assistance system in different age brackets. It includes 9 males and 3 females, which are not equal in gender balance. This can threaten the validity of the results, as there could be differences between the genders regarding interaction with the driving simulator. Diversity of educational background enriches it, but at the same time, this may introduce variability in the results. Participants with higher educational levels might perform differently compared to those with lower educational levels, affecting the overall findings. The participants' driving experience varies, with some having no prior experience and others having up to three years of experience. This variability could influence the results, as more experienced drivers might perform better in the simulator. This is a double-edged sword since most of them are relatively naive when it comes to driving simulators, which will enable the assessment of the learning curve but perhaps introduce some variability into the results as participants get used to it at different rates. Self-reported data may also introduce demographic biases or may not correctly state driving experience, VR experience, and other demographic information. Participants' motivation and engagement levels might vary, affecting their performance in the simulator. Some participants might be more interested and engaged, leading to better performance, while others might not take the study as seriously. The design of the efficiency and effectiveness questionnaires might not fully capture all aspects of driving performance and usability. The non-normal distribution of data in some cases might affect the robustness of the statistical tests used. Participants might experience simulator sickness, which could affect their performance and the overall results. This could lead to variability in the data that is not related to the effectiveness of the assistance system. Any technical glitches or malfunctions in the simulator or assistance system could impact the participants' performance and the validity of the results. An example on that is the technical issues faced the application caused by the Logitech SDK (Software Development Kit) integration of the Logitech Gaming Driving kit. The compatible devices in the documentation of this SDK, mentioned the Logitech G25 as a compatible device which was not true as we tested the application with the Logitech G25 driving setup. Extended use of the simulator might lead to participant fatigue, which could affect their performance in later sessions. Participants might alter their behavior simply because they know they are being observed, which could affect the validity of the results. Inconsistent calibration of the simulator or assistance system between sessions could lead to variability in the results. The driving scenarios used in the

simulator might not cover all possible real-world situations, which could limit the validity of the findings.

7 Conclusion

This chapter provides a general conclusion to the research and prototype discussed in this thesis. It summarizes the key findings, compares this study with its related works, and offers recommendations for future research. The chapter is structured as follows: Section 7.1 presents a summary of the research findings, while Section 7.2 provides a comparative analysis of the prototype developed in this thesis with related works. Finally, Section 7.3 offers recommendations for future research and development in the field of VR-based driving simulation.

7.1 Summary

The goal of this thesis was to examine the effectiveness, efficiency, and user satisfaction of an assistance system integrated into a VR-based driving simulator. It also aimed to evaluate how the shift from non-VR to VR advanced driving assistance system influences performance and usability metrics. These research questions have been developed to figure out if this support system could enhance driving performance and be independently used without the presence of a driving instructor, and also to appraise the effects of VR on performance ratings and its usability. This project is initiated with the conceptual framework and the prototype development, followed by a user study which appraises the effectiveness, efficiency, and usability of the support system in a virtual environment. The results and statistics of the user study, which are shown in Section Evaluation 6, indicate that using a VR-based driving simulator equipped with a support system can enhance driving performance, especially on the highway map. In this scenario, the VR mode demonstrated a significant improvement in effectiveness scores compared to the non-VR mode. However, there are only minor differences between the performance rankings of VR and non-VR modes for Training Area and Paderborn City maps, with the VR mode showing slight improvement. Although participants encountered initial difficulties in executing tasks, the transition to VR mode resulted in enhanced efficiency during usability evaluation. Yet, limitations are realized in the small sample size and demographic imbalances that could impact the reliability of the results. Conclusions derived from the evaluation of performance rating and usability criteria after switching to VR can also be contributed to the field of driving simulation and VR technology. The study emphasizes the capacity of VR technology to improve driving performance and the usability of driving simulators, with implications for driving education and training. The findings have important implications for future research and development in the field of VR-based driving simulation with an integrated assistance system.

7.2 Comparative Analysis

The comparison of the four research prototypes, VR-OOM, DREYEVR, Automated Driving, and AR-ADAS, provided relevant insight into the future outlook of driving assistance systems in driving simulators. This uniform integration of support for HMDs and Six DoF Head Movement from all the previous studies underlines the requirement for immersive, realistic simulation experiences. These aspects are basic requirements for simulating reality in a virtual setting, which is needed for any kind of efficient training and research. The addition of Assistance Systems in each prototype also serves the purpose of underlining the need for providing support to users through the simulation so that they can easily interact with the virtual environment as well as receive advice where necessary, hence improving the user experience. Unique features in some prototypes reflect various goals and target scenarios. For example, automated driving [77] only includes Highway Driving and Lane Changes, thereby becoming more and more tailored for complex driving scenarios. On the contrary, DREYEVR [24] is unique with its City Driving, Traffic Signals, and Speed Limits, a very holistic approach to the simulation of driving in a city. Among all prototypes, Parking Training, Night Driving, and Accidents have not been implemented, which points to a very important area of future development. All these features will greatly enhance the realism and comprehensiveness of the driving simulators to give complete training to the users. The cost consideration is also reflected in that three out of the four prototypes must feature the Low-Cost Simulator for widespread use and easy accessibility. Building on these, our prototype tries to fill a host of gaps identified in the related works. The main problem in these previous works was, that there was no single driving simulator prototype, which combined all the features mentioned in the features list provided in Section 3.1. Our prototype, however, tries to combine all these features in a single driving simulator application. The VR-based driving simulator application developed in this thesis integrates a variety of features, including different maps, daytime settings, car models, and driving durations. The application also includes an assistance system that provides real-time feedback and guidance to users during driving scenarios. For example, there was no previous work that developed parking training tasks, night driving mode, and accident prevention scenarios. Some of the previous works did not integrate highway driving, city driving, traffic signals or street lights detection, lane change detection, and speed limit for each street. Therefore, our prototype focuses on developing the missing features in realted works. Doing this allows us to have a complete driving simulator application that has as many realistic features as possible to make the VR-based car driving experience very realistic and immersive. Table 7.1 shows a summary of the features developed in previous works and in our prototype.

Features	VR-OOM [76]	DREYEVR [24]	Automated Driving [77]	AR-ADAS [78]	This Work
Support for HMDs	✓	✓	✓	✓	✓
Six DoF Head Movement	✓	✓	✓	✓	✓
Parking Training	✗	✗	✗	✗	✓
Highway Driving	✗	✗	✓	✗	✓
City Driving	✗	✓	✗	✗	✓
Night Driving	✗	✗	✗	✗	✓
Assistance System	✓	✓	✓	✓	✓
Traffic Signals	✗	✓	✗	✗	✓
Lane Changes	✗	✗	✓	✓	✓
Speed Limits	✗	✓	✓	✗	✓
Accidents	✗	✗	✗	✗	✓
Low-Cost Simulator	✓	✓	✗	✓	✓

Legend
✓ - Supported
✗ - Not Supported

Table 7.1: Comparison of Related Work Support for Various Features.

7.3 Future Work

The results of this thesis leave room for new opportunities to take the research further. Some proposals for future study and development in VR-based driving simulation integrated with an assistance system will be as follows:

- **Larger Sample Size:** The future studies should allow for an increased sample size to advance the robustness and generalisability of the results. A larger group of participants would provide a broader view and deeper insight into the effectiveness and ease of use of the assistance systems in VR-based driving simulators.
- **Varied Demographics:** The involvement of more varied demographics in user tests could allow different age groups and educational backgrounds to try out and work with the VR-based driving simulator and assistance systems. The variation would possibly enable the detection of variations in performance and usability across different user groups.
- **Sophisticated Driving Scenarios:** Development of more complex scenarios could simulate difficult real-life conditions, possibly allowing Improve the

realism and effectiveness of the virtual driving experience. For example, addition of the factors such as bad weather conditions, complex scenes of traffic, and situations leading to emergencies would give better training for the users.

- **Enhanced Graphics and Realism:** Advancing the resolution of the textures and the realism of the VR-based driving simulator can give users a better immersive experience and more interaction. Improvement of the visual detail and environment, details, and car models would create a much more realistic and fascinating simulation environment.
- **Adaptive Assistance Systems:** More complex-scenario development would create situations that would mimic challenging real-life conditions which, in turn, could enable the driving practice in the virtual environment should be made more realistic and effective. For example, adding the factors of bad weather conditions, complicated scenes of traffic flow, and emergency situations would better train the users.
- **Integration with Real-world Driving:** The integration of real driving experiences with VR driving simulators may be able to create a wholesome effect: there is no gap in virtual and physical driver training. Combining on-road practice with VR driving simulations can provide all-encompassing training, which will take users through different scenarios. It could also be put into practice as a pre-license-exam training tool in driving.
- **Using AI for Assistance:** The use of AI-driving-assistance can improve the ability of the driving assistance system and give real-time feedback to the user. One can do this by taking into consideration foretold patterns of user behavior and providing feedback based on that using machine learning algorithms.

Bibliography

[1] JM Zheng, KW Chan, and Ian Gibson. Virtual reality. *Ieee Potentials*, 17(2):20–23, 1998.

[2] Nicolae-Adrian Tocu, Arpad Gellert, Ioana-Ramona Stefan, Teodor-Marian Nitescu, and Gabriela-Alexandra Luca. The impact of virtual reality simulators in manufacturing industry. In *EDULEARN20 Proceedings*, pages 3084–3093. IATED, 2020.

[3] Yuejun Xu and Makio Ishihara. A pilot study on progress of driving skills with immersive vr driving simulator. In *HCI International 2021-Late Breaking Papers: Multimodality, eXtended Reality, and Artificial Intelligence: 23rd HCI International Conference, HCII 2021, Virtual Event, July 24–29, 2021, Proceedings 23*, pages 356–365. Springer, 2021.

[4] Julien Monteil, Niall O'Hara, Vinny Cahill, and Melanie Bouroche. Real-time estimation of drivers' behaviour. In *2015 IEEE 18th International Conference on Intelligent Transportation Systems*, pages 2046–2052. IEEE, 2015.

[5] Arun Tigadi, Rudrappa Gujanatti, Anil Gonchi, and Belagavi Klemsscet. Advanced driver assistance systems. *International Journal of Engineering Research and General Science*, 4(3):151–158, 2016.

[6] Quinatei Lhemedu-Steinke, Gerrit Meixner, and Michael Weber. Comparing vr display with conventional displays for user evaluation experiences. In *2018 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pages 583–584. IEEE, 2018.

[7] David F McAllister. Display technology: stereo & 3d display technologies. *Encyclopedia of imaging science and technology*, 2:1327–1344, 2002.

[8] Tyler A Jost, Grant Drewelow, Scott Koziol, and Jonathan Rylander. A quantitative method for evaluation of 6 degree of freedom virtual reality systems. *Journal of biomechanics*, 97:109379, 2019.

[9] Alexey Dosovitskiy, German Ros, Felipe Codevilla, Antonio Lopez, and Vladlen Koltun. Carla: An open urban driving simulator. In *Conference on robot learning*, pages 1–16. PMLR, 2017.

[10] Quinate Chioma Ihemedu-Steinke, Rainer Erbach, Prashanth Halady, Gerrit Meixner, and Michael Weber. *Virtual Reality Driving Simulator Based on Head-Mounted Displays*, pages 401–428. Springer International Publishing, Cham, 2017.

[11] Tomasz Kaminski, Michal Niezgoda, Joanna Gacparska, Rafal Grzeszczyk, and Przemysław Filipek. Advanced car driving simulator-as 1200-6. *Journal of KONES*, 18:173–178, 2011.

[12] Takeki Ogitsu and Hiroshi Mizoguchi. A study on driver training on advanced driver assistance systems by using a driving simulator. In *2015 International Conference on Connected Vehicles and Expo (ICCVE)*, pages 352–353, 2015.

[13] Clara Marina Martinez, Mira Heucke, Fei-Yue Wang, Bo Gao, and Dongpu Cao. Driving style recognition for intelligent vehicle control and advanced driver assistance: A survey. *IEEE Transactions on Intelligent Transportation Systems*, 19(3):666–676, 2017.

[14] Harpreet Singh and Ankit Kathuria. Analyzing driver behavior under naturalistic driving conditions: A review. *Accident Analysis & Prevention*, 150:105908, 2021.

[15] Alessandro Calvi, Fabrizio D'Amico, Chiara Ferrante, and Luca Bianchini Ciampoli. Effectiveness of augmented reality warnings on driving behaviour whilst approaching pedestrian crossings: A driving simulator study. *Accident Analysis & Prevention*, 147:105760, 2020.

[16] Quinate Chioma Ihemedu-Steinke, Stanislava Rangelova, Michael Weber, Rainer Erbach, Gerrit Meixner, and Nicola Marsden. Simulation sickness related to virtual reality driving simulation. In *Virtual, Augmented and Mixed Reality: 9th International Conference, VAMR 2017, Held as Part of HCI International 2017, Vancouver, BC, Canada, July 9-14, 2017, Proceedings 9*, pages 521–532. Springer, 2017.

[17] Donald Shoup. *High cost of free parking*. Routledge, 2021.

[18] MarketsandMarkets. Parking management market by offering, application, region - global forecast to 2032, 2022.

[19] Driving in the dark - tips for safe and confident night driving. *SafetyConnect*, 2023.

[20] City driving hazards: The challenges of navigating urban roads. *ePermitTest*, 2020.

[21] Judie Nuskey. How to safely handle common highway driving challenges. *Automotive Fleet*, 2023.

[22] Meta. Meta quest 3. `https://www.meta.com/de/quest/quest-3/`. Accessed: 2024-08-09.

[23] Logitech. Logitech g923 trueforce sim racing wheel. `https://www.logitechg.com/de-de/products/driving/g923-trueforce-sim-racing-wheel.html`. Accessed: 2024-08-09.

[24] Gustavo Silvera, Abhijat Biswas, and Henny Admoni. Dreye vr: Democratizing virtual reality driving simulation for behavioural & interaction research. In *2022 17th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, pages 639–643. IEEE, 2022.

[25] Isabell Wohlgenannt, Alexander Simons, and Stefan Stieglitz. Virtual reality. *Business & Information Systems Engineering*, 62:455–461, 2020.

[26] Ivan E Sutherland et al. The ultimate display. In *Proceedings of the IFIP Congress*, number 506-508 in IFIP Congress Series, pages 506–508. New York, 1965.

[27] Lucia Valmaggia. The use of virtual reality in psychosis research and treatment. *World Psychiatry*, 16(3):246, 2017.

[28] Lisa Rebenitsch. Managing cybersickness in virtual reality. *XRDS: Crossroads, The ACM Magazine for Students*, 22(1):46–51, 2015.

[29] Carolina Cruz-Neira, Daniel J Sandin, and Thomas A DeFanti. Surround-screen projection-based virtual reality: the design and implementation of the cave. In *Seminal Graphics Papers: Pushing the Boundaries, Volume 2*, pages 51–58. usa, 2023.

[30] Muhanna A Muhanna. Virtual reality and the cave: Taxonomy, interaction challenges and research directions. *Journal of King Saud University-Computer and Information Sciences*, 27(3):344–361, 2015.

[31] Doug A Bowman and Ryan P McMahan. Virtual reality: how much immersion is enough? *Computer*, 40(7):36–43, 2007.

[32] Woodrow Barfield, David Zeltzer, Thomas Sheridan, and Mel Slater. Presence and performance within virtual environments. In *Virtual environments and advanced interface design*, pages 473–513. IEEE, 1995.

[33] Carina Girvan. What is a virtual world? definition and classification. *Educational Technology Research and Development*, 66(5):1087–1100, 2018.

[34] Ulrike Schultze. Embodiment and presence in virtual worlds: a review. *Journal of Information Technology*, 25(4):434–449, 2010.

[35] Kenneth R Walsh and Suzanne D Pawlowski. Virtual reality: A technology in need of is research. *Communications of the Association for Information Systems*, 8(1):20, 2002.

[36] Maria V Sanchez-Vives and Mel Slater. From presence to consciousness through virtual reality. *Nature reviews neuroscience*, 6(4):332–339, 2005.

[37] Jonathan Steuer, Frank Biocca, Mark R Levy, et al. Defining virtual reality: Dimensions determining telepresence. *Communication in the age of virtual reality*, 33(37-39):1, 1995.

[38] Niels Chr Nilsson, Rolf Nordahl, and Stefania Serafin. Immersion revisited: A review of existing definitions of immersion and their relation to different theories of presence. *Human technology*, 12(2):108–134, 2016.

[39] Mel Slater and Sylvia Wilbur. A framework for immersive virtual environments (five): Speculations on the role of presence in virtual environments. *Presence: Teleoperators & Virtual Environments*, 6(6):603–616, 1997.

[40] Staffan Bjork and Jussi Holopainen. *Patterns in game design*, volume 11. Charles River Media Hingham, 2005.

[41] Laura Freina and Michela Ott. A literature review on immersive virtual reality in education: state of the art and perspectives. In *The international scientific conference elearning and software for education*, number 133 in 2, pages 10–1007, 2015.

[42] Abbie Brown and Tim Green. Virtual reality: Low-cost tools and resources for the classroom. *TechTrends*, 60:517–519, 2016.

[43] King M Roberts. The fhwa highway driving simulator. *Public Roads*, 44(3):97–102, 1980.

[44] R Wade Allen, Richard H Klein, and Kenneth Ziedman. Automobile research simulators: a review and new approaches. *Transportation research record*, 706:9–15, 1979.

[45] Evi Blana. Driving simulator validation studies: a literature review. *Public Roads*, 1996.

[46] Lena Nilsson. The vti driving simulator. description of a research tool. *VTI särtryck*, 1(150), 1989.

[47] Johannes Drosdol and Ferdinand Panik. The daimler-benz driving simulator a tool for vehicle development. *SAE Transactions*, pages 981–997, 1985.

[48] PJ Zwaneveld, Bart Van Arem, E Bastiaensen, H Soeteman, and B Ulmer. Deployment scenarios for advanced driver assistance systems in europe. In *PROCEEDINGS OF 6TH WORLD CONGRESS ON INTELLIGENT TRANSPORT SYSTEMS (ITS), HELD TORONTO, CANADA, NOVEMBER 8-12, 1999*, 1999.

[49] OMJ Carsten and Lena Nilsson. Safety assessment of driver assistance systems. *European Journal of Transport and Infrastructure Research*, 1(3):225–243, 2001.

[50] Buse Asena Koca, Burakhan Çubukçu, and Uğur Yüzgeç. Augmented reality application for preschool children with unity 3d platform. In *2019 3rd International Symposium on Multidisciplinary Studies and Innovative Technologies (ISMSIT)*, pages 1–4. IEEE, 2019.

[51] Create & Learn. Top games made with unity, 2024. Accessed: 2024-08-10.

[52] Sung Lae Kim, Hae Jung Suk, Jeong Hwa Kang, Jun Mo Jung, Teemu H Laine, and Joonas Westlin. Using unity 3d to facilitate mobile augmented reality game development. In *2014 IEEE World Forum on Internet of Things (WF-IoT)*, pages 21–26. IEEE, 2014.

[53] Swati Singh and Amanpreet Kaur. Game development using unity game engine. In *2022 3rd International Conference on Computing, Analytics and Networks (ICAN)*, pages 1–6. IEEE, 2022.

[54] Ismail Buyuksalih, Serdar Bayburt, Gurcan Buyuksalih, AP Baskaraca, Hairi Karim, and Alias Abdul Rahman. 3d modelling and visualization based on the unity game engine–advantages and challenges. *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, 4:161–166, 2017.

[55] Farouk Messaoudi, Gwendal Simon, and Adlen Ksentini. Dissecting games engines: The case of unity3d. In *2015 international workshop on network and systems support for games (NetGames)*, pages 1–6. IEEE, 2015.

[56] Lav Soni, Amanpreet Kaur, and Avinash Sharma. A review on different versions and interfaces of blender software. In *2023 7th International Conference on Trends in Electronics and Informatics (ICOEI)*, pages 882–887. IEEE, 2023.

[57] Blender. `https://www.blender.org/`.

[58] Brian R Kent. Visualizing astronomical data with blender. *Publications of the Astronomical Society of the Pacific*, 125(928):731, 2013.

[59] David M Johnson. Introduction to and review of simulator sickness research. *Sensors*, 2005.

[60] Robert S Kennedy, Julie Drexler, and Robert C Kennedy. Research in visually induced motion sickness. *Applied ergonomics*, 41(4):494–503, 2010.

[61] Robert S Kennedy and Jennifer E Fowlkes. Simulator sickness is polygenic and polysymptomatic: implications for research. In *Simulation in Aviation Training*, pages 167–182. Routledge, 2017.

[62] Guanghan Zhao, Jason Orlosky, Steven Feiner, Photchara Ratsamee, and Yuki Uranishi. Mitigation of vr sickness during locomotion with a motion-based dynamic vision modulator. *IEEE transactions on visualization and computer graphics*, 2022.

[63] Judy Jeng. What is usability in the context of the digital library and how can it be measured? *Information technology and libraries*, 24(2):3, 2005.

[64] Kyunghye Kim. *A model of digital library information seeking process (DLISP model) as a frame for classifying usability problems.* Rutgers The State University of New Jersey, School of Graduate Studies, 2002.

[65] Myke Gluck. *A descriptive study of the usability of geospatial metadata*. OCLC, 1998.

[66] Thomas K Landauer. *The trouble with computers: Usefulness, usability, and productivity*. MIT press, 1995.

[67] Nigel Bevan. International standards for hci and usability. *International journal of human-computer studies*, 55(4):533–552, 2001.

[68] Jakob Nielsen. *Usability engineering*. Morgan Kaufmann, 1994.

[69] Tom Brinck, Darren Gergle, and Scott D Wood. *Designing Web sites that work: Usability for the Web*. Morgan Kaufmann, 2002.

[70] Paul Booth. *An introduction to human-computer interaction (psychology revivals)*. Psychology Press, 2014.

[71] Deborah Hix and H Rex Hartson. *Developing user interfaces: ensuring usability through product & process*. John Wiley & Sons, Inc., 1993.

[72] Patrick Harms. Automated usability evaluation of virtual reality applications. *ACM Transactions on Computer-Human Interaction (TOCHI)*, 26(3):1–36, 2019.

[73] Doug A Bowman, Joseph L Gabbard, and Deborah Hix. A survey of usability evaluation in virtual environments: classification and comparison of methods. *Presence: Teleoperators & Virtual Environments*, 11(4):404–424, 2002.

[74] DIN Deutsches Institut für Normung e.V. Mensch-system-interaktion, 2024. Accessed: 2024-08-10.

[75] Jakob Nielsen et al. Usability 101: Introduction to usability. *Sensors*, 2012.

[76] David Goedicke, Jamy Li, Vanessa Evers, and Wendy Ju. Vr-oom: Virtual reality on-road driving simulation. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, CHI '18, page 1–11, New York, NY, USA, 2018. Association for Computing Machinery.

[77] Daniele Sportillo, Alexis Paljic, and Luciano Ojeda. Get ready for automated driving using virtual reality. *Accident Analysis and Prevention*, 118:102–113, 2018.

[78] Ziran Wang, Kyungtae Han, and Prashant Tiwari. Augmented reality-based advanced driver-assistance system for connected vehicles. In *2020 ieee international conference on systems, man, and cybernetics (SMC)*, pages 752–759. IEEE, 2020.

[79] Road to VR. Htc teases 6dof mode for vive focus controller, optical hand tracking for gesture input. *Road to VR*, 2018.

[80] Logitech g developer lab - software development kits. `https://www.logitechg.com/en-us/innovation/developer-lab.html`. Accessed: 2024-08-25.

[81] Olaf Gietelink, Jeroen Ploeg, Bart De Schutter, and Michel Verhaegen. Development of advanced driver assistance systems with vehicle hardware-in-the-loop simulations. *Vehicle System Dynamics*, 44(7):569–590, 2006.

[82] Dean A Pomerleau. Alvinn: An autonomous land vehicle in a neural network. *Advances in neural information processing systems*, 1, 1988.

[83] B Abdulbaqi Jumaa, A Mousa Abdulhassan, and A Mousa Abdulhass. Advanced driver assistance system (adas): A review of systems and technologies. *International Journal of Advanced Research in Computer Engineering & Technology (IJARCET) vol*, 8:2278–1323, 2019.

[84] Emma Raymer, Áine MacDermott, and Alex Akinbi. Virtual reality forensics: Forensic analysis of meta quest 2. *Forensic Science International: Digital Investigation*, 47:301658, 2023.

[85] Sketchfab. `https://sketchfab.com/feed`.

[86] Shen Lin and Brian W. Kernighan. An effective heuristic algorithm for the travelling-salesman problem. *Operations Research*, 21:498–516, 1973.

[87] GIMP Development Team. Gimp, 2024. Accessed: 2024-08-30.

[88] SC Paderborn 07. Home deluxe arena, 2024. Accessed: 2024-08-30.

[89] Microsoft. Visual studio code - download, 2024. Accessed: 2024-08-30.

[90] ElevenLabs. Elevenlabs, 2024. Accessed: 2024-08-30.

[91] Ahmed Sabbir Arif. University of california. a brief note on selecting & reporting the right statistical test, 2021.

[92] Bangor Aaron. Determining what individual sus scores mean: adding an adjective rating scale. *Journal of usability studies*, 4:3, 2009.

[93] Jacinto Vasconcelos-Raposo, Maximino Bessa, Miguel Melo, Luis Barbosa, Rui Rodrigues, Carla Maria Teixeira, Luciana Cabral, and António Augusto Sousa. Adaptation and validation of the igroup presence questionnaire (ipq) in a portuguese sample. *Presence*, 25(3):191–203, 2016.

[94] Tomoko Nemoto and David Beglar. Likert-scale questionnaires. In *JALT 2013 conference proceedings*, volume 108, pages 1–6, 2014.

Appendix A

A.1 Prototype Code

The source code used in the C# scripts for the Unity project can be found in the following GitHub repository: `https://github.com/salahseidan/Behind-the-VR-Wheel`. Links to clone the repository are available on this site. Alternatively, the following links can be used for cloning via HTTPS: `https://github.com/salahseidan/Behind-the-VR-Wheel.git`, or SSH: `git@github.com:salahseidan/Behind-the-VR-Wheel.git`. Furthermore, you can find the driving simulator VR based prototype on the following link: `https://mega.nz/file/sExhnYIB#LaFwXH46UTDdKchahEKRM_r_JKuYaZM4zF1QVk_cviw`. To be able to run the simulator, you need to have a VR headset (Meta Quest Series preferred), Pc Monitor, and a PC with the following minimum requirements: Windows 11, Intel i5-4590/AMD Ryzen 5 1500X, 8 GB RAM, NVIDIA GTX 970/AMD Radeon R9 290, and 1 GB of free space. Also you need to have a Logitech compatible gaming steering wheel setup. A list of compatible steering wheels can be found in the table A.1 below. There is no garanty that the simulator will work with all the steering wheels in the list. We have this list from the Logitech SDK documentation. We tested the simulator with the Logitech G29 and G923 and it worked perfectly.

Logitech Model
LOGI MODEL G27
LOGI MODEL G29
LOGI MODEL G920
LOGI MODEL G923
LOGI MODEL DRIVING FORCE GT
LOGI MODEL MOMO FORCE FEEDBACK
LOGI MODEL MOMO RACING
LOGI MODEL MOMO FORCE
LOGI MODEL DRIVING FORCE PRO
LOGI MODEL DRIVING FORCE
LOGI MODEL NASCAR RACING WHEEL
LOGI MODEL FORMULA FORCE
LOGI MODEL FORMULA FORCE GP
LOGI MODEL FORCE 3D PRO
LOGI MODEL EXTREME 3D PRO

Table A.1: List Of The Compatible Logitech Models

A.2 Install Unity

To be able to run the simulator, you need to have Unity installed on your PC. You can download Unity from the following link: `https://unity3d.com/get-unity/download`. You need to have a Unity account to be able to download the software. After downloading the software, you can install it on your PC. The simulator was developed using Unity 2022.3.20f1 version. You can download this version in Unity Hub by selecting the version from the list of available versions. Figure 49 shows the Unity Hub window and where you can select the version of Unity to install. First click on the "Installs" tab, then click on the "Add" button, and select the version of Unity you want to install. After selecting the version, click on the "Install" button and follow the instructions to install the Version. For our prototype we only used the PCVR to test it in VR mode, so we don not need to install the Android Build Support. If you want to test the simulator in VR mode, you need to have a VR headset connected to your PC. You can use the Meta Quest Series or any other VR headset that is compatible with Unity.

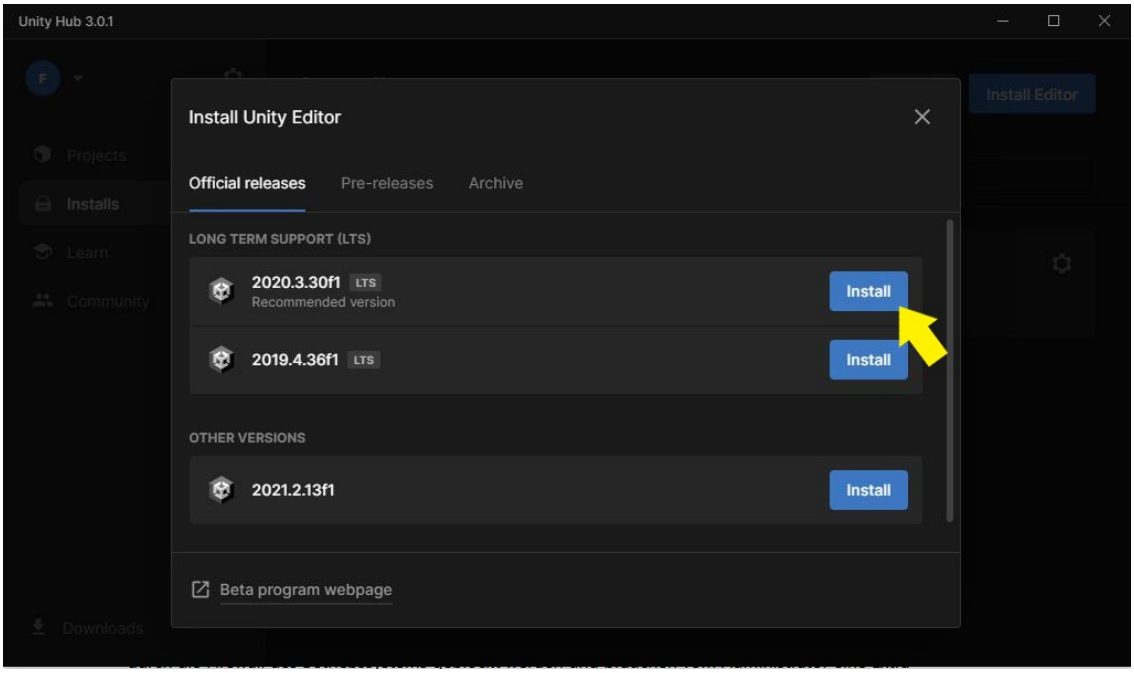


Figure 49: Unity Version Install Window.

A.3 Build Settings

To be able to run the simulator in VR mode, you need to change the build settings in Unity. To do this, you need to open the Unity project and go to the "File" menu, then click on "Build Settings". In the build settings window, you need to select the platform you want to build the project for. In our case, we selected the "Windows, Mac, Linux" platform for PCVR app. After selecting the platform,

you need to click on the "Switch Platform" button. After switching the platform, you need to click on the "Player Settings" button. In the player settings window, you need to change the "XR Plugin Management" settings. You need to enable the "OpenVR" plugin and disable the "Oculus" plugin as shown in Figure 50. In OpenXR Window you need to add some settings like Meta XR so that the Meta Quest headsets work correctly in the application as shown in Figure 51. After changing the settings, you need to click on the "Build" button to build the project. After building the project, you can run the simulator in VR mode with the VR headsets, or take off the VR headsets and test the application on PC monitor in non-VR mode. It also possibil to test the application in XR Simulator tool in Unity if you don't have a VR headset, how to set the XR Simulator settings in Unity is shown in Figure 52. Finally, you will execute an application like this one provided in the following link `https://mega.nz/file/sExhnYIB#LaFwXH46UTDdKchahEKRM_r_JKuYaZM4zF1QVk_cviw`. Figure 53 shows the build settings window and where you can change the settings to run the simulator in VR mode.

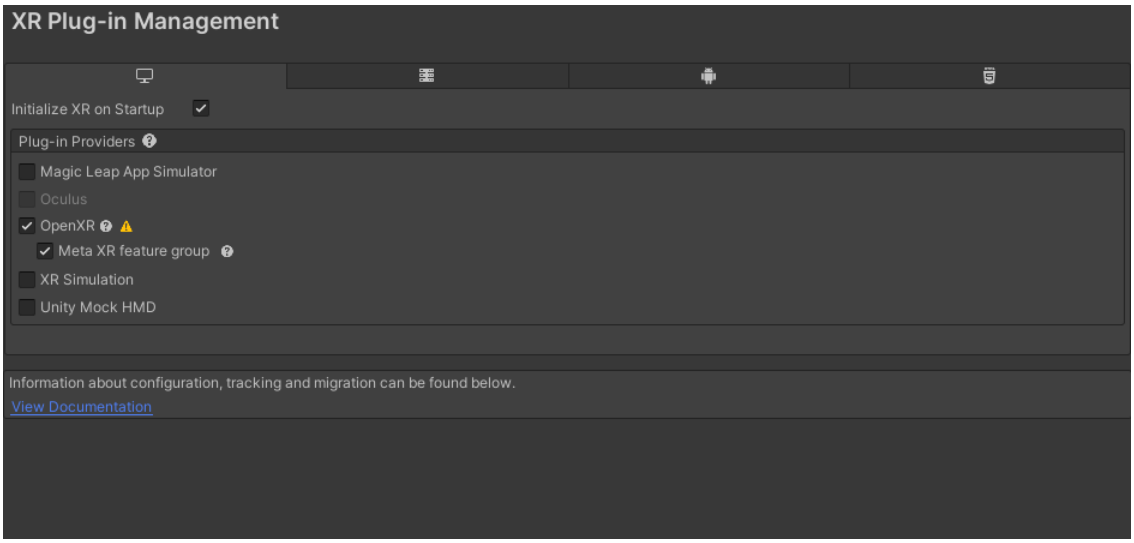


Figure 50: XR Management Settings.

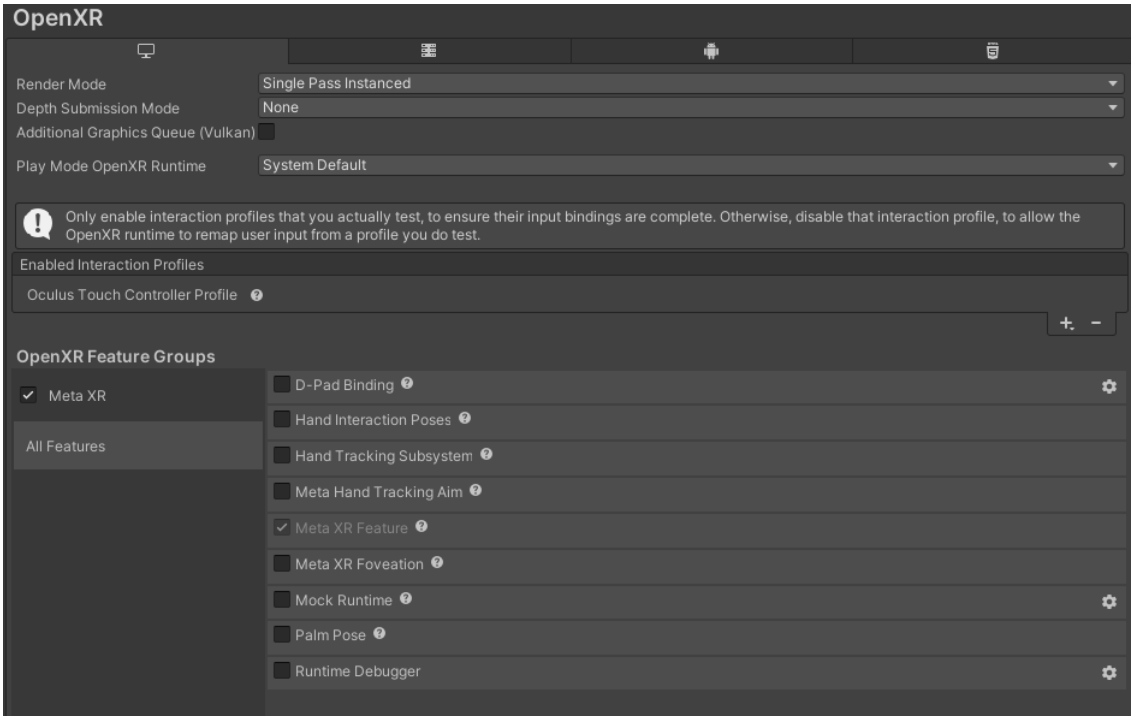


Figure 51: Open XR Settings.

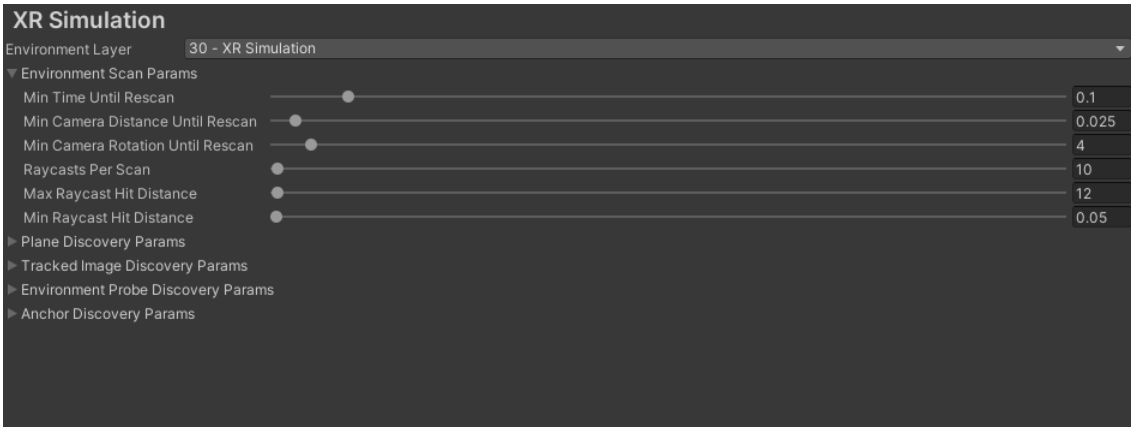


Figure 52: XR Simulator Settings.

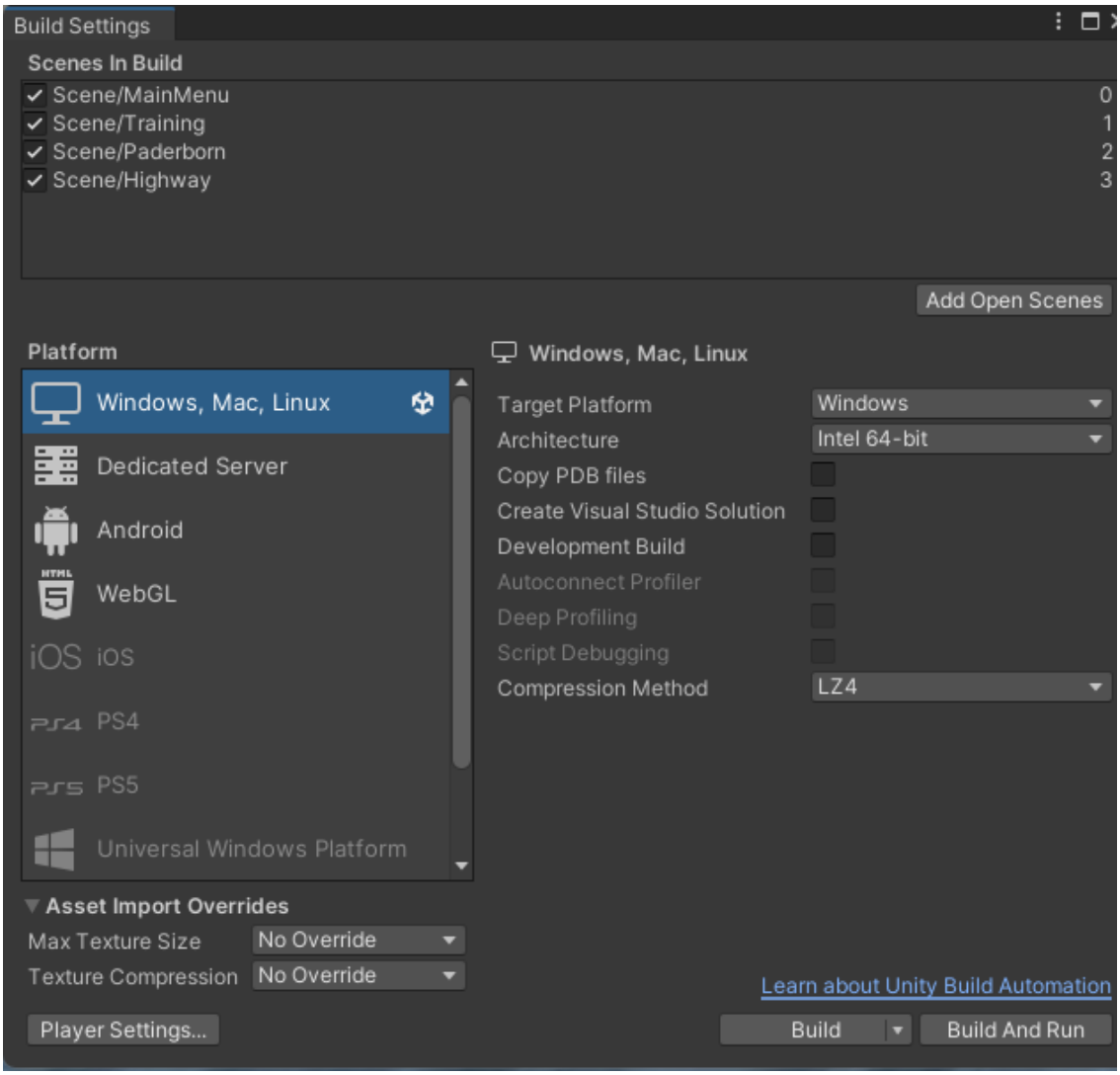


Figure 53: Build Settings.

Appendix B

Usability Study Questionnaire

Driving Simulator Usability Scale Questionnaire

Paderborn August 2, 2024

Demographic questions

1. **Name:** _____
2. **Age:** _____
3. **Gender:** _____
4. **Education Level:**
   - ____ High School
   - ____ Bachelor's Degree
   - ____ Master's Degree
   - ____ Doctorate
   - ____ Other: _____
5. **Driving Experience:**
   - ____ None
   - ____ Less than 1 year
   - ____ 1-3 years
   - ____ 3-5 years
   - ____ More than 5 years
6. **Frequency of Driving:**
   - ____ Daily
   - ____ Weekly
   - ____ Monthly
   - ____ Rarely
7. **Experience with Driving Simulators:**
   - ____ None
   - ____ Some
   - ____ Extensive
8. **Experience with Virtual Reality (VR):**
   - ____ None
   - ____ Some
   - ____ Extensive
9. **Susceptibility to Motion Sickness:**
   - ____ Not at all
   - ____ Slightly
   - ____ Moderately
   - ____ Severely

Training Area Non-VR

Effectiveness	**1 Strongly Disagree**	**2**	**3**	**4**	**5 Strongly Agree**
I found the driving tasks easy to understand.	[ ]	[ ]	[ ]	[ ]	[ ]
The Assistance System has improved my driving performance.	[ ]	[ ]	[ ]	[ ]	[ ]
The Assistance System gave me good feedback.	[ ]	[ ]	[ ]	[ ]	[ ]
The simulator has helped me with the use of the vehicle control devices (gearshift,Steering Wheel and Driving Pedals)	[ ]	[ ]	[ ]	[ ]	[ ]
I have a better understanding of how to act according to different german street rules.	[ ]	[ ]	[ ]	[ ]	[ ]
The Assistance System has aided me in driving more carefully.	[ ]	[ ]	[ ]	[ ]	[ ]
Efficiency	**1 Strongly Disagree**	**2**	**3**	**4**	**5 Strongly Agree**
It was easy to control the vehicle.	[ ]	[ ]	[ ]	[ ]	[ ]
The assistance system in this mode was easy to understand.	[ ]	[ ]	[ ]	[ ]	[ ]
I had enough driving time.	[ ]	[ ]	[ ]	[ ]	[ ]

Score:

Training Area VR

Effectiveness	**1 Strongly Disagree**	**2**	**3**	**4**	**5 Strongly Agree**
I found the driving tasks easy to understand.	[ ]	[ ]	[ ]	[ ]	[ ]
The Assistance System has improved my driving performance.	[ ]	[ ]	[ ]	[ ]	[ ]
The Assistance System gave me good feedback.	[ ]	[ ]	[ ]	[ ]	[ ]
The simulator has helped me with the use of the vehicle control devices (gearshift,Steering Wheel and Driving Pedals)	[ ]	[ ]	[ ]	[ ]	[ ]
I have a better understanding of how to act to different german street rules.	[ ]	[ ]	[ ]	[ ]	[ ]
The Assistance System has aided me in driving more carefully.	[ ]	[ ]	[ ]	[ ]	[ ]
My performance has improved after using the VR mode .	[ ]	[ ]	[ ]	[ ]	[ ]
Efficiency	**1 Strongly Disagree**	**2**	**3**	**4**	**5 Strongly Agree**
It was easy to control the vehicle.	[ ]	[ ]	[ ]	[ ]	[ ]
The assistance system in the last mode was very challenging and hard to understand.	[ ]	[ ]	[ ]	[ ]	[ ]
I had enough driving time in the last mode.	[ ]	[ ]	[ ]	[ ]	[ ]

Score:

Highway Non-VR

Effectiveness	**1 Strongly Disagree**	**2**	**3**	**4**	**5 Strongly Agree**
I found the driving tasks easy to understand.	[ ]	[ ]	[ ]	[ ]	[ ]
The Assistance System has improved my driving performance.	[ ]	[ ]	[ ]	[ ]	[ ]
The Assistance System gave me good feedback.	[ ]	[ ]	[ ]	[ ]	[ ]
The simulator has helped me with the use of the vehicle control devices (gearshift,Steering Wheel and Driving Pedals)	[ ]	[ ]	[ ]	[ ]	[ ]
I have a better understanding of how to act according to different german street rules.	[ ]	[ ]	[ ]	[ ]	[ ]
The Assistance System has aided me in driving more carefully.	[ ]	[ ]	[ ]	[ ]	[ ]
Efficiency	**1 Strongly Disagree**	**2**	**3**	**4**	**5 Strongly Agree**
It was easy to control the vehicle.	[ ]	[ ]	[ ]	[ ]	[ ]
The assistance system in this mode was easy to understand.	[ ]	[ ]	[ ]	[ ]	[ ]
I had enough driving time.	[ ]	[ ]	[ ]	[ ]	[ ]

Score:

Highway VR

Effectiveness	**1 Strongly Disagree**	**2**	**3**	**4**	**5 Strongly Agree**
I found the driving tasks easy to understand.	[ ]	[ ]	[ ]	[ ]	[ ]
The Assistance System has improved my driving performance.	[ ]	[ ]	[ ]	[ ]	[ ]
The Assistance System gave me good feedback.	[ ]	[ ]	[ ]	[ ]	[ ]
The simulator has helped me with the use of the vehicle control devices (gearshift,Steering Wheel and Driving Pedals)	[ ]	[ ]	[ ]	[ ]	[ ]
I have a better understanding of how to act to different german street rules.	[ ]	[ ]	[ ]	[ ]	[ ]
The Assistance System has aided me in driving more carefully.	[ ]	[ ]	[ ]	[ ]	[ ]
My performance has improved after using the VR mode .	[ ]	[ ]	[ ]	[ ]	[ ]
Efficiency	**1 Strongly Disagree**	**2**	**3**	**4**	**5 Strongly Agree**
It was easy to control the vehicle.	[ ]	[ ]	[ ]	[ ]	[ ]
The assistance system in the last mode was very challenging and hard to understand.	[ ]	[ ]	[ ]	[ ]	[ ]
I had enough driving time in the last mode.	[ ]	[ ]	[ ]	[ ]	[ ]

Score:

Paderborn City Non-VR

Effectiveness	**1 Strongly Disagree**	**2**	**3**	**4**	**5 Strongly Agree**
I found the driving tasks easy to understand.	[ ]	[ ]	[ ]	[ ]	[ ]
The Assistance System has improved my driving performance.	[ ]	[ ]	[ ]	[ ]	[ ]
The Assistance System gave me good feedback.	[ ]	[ ]	[ ]	[ ]	[ ]
The simulator has helped me with the use of the vehicle control devices (gearshift,Steering Wheel and Driving Pedals)	[ ]	[ ]	[ ]	[ ]	[ ]
I have a better understanding of how to act according to different german street rules.	[ ]	[ ]	[ ]	[ ]	[ ]
The Assistance System has aided me in driving more carefully.	[ ]	[ ]	[ ]	[ ]	[ ]
Efficiency	**1 Strongly Disagree**	**2**	**3**	**4**	**5 Strongly Agree**
It was easy to control the vehicle.	[ ]	[ ]	[ ]	[ ]	[ ]
The assistance system in this mode was easy to understand.	[ ]	[ ]	[ ]	[ ]	[ ]
I had enough driving time.	[ ]	[ ]	[ ]	[ ]	[ ]

Score:

Paderborn City VR

Effectiveness	**1 Strongly Disagree**	**2**	**3**	**4**	**5 Strongly Agree**
I found the driving tasks easy to understand.	[ ]	[ ]	[ ]	[ ]	[ ]
The Assistance System has improved my driving performance.	[ ]	[ ]	[ ]	[ ]	[ ]
The Assistance System gave me good feedback.	[ ]	[ ]	[ ]	[ ]	[ ]
The simulator has helped me with the use of the vehicle control devices (gearshift,Steering Wheel and Driving Pedals)	[ ]	[ ]	[ ]	[ ]	[ ]
I have a better understanding of how to act to different german street rules.	[ ]	[ ]	[ ]	[ ]	[ ]
The Assistance System has aided me in driving more carefully.	[ ]	[ ]	[ ]	[ ]	[ ]
My performance has improved after using the VR mode .	[ ]	[ ]	[ ]	[ ]	[ ]
Efficiency	**1 Strongly Disagree**	**2**	**3**	**4**	**5 Strongly Agree**
It was easy to control the vehicle.	[ ]	[ ]	[ ]	[ ]	[ ]
The assistance system in the last mode was very challenging and hard to understand.	[ ]	[ ]	[ ]	[ ]	[ ]
I had enough driving time in the last mode.	[ ]	[ ]	[ ]	[ ]	[ ]

Score:

General Questions

	1 Strongly Disagree	**2**	**3**	**4**	**5 Strongly Agree**
The driving simulator could be improved by adding more realistic training scenarios.	[ ]	[ ]	[ ]	[ ]	[ ]
I suggest enhancing the graphics.	[ ]	[ ]	[ ]	[ ]	[ ]
Overall, the assistance system is an excellent tool for car driving education.	[ ]	[ ]	[ ]	[ ]	[ ]
The simulatorgave you a good example of what it is like to drive in different daytimes?	[ ]	[ ]	[ ]	[ ]	[ ]
It was easy to navigate between different UI elements.	[ ]	[ ]	[ ]	[ ]	[ ]

Which transmission mode do you find better for you?	
automatic transmission	manual transmission

Which Mode do you feel was the most beneficial for you?		
Training Area	Highway	Paderborn City

Which daytime do you feel was the most beneficial for you?	
Night	Day

Which Car did you like to drive the most?		
BMW e92	Volkswagen Mk1	Fiat Spider

IPQ SURVEY ON EXPERIENCES IN VIRTUAL WORLDS

How aware were you of the real world surrounding while navigating in the virtual world? (i.e. sounds, room temperature, other people, etc.)?	**-3** **extremely aware**	**-2**	**-1**	**0**	**+1**	**+2**	**+3** **not aware at all**
How real did the virtual world seem to you?	**-3** **completely real**	**-2**	**-1**	**0**	**+1**	**+2**	**+3** **not real at all**
I had a sense of acting in the virtual space, rather than operating something from outside.	**-3** **fully disagree**	**-2**	**-1**	**0**	**+1**	**+2**	**+3** **fully agree**
How much did your experience in the virtual environment seem consistent with your real world experience ?	**-3** **not consistent**	**-2**	**-1**	**0**	**+1**	**+2**	**+3** **very consistent**
How real did the virtual world seem to you?	**-3** **about as real as an imagined world**	**-2**	**-1**	**0**	**+1**	**+2**	**+3** **indistinguishable from the real world**
I did not feel present in the virtual space.	**-3** **did not feel**	**-2**	**-1**	**0**	**+1**	**+2**	**+3** **felt present**
I was not aware of my real environment.	**-3** **fully disagree**	**-2**	**-1**	**0**	**+1**	**+2**	**+3** **fully agree**
In the computer generated world, I had a sense of "being there."	**-3** **not at all**	**-2**	**-1**	**0**	**+1**	**+2**	**+3** **very much**

Somehow, I felt that the virtual world surrounded me.	**-3** **fully disagree**	**-2**	**-1**	**0**	**+1**	**+2**	**+3** **fully agree**
I felt present in the virtual space.	**-3** **fully disagree**	**-2**	**-1**	**0**	**+1**	**+2**	**+3** **fully agree**
I still paid attention to the real environment.	**-3** **fully disagree**	**-2**	**-1**	**0**	**+1**	**+2**	**+3** **fully agree**
The virtual world seemed more realistic than the real world.	**-3** **fully disagree**	**-2**	**-1**	**0**	**+1**	**+2**	**+3** **fully agree**
I felt like I was just perceiving pictures.	**-3** **fully disagree**	**-2**	**-1**	**0**	**+1**	**+2**	**+3** **fully agree**
I was completely captivated by the virtual world.	**-3** **fully disagree**	**-2**	**-1**	**0**	**+1**	**+2**	**+3** **fully agree**

1. How long did you interact with the virtual world? _____ min
2. Which video hardware did you use? _____
3. What audio output was available? _____
4. What type of software did you use? _____

Figure 8.6

NASA Task Load Index

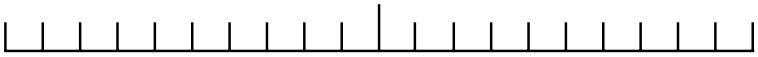
Hart and Staveland's NASA Task Load Index (TLX) method assesses work load on five 7-point scales. Increments of high, medium and low estimates for each point result in 21 gradations on the scales.

Name	Task	Date

Mental Demand — How mentally demanding was the task?

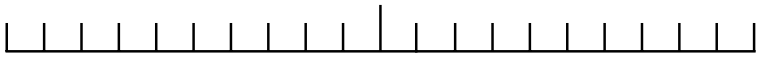
Very Low — Very High

Physical Demand — How physically demanding was the task?



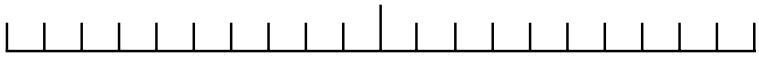
Very Low — Very High

Temporal Demand — How hurried or rushed was the pace of the task?



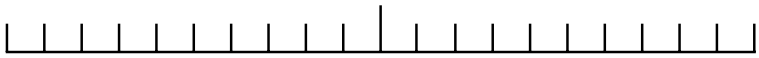
Very Low — Very High

Performance — How successful were you in accomplishing what you were asked to do?



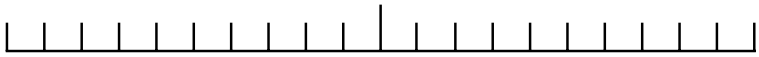
Perfect — Failure

Effort — How hard did you have to work to accomplish your level of performance?



Very Low — Very High

Frustration — How insecure, discouraged, irritated, stressed, and annoyed wereyou?



Very Low — Very High

System Usability Scale

© Digital Equipment Corporation, 1986.

	Strongly disagree				Strongly agree
1. I think that I would like to use this system frequently	1	2	3	4	5
2. I found the system unnecessarily complex	1	2	3	4	5
3. I thought the system was easy to use	1	2	3	4	5
4. I think that I would need the support of a technical person to be able to use this system	1	2	3	4	5
5. I found the various functions in this system were well integrated	1	2	3	4	5
6. I thought there was too much inconsistency in this system	1	2	3	4	5
7. I would imagine that most people would learn to use this system very quickly	1	2	3	4	5
8. I found the system very cumbersome to use	1	2	3	4	5
9. I felt very confident using the system	1	2	3	4	5
10. I needed to learn a lot of things before I could get going with this system	1	2	3	4	5